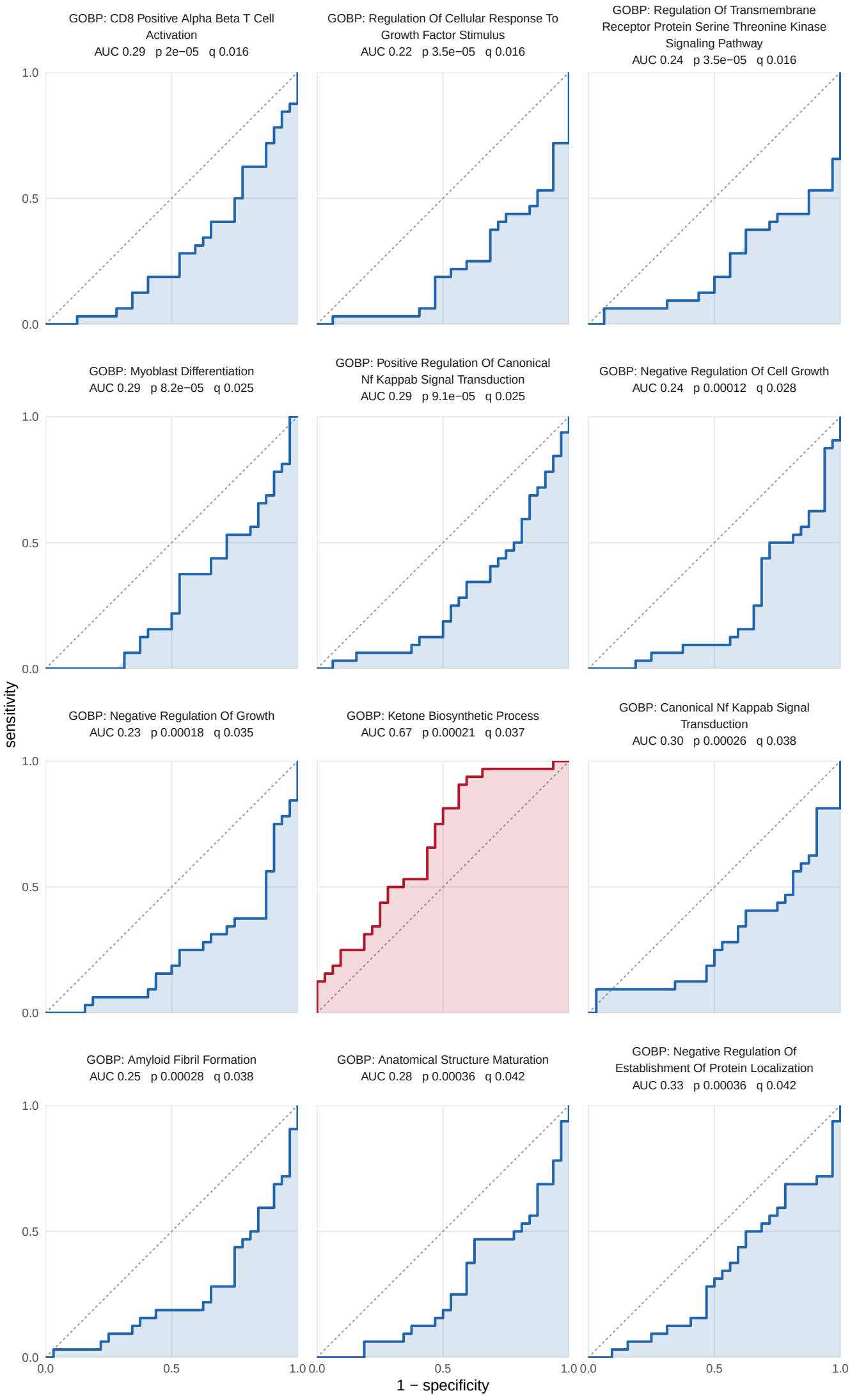
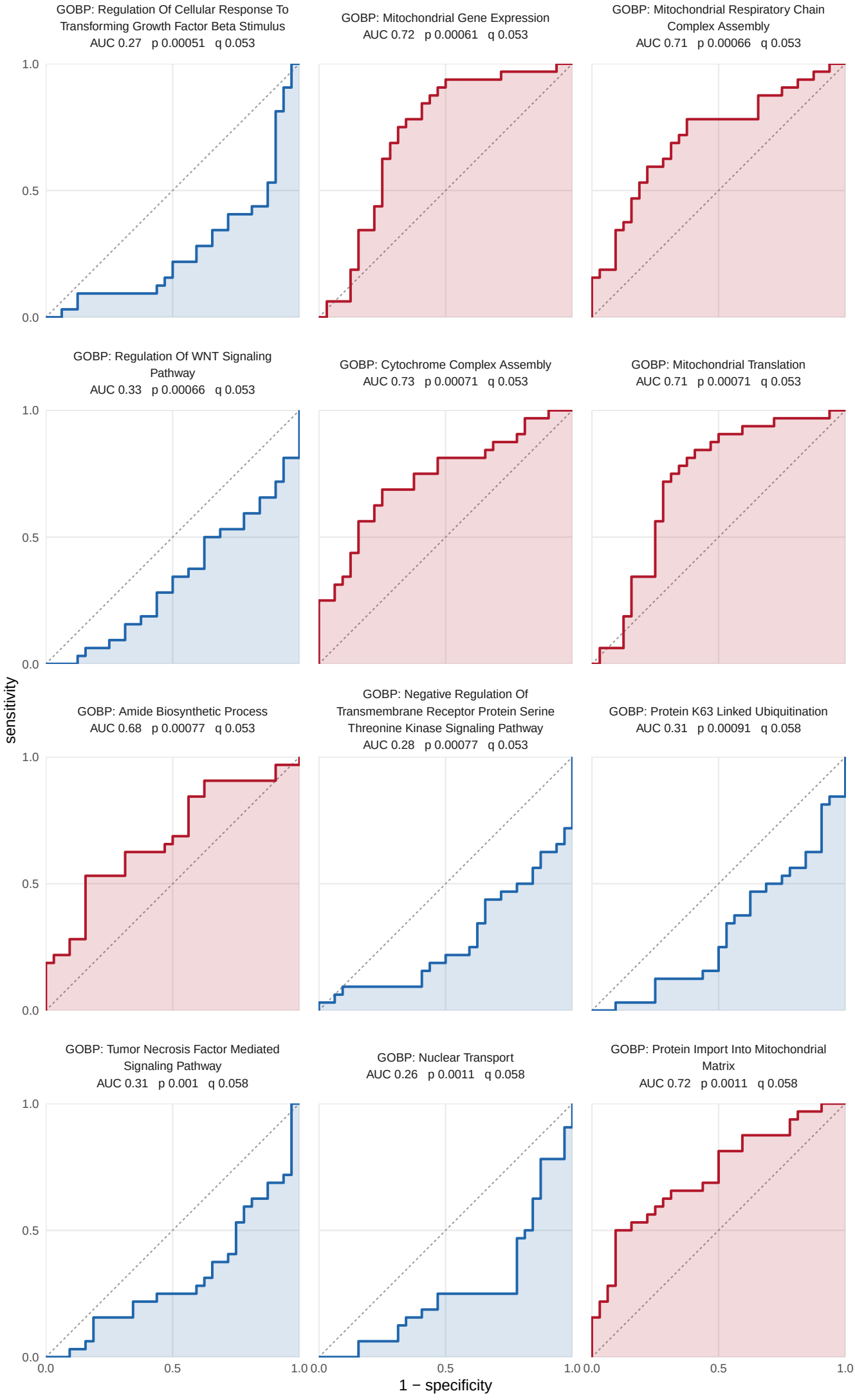
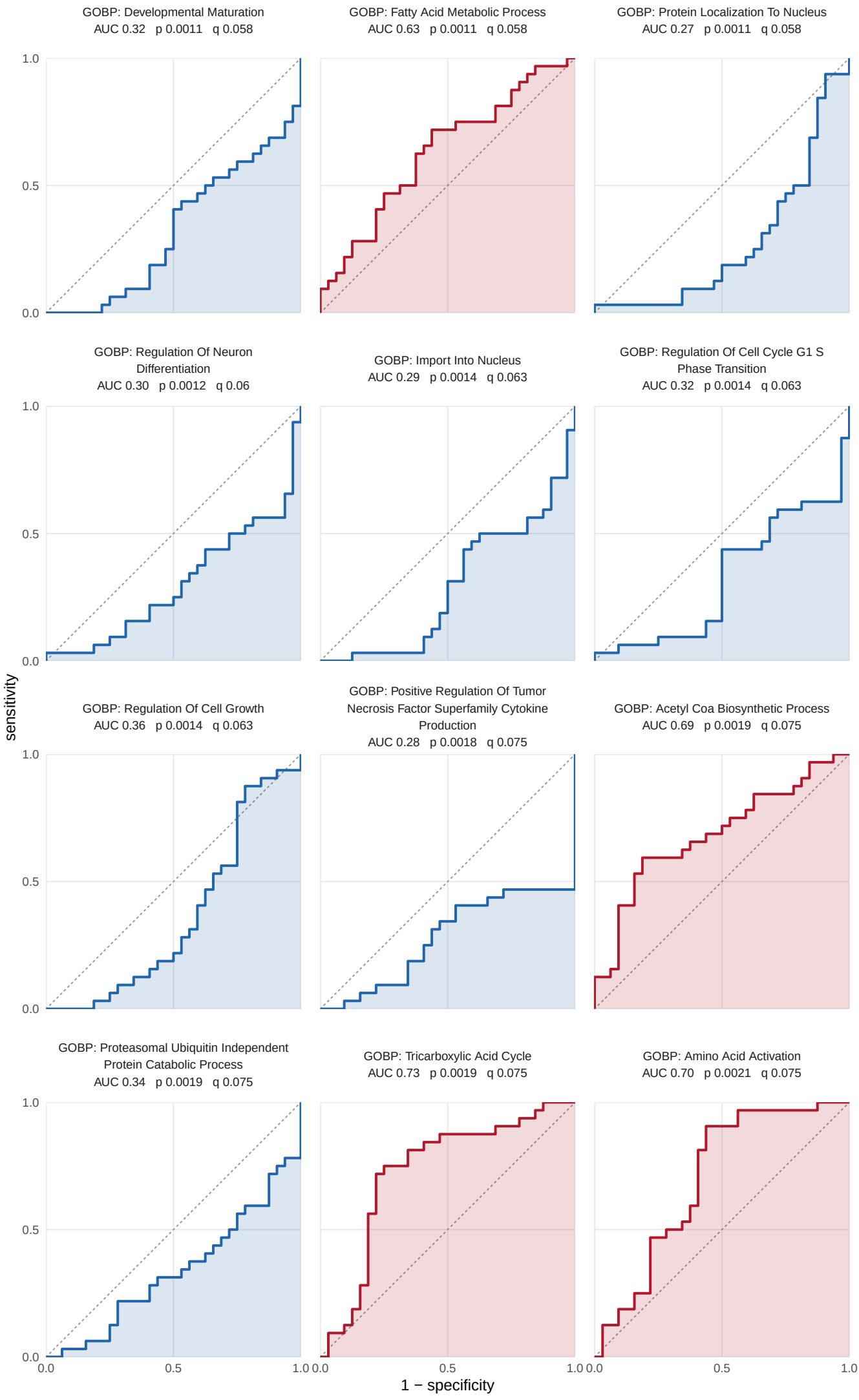
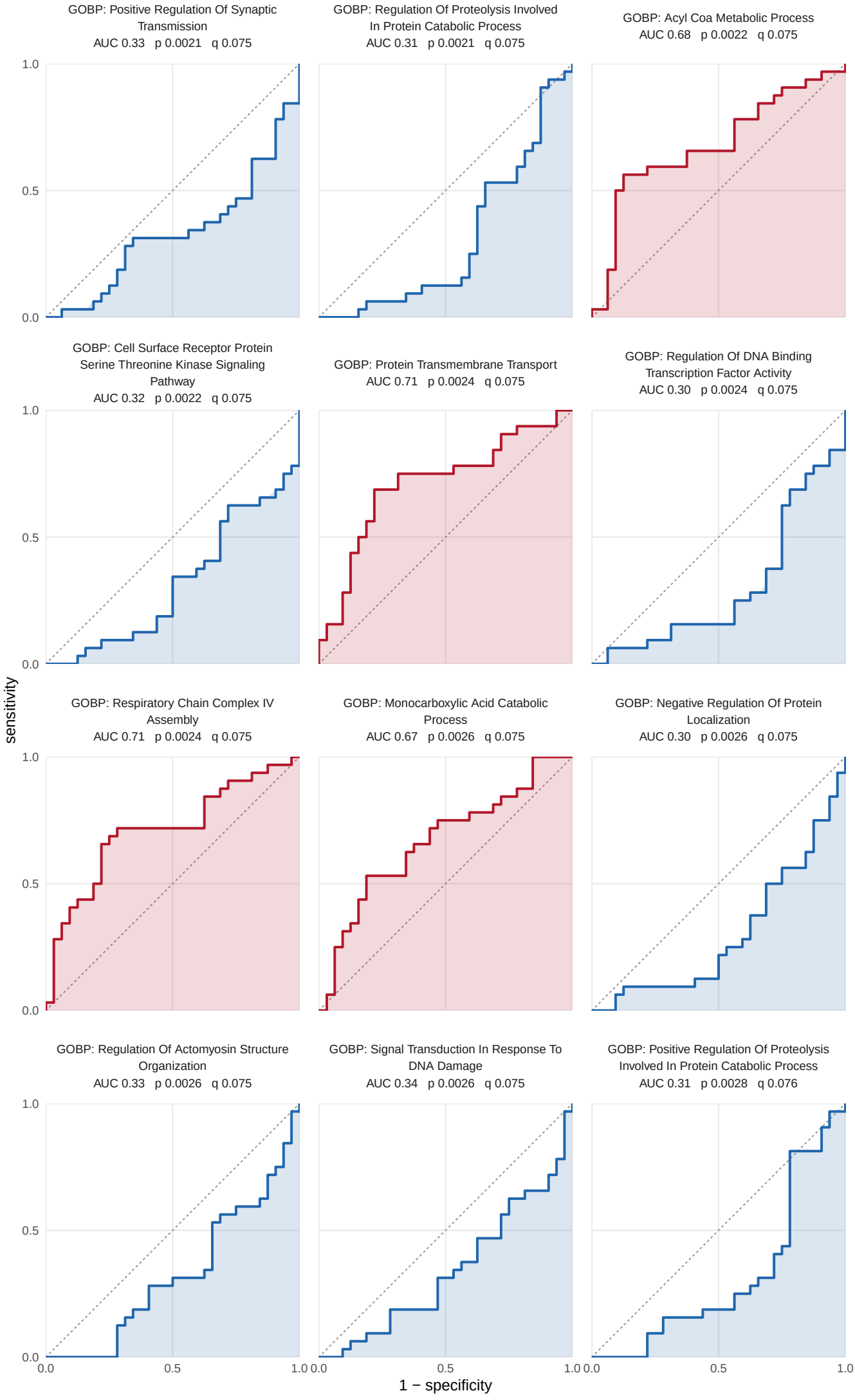
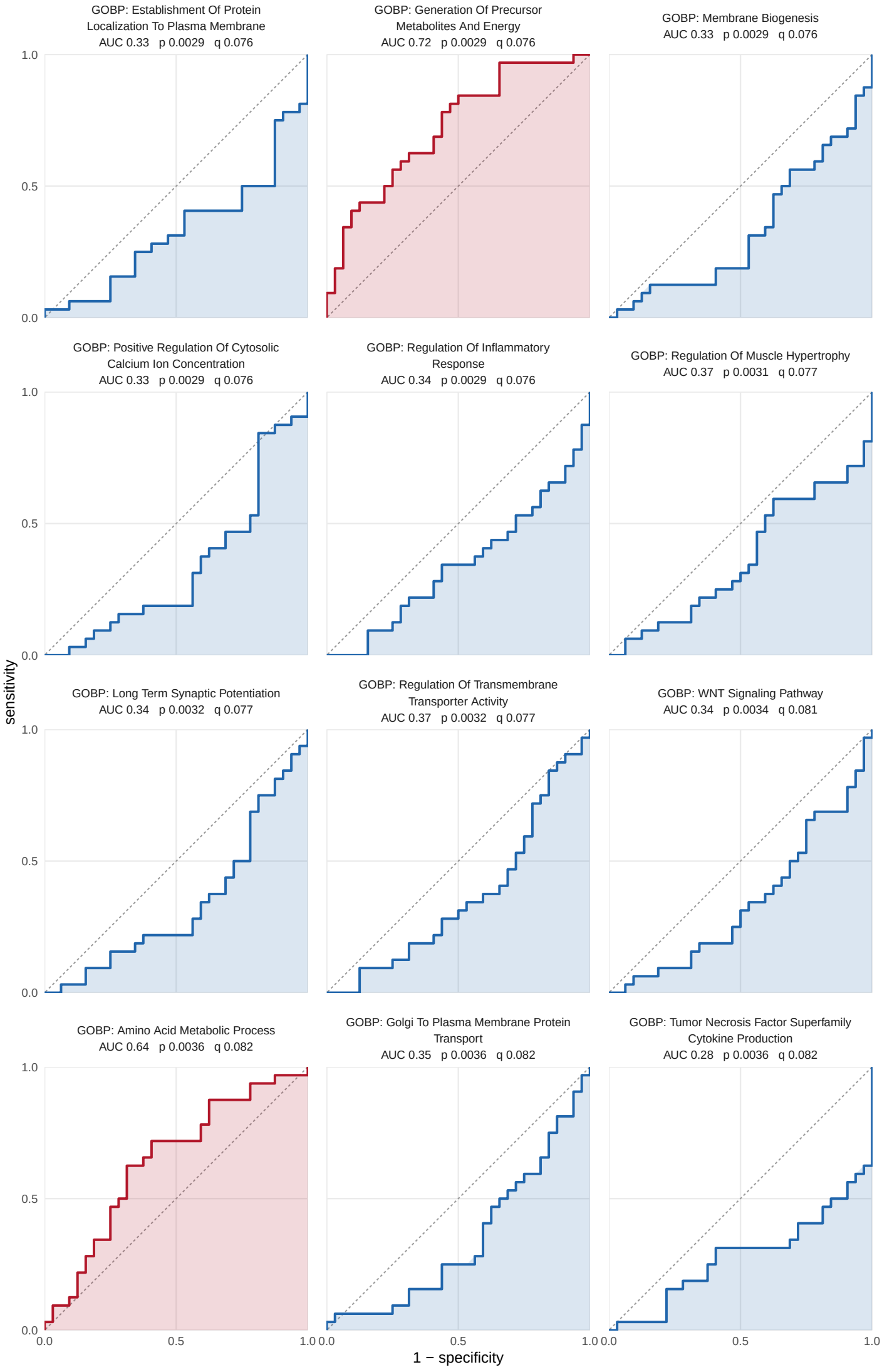
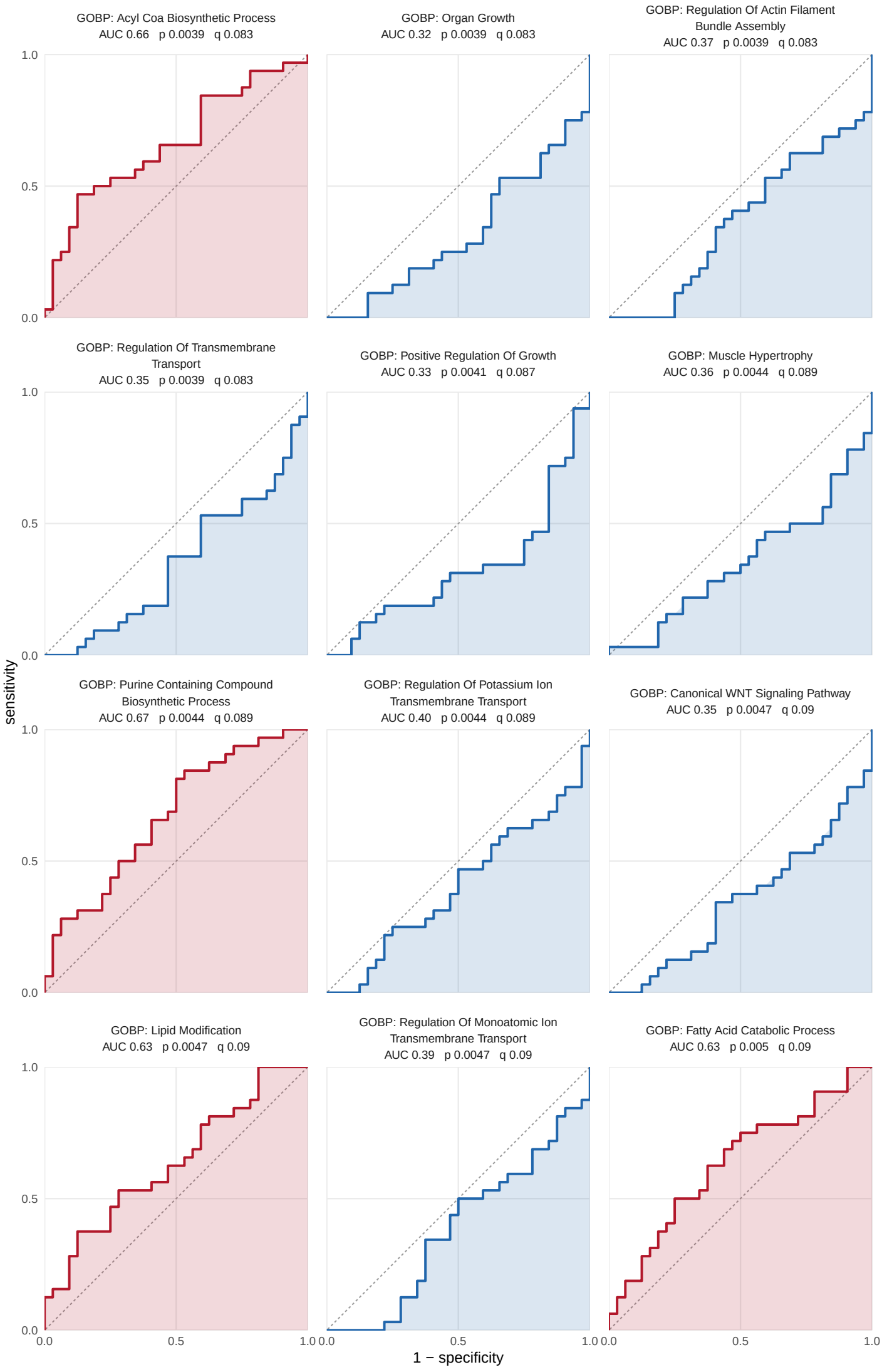

GOBP: CD8 Positive Alpha Beta T Cell Activation
AUC 0.29 p 2e−05 q 0.016
GOBP: Regulation Of Cellular Response To Growth Factor Stimulus
AUC 0.22 p 3.5e−05 q 0.016
GOBP: Regulation Of Transmembrane Receptor Protein Serine Threonine Kinase Signaling Pathway
AUC 0.24 p 3.5e−05 q 0.016
GOBP: Myoblast Differentiation
AUC 0.29 p 8.2e−05 q 0.025
GOBP: Positive Regulation Of Canonical Nf Kappab Signal Transduction
AUC 0.29 p 9.1e−05 q 0.025
GOBP: Negative Regulation Of Cell Growth
AUC 0.24 p 0.00012 q 0.028
GOBP: Negative Regulation Of Growth
AUC 0.23 p 0.00018 q 0.035
GOBP: Ketone Biosynthetic Process
AUC 0.67 p 0.00021 q 0.037
GOBP: Canonical Nf Kappab Signal Transduction
AUC 0.30 p 0.00026 q 0.038
GOBP: Amyloid Fibril Formation
AUC 0.25 p 0.00028 q 0.038
GOBP: Anatomical Structure Maturation
AUC 0.28 p 0.00036 q 0.042
GOBP: Negative Regulation Of Establishment Of Protein Localization
AUC 0.33 p 0.00036 q 0.042
sensitivity
1 − specificity
0.0
0.5
1.0

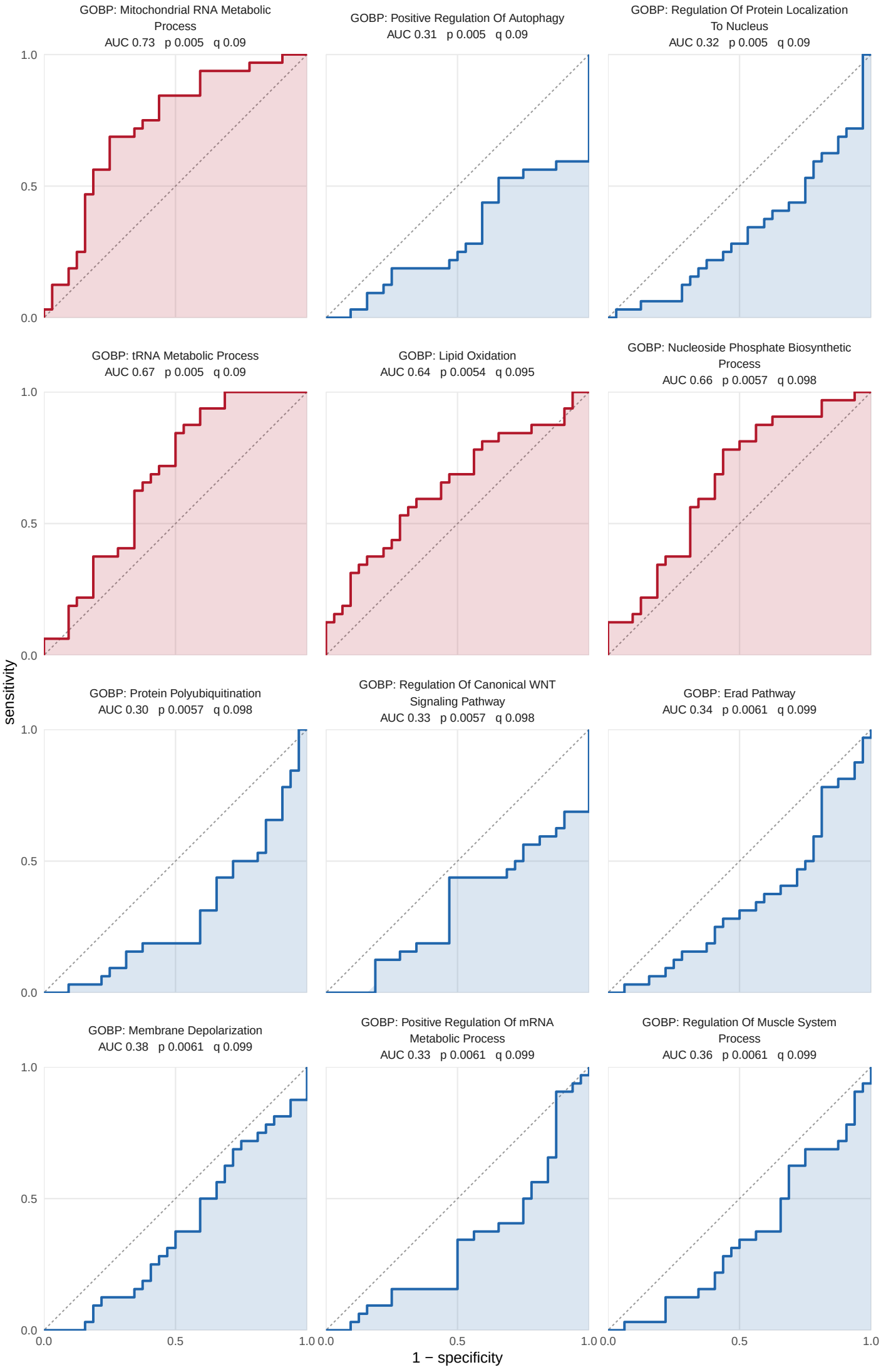

GOBP: Regulation Of Cellular Response To Transforming Growth Factor Beta Stimulus
AUC 0.27 p 0.00051 q 0.053
GOBP: Mitochondrial Gene Expression
AUC 0.72 p 0.00061 q 0.053
GOBP: Mitochondrial Respiratory Chain Complex Assembly
AUC 0.71 p 0.00066 q 0.053
GOBP: Regulation Of WNT Signaling Pathway
AUC 0.33 p 0.00066 q 0.053
GOBP: Cytochrome Complex Assembly
AUC 0.73 p 0.00071 q 0.053
GOBP: Mitochondrial Translation
AUC 0.71 p 0.00071 q 0.053
GOBP: Amide Biosynthetic Process
AUC 0.68 p 0.00077 q 0.053
GOBP: Negative Regulation Of Transmembrane Receptor Protein Serine Threonine Kinase Signaling Pathway
AUC 0.28 p 0.00077 q 0.053
GOBP: Protein K63 Linked Ubiquitination
AUC 0.31 p 0.00091 q 0.058
GOBP: Tumor Necrosis Factor Mediated Signaling Pathway
AUC 0.31 p 0.001 q 0.058
GOBP: Nuclear Transport
AUC 0.26 p 0.0011 q 0.058
GOBP: Protein Import Into Mitochondrial Matrix
AUC 0.72 p 0.0011 q 0.058
sensitivity
1 − specificity
0.0
0.5
1.0

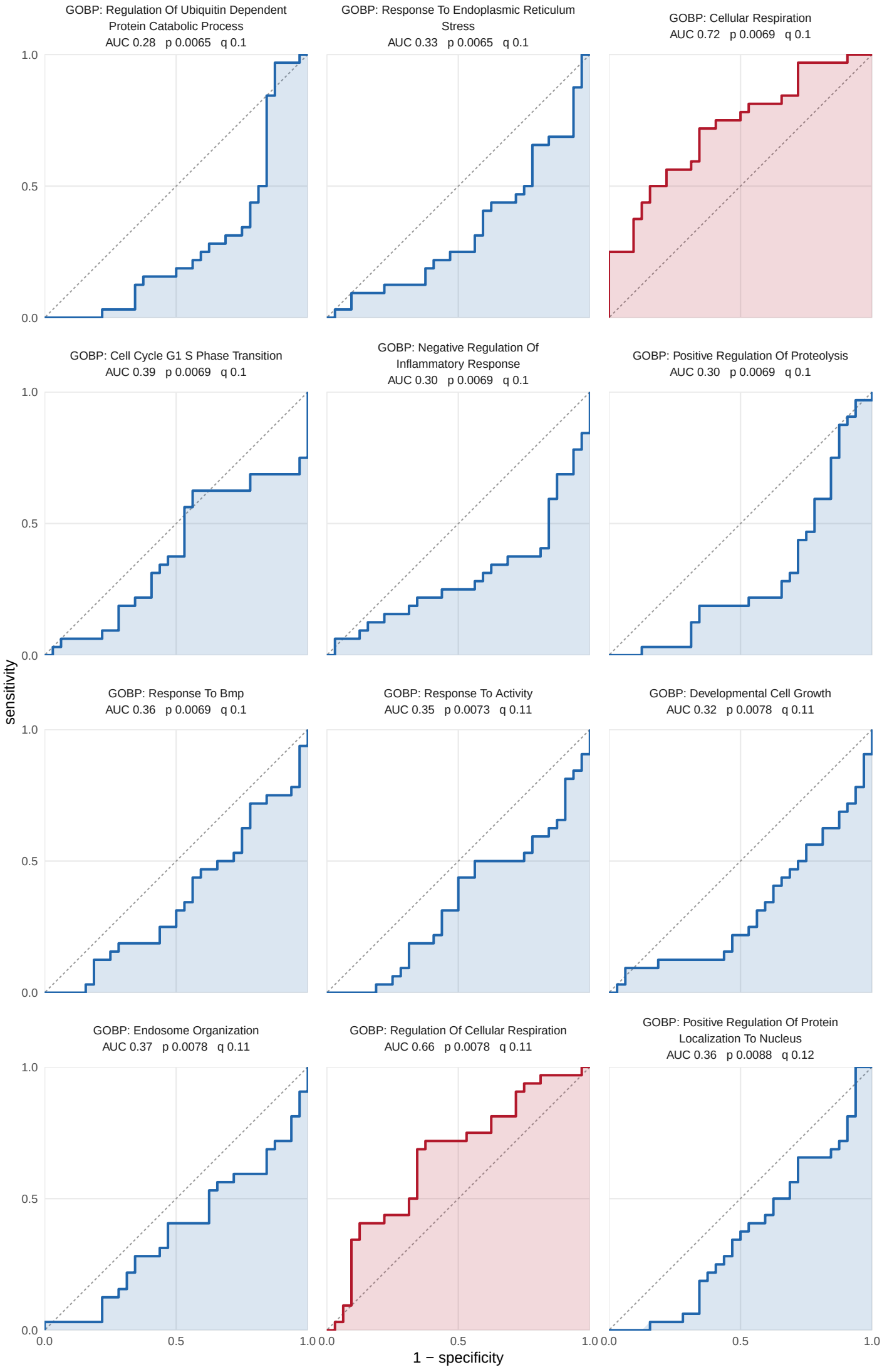

GOBP: Developmental Maturation
AUC 0.32 p 0.0011 q 0.058
GOBP: Fatty Acid Metabolic Process
AUC 0.63 p 0.0011 q 0.058
GOBP: Protein Localization To Nucleus
AUC 0.27 p 0.0011 q 0.058
GOBP: Regulation Of Neuron Differentiation
AUC 0.30 p 0.0012 q 0.06
GOBP: Import Into Nucleus
AUC 0.29 p 0.0014 q 0.063
GOBP: Regulation Of Cell Cycle G1 S Phase Transition
AUC 0.32 p 0.0014 q 0.063
GOBP: Regulation Of Cell Growth
AUC 0.36 p 0.0014 q 0.063
GOBP: Positive Regulation Of Tumor Necrosis Factor Superfamily Cytokine Production
AUC 0.28 p 0.0018 q 0.075
GOBP: Acetyl Coa Biosynthetic Process
AUC 0.69 p 0.0019 q 0.075
GOBP: Proteasomal Ubiquitin Independent Protein Catabolic Process
AUC 0.34 p 0.0019 q 0.075
GOBP: Tricarboxylic Acid Cycle
AUC 0.73 p 0.0019 q 0.075
GOBP: Amino Acid Activation
AUC 0.70 p 0.0021 q 0.075
sensitivity
1 − specificity
0.0
0.5
1.0

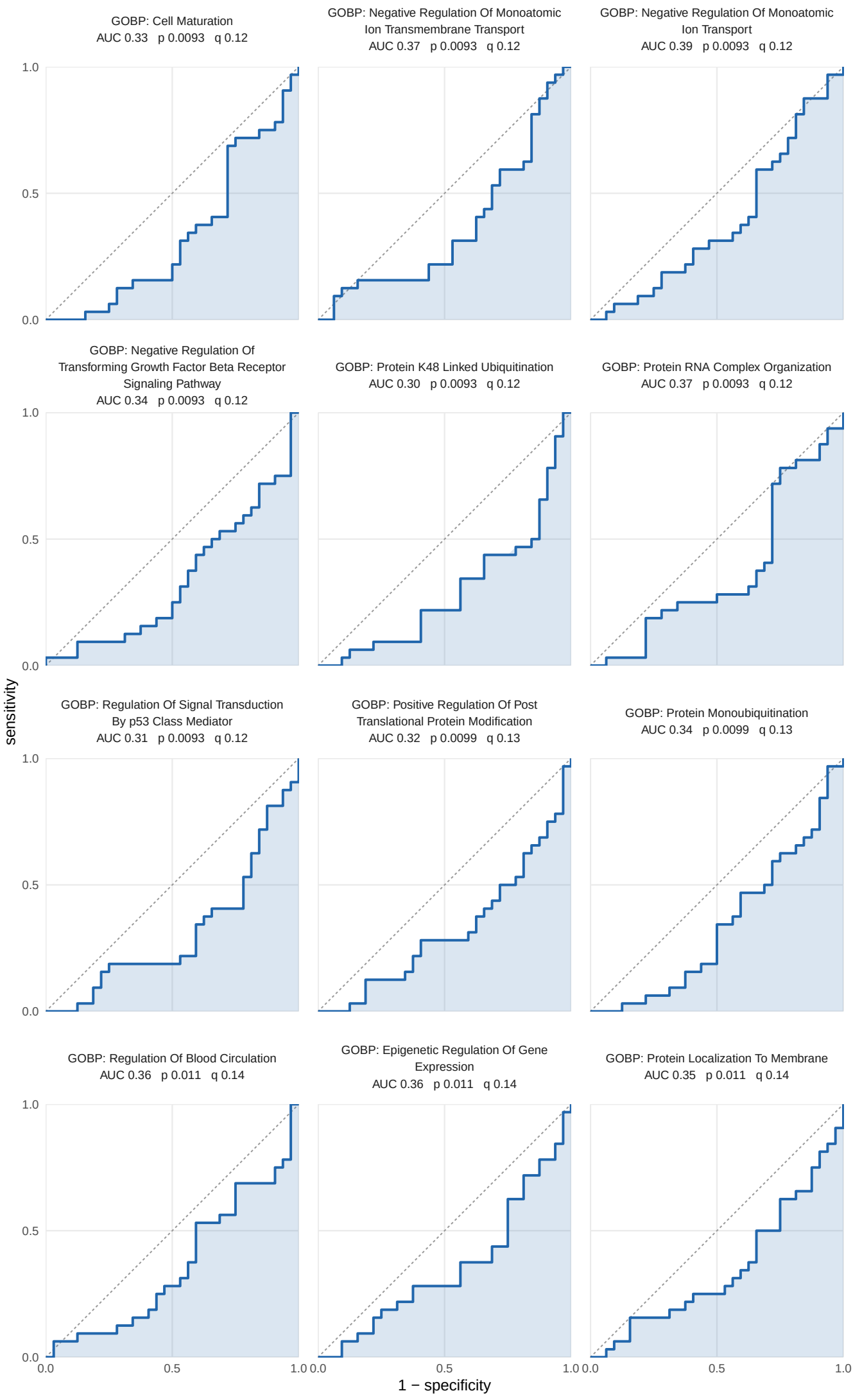

GOBP: Positive Regulation Of Synaptic Transmission
AUC 0.33 p 0.0021 q 0.075
GOBP: Regulation Of Proteolysis Involved In Protein Catabolic Process
AUC 0.31 p 0.0021 q 0.075
GOBP: Acyl Coa Metabolic Process
AUC 0.68 p 0.0022 q 0.075
GOBP: Cell Surface Receptor Protein Serine Threonine Kinase Signaling Pathway
AUC 0.32 p 0.0022 q 0.075
GOBP: Protein Transmembrane Transport
AUC 0.71 p 0.0024 q 0.075
GOBP: Regulation Of DNA Binding Transcription Factor Activity
AUC 0.30 p 0.0024 q 0.075
GOBP: Respiratory Chain Complex IV Assembly
AUC 0.71 p 0.0024 q 0.075
GOBP: Monocarboxylic Acid Catabolic Process
AUC 0.67 p 0.0026 q 0.075
GOBP: Negative Regulation Of Protein Localization
AUC 0.30 p 0.0026 q 0.075
GOBP: Regulation Of Actomyosin Structure Organization
AUC 0.33 p 0.0026 q 0.075
GOBP: Signal Transduction In Response To DNA Damage
AUC 0.34 p 0.0026 q 0.075
GOBP: Positive Regulation Of Proteolysis Involved In Protein Catabolic Process
AUC 0.31 p 0.0028 q 0.076
sensitivity
1 − specificity
0.0
0.5
1.0

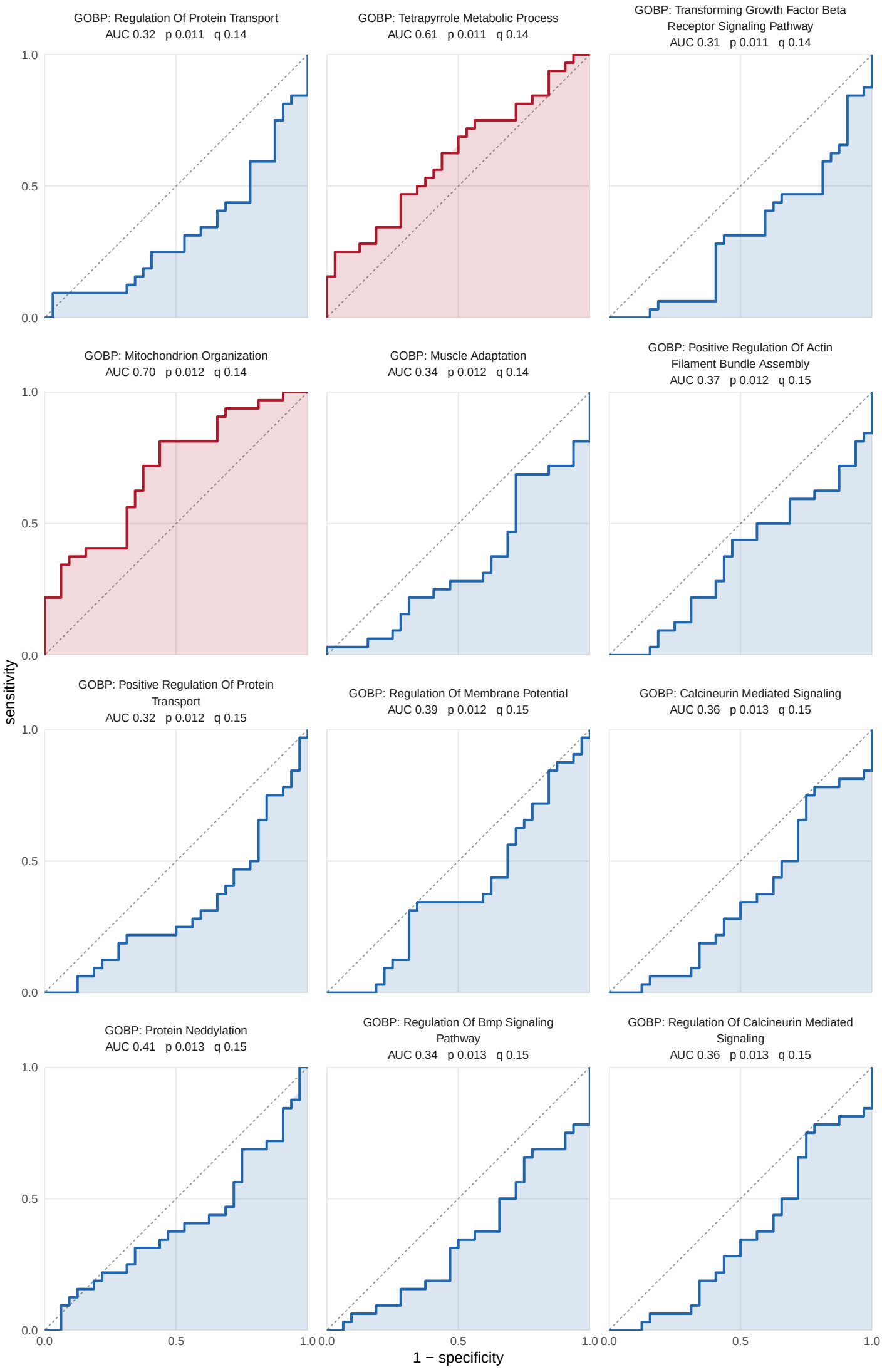

GOBP: Establishment Of Protein Localization To Plasma Membrane
AUC 0.33 p 0.0029 q 0.076
GOBP: Generation Of Precursor Metabolites And Energy
AUC 0.72 p 0.0029 q 0.076
GOBP: Membrane Biogenesis
AUC 0.33 p 0.0029 q 0.076
GOBP: Positive Regulation Of Cytosolic Calcium Ion Concentration
AUC 0.33 p 0.0029 q 0.076
GOBP: Regulation Of Inflammatory Response
AUC 0.34 p 0.0029 q 0.076
GOBP: Regulation Of Muscle Hypertrophy
AUC 0.37 p 0.0031 q 0.077
GOBP: Long Term Synaptic Potentiation
AUC 0.34 p 0.0032 q 0.077
GOBP: Regulation Of Transmembrane Transporter Activity
AUC 0.37 p 0.0032 q 0.077
GOBP: WNT Signaling Pathway
AUC 0.34 p 0.0034 q 0.081
GOBP: Amino Acid Metabolic Process
AUC 0.64 p 0.0036 q 0.082
GOBP: Golgi To Plasma Membrane Protein Transport
AUC 0.35 p 0.0036 q 0.082
GOBP: Tumor Necrosis Factor Superfamily Cytokine Production
AUC 0.28 p 0.0036 q 0.082
sensitivity
1 − specificity
0.0
0.5
1.0

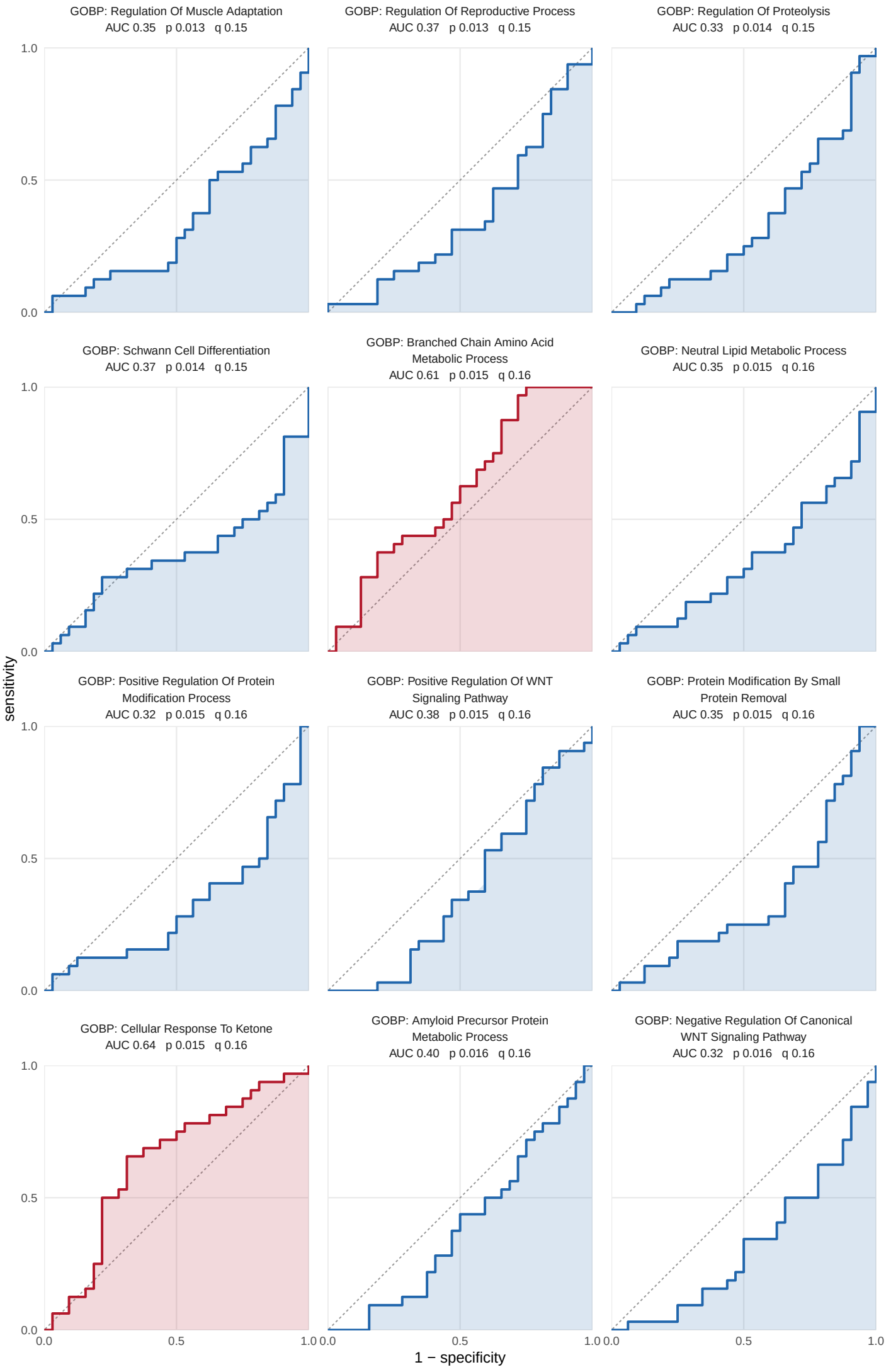

GOBP: Acyl Coa Biosynthetic Process
AUC 0.66 p 0.0039 q 0.083
GOBP: Organ Growth
AUC 0.32 p 0.0039 q 0.083
GOBP: Regulation Of Actin Filament Bundle Assembly
AUC 0.37 p 0.0039 q 0.083
GOBP: Regulation Of Transmembrane Transport
AUC 0.35 p 0.0039 q 0.083
GOBP: Positive Regulation Of Growth
AUC 0.33 p 0.0041 q 0.087
GOBP: Muscle Hypertrophy
AUC 0.36 p 0.0044 q 0.089
GOBP: Purine Containing Compound Biosynthetic Process
AUC 0.67 p 0.0044 q 0.089
GOBP: Regulation Of Potassium Ion Transmembrane Transport
AUC 0.40 p 0.0044 q 0.089
GOBP: Canonical WNT Signaling Pathway
AUC 0.35 p 0.0047 q 0.09
GOBP: Lipid Modification
AUC 0.63 p 0.0047 q 0.09
GOBP: Regulation Of Monoatomic Ion Transmembrane Transport
AUC 0.39 p 0.0047 q 0.09
GOBP: Fatty Acid Catabolic Process
AUC 0.63 p 0.005 q 0.09
sensitivity
1 − specificity
0.0
0.5
1.0

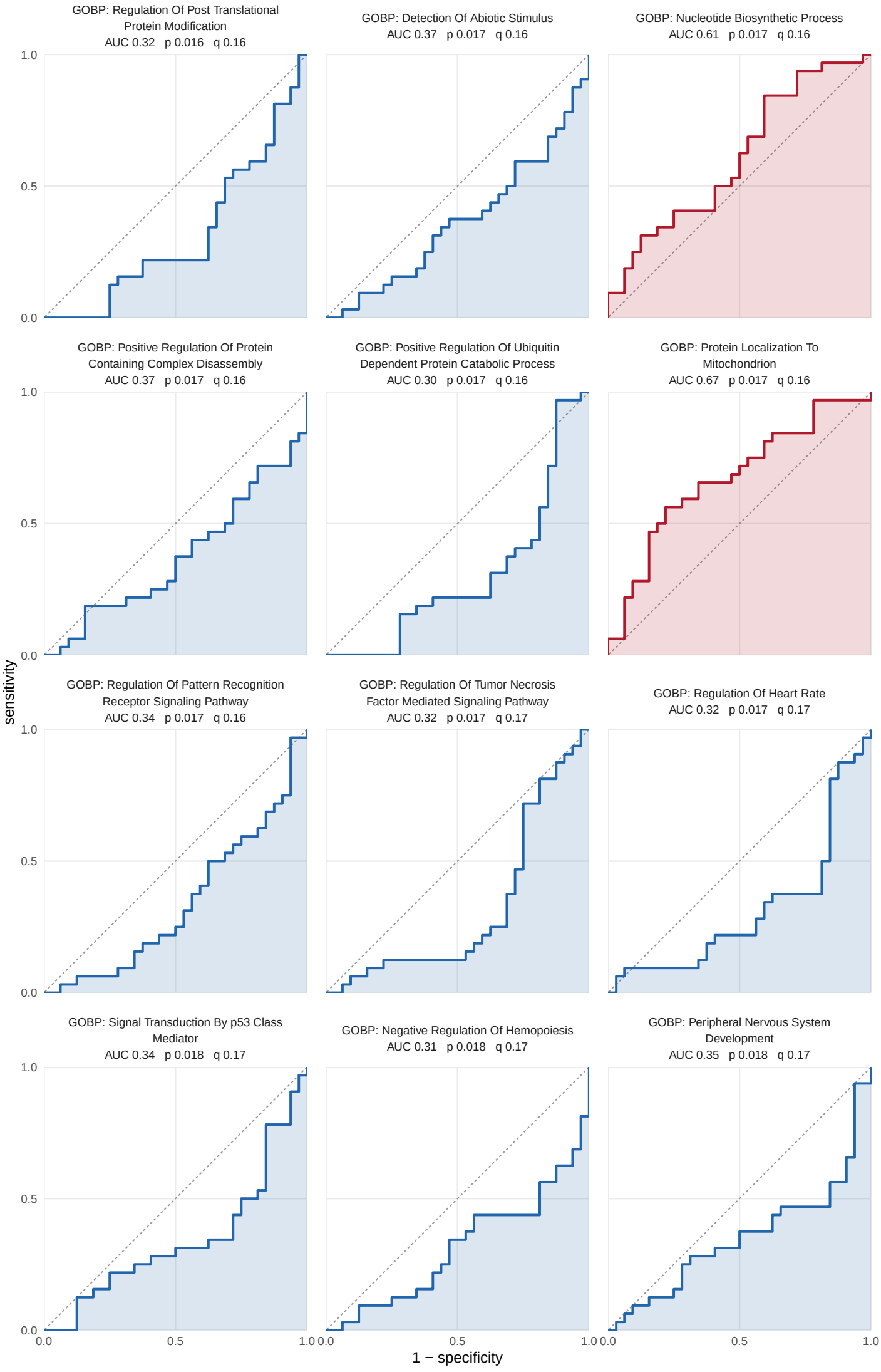

GOBP: Mitochondrial RNA Metabolic Process
AUC 0.73 p 0.005 q 0.09
GOBP: Positive Regulation Of Autophagy
AUC 0.31 p 0.005 q 0.09
GOBP: Regulation Of Protein Localization To Nucleus
AUC 0.32 p 0.005 q 0.09
GOBP: tRNA Metabolic Process
AUC 0.67 p 0.005 q 0.09
GOBP: Lipid Oxidation
AUC 0.64 p 0.0054 q 0.095
GOBP: Nucleoside Phosphate Biosynthetic Process
AUC 0.66 p 0.0057 q 0.098
GOBP: Protein Polyubiquitination
AUC 0.30 p 0.0057 q 0.098
GOBP: Regulation Of Canonical WNT Signaling Pathway
AUC 0.33 p 0.0057 q 0.098
GOBP: Erad Pathway
AUC 0.34 p 0.0061 q 0.099
GOBP: Membrane Depolarization
AUC 0.38 p 0.0061 q 0.099
GOBP: Positive Regulation Of mRNA Metabolic Process
AUC 0.33 p 0.0061 q 0.099
GOBP: Regulation Of Muscle System Process
AUC 0.36 p 0.0061 q 0.099
sensitivity
1 − specificity
0.0
0.5
1.0

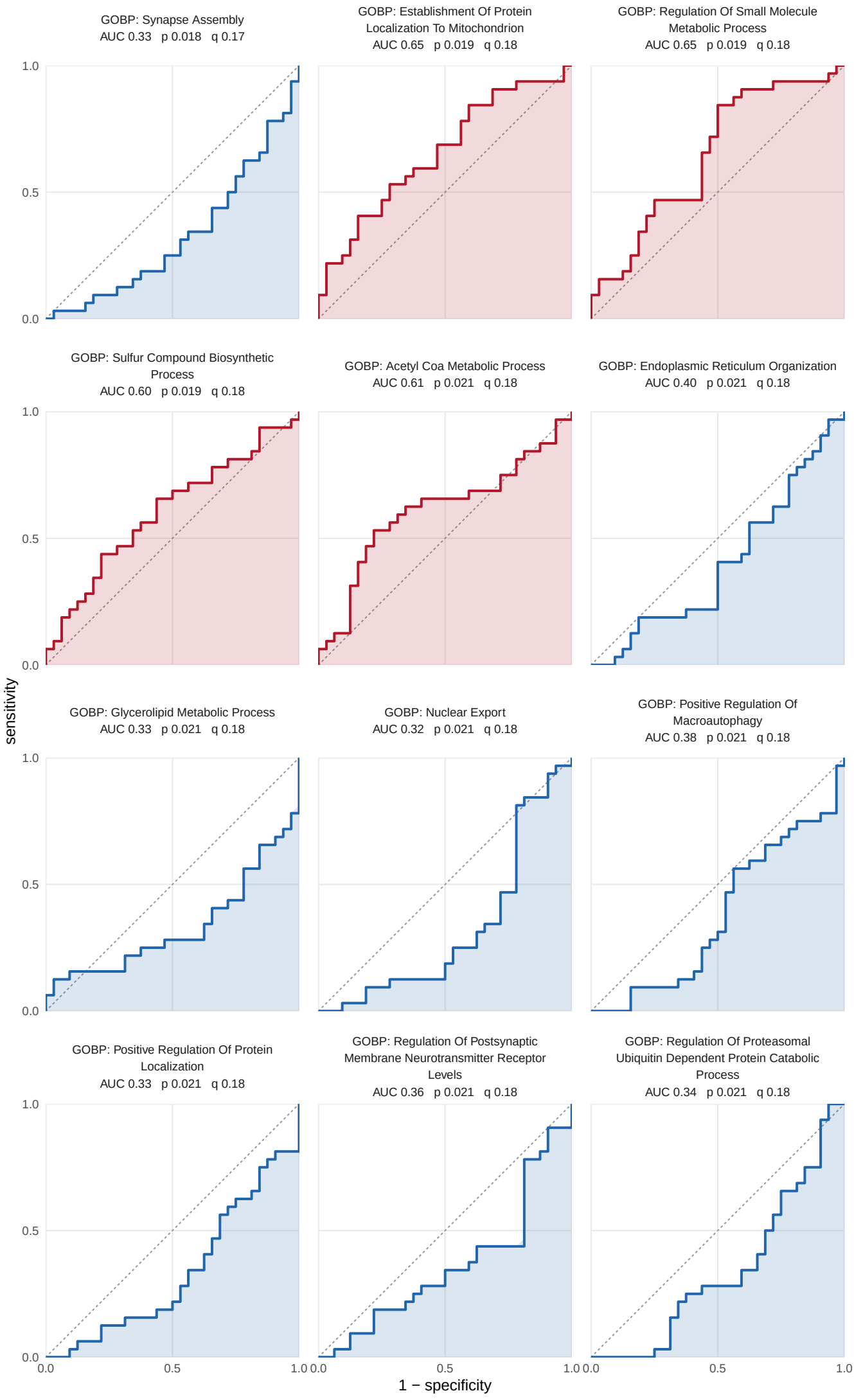

GOBP: Regulation Of Ubiquitin Dependent Protein Catabolic Process
AUC 0.28 p 0.0065 q 0.1
GOBP: Response To Endoplasmic Reticulum Stress
AUC 0.33 p 0.0065 q 0.1
GOBP: Cellular Respiration
AUC 0.72 p 0.0069 q 0.1
GOBP: Cell Cycle G1 S Phase Transition
AUC 0.39 p 0.0069 q 0.1
GOBP: Negative Regulation Of Inflammatory Response
AUC 0.30 p 0.0069 q 0.1
GOBP: Positive Regulation Of Proteolysis
AUC 0.30 p 0.0069 q 0.1
GOBP: Response To Bmp
AUC 0.36 p 0.0069 q 0.1
GOBP: Response To Activity
AUC 0.35 p 0.0073 q 0.11
GOBP: Developmental Cell Growth
AUC 0.32 p 0.0078 q 0.11
GOBP: Endosome Organization
AUC 0.37 p 0.0078 q 0.11
GOBP: Regulation Of Cellular Respiration
AUC 0.66 p 0.0078 q 0.11
GOBP: Positive Regulation Of Protein Localization To Nucleus
AUC 0.36 p 0.0088 q 0.12
sensitivity
1 − specificity
0.0
0.5
1.0

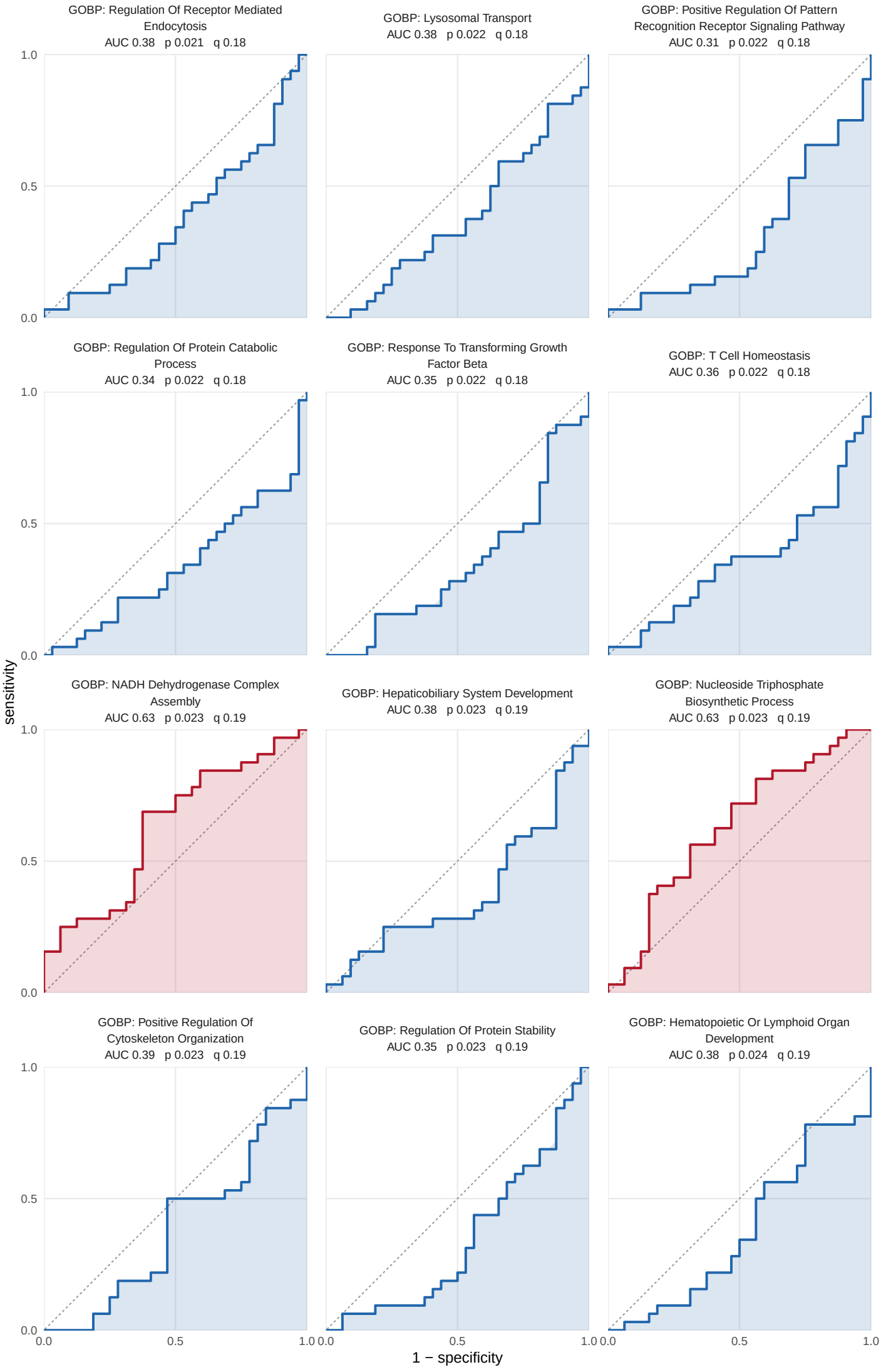

GOBP: Cell Maturation
AUC 0.33 p 0.0093 q 0.12
GOBP: Negative Regulation Of Monoatomic Ion Transmembrane Transport
AUC 0.37 p 0.0093 q 0.12
GOBP: Negative Regulation Of Monoatomic Ion Transport
AUC 0.39 p 0.0093 q 0.12
GOBP: Negative Regulation Of Transforming Growth Factor Beta Receptor Signaling Pathway
AUC 0.34 p 0.0093 q 0.12
GOBP: Protein K48 Linked Ubiquitination
AUC 0.30 p 0.0093 q 0.12
GOBP: Protein RNA Complex Organization
AUC 0.37 p 0.0093 q 0.12
GOBP: Regulation Of Signal Transduction By p53 Class Mediator
AUC 0.31 p 0.0093 q 0.12
GOBP: Positive Regulation Of Post Translational Protein Modification
AUC 0.32 p 0.0099 q 0.13
GOBP: Protein Monoubiquitination
AUC 0.34 p 0.0099 q 0.13
GOBP: Regulation Of Blood Circulation
AUC 0.36 p 0.011 q 0.14
GOBP: Epigenetic Regulation Of Gene Expression
AUC 0.36 p 0.011 q 0.14
GOBP: Protein Localization To Membrane
AUC 0.35 p 0.011 q 0.14
sensitivity
1 − specificity
0.0
0.5
1.0

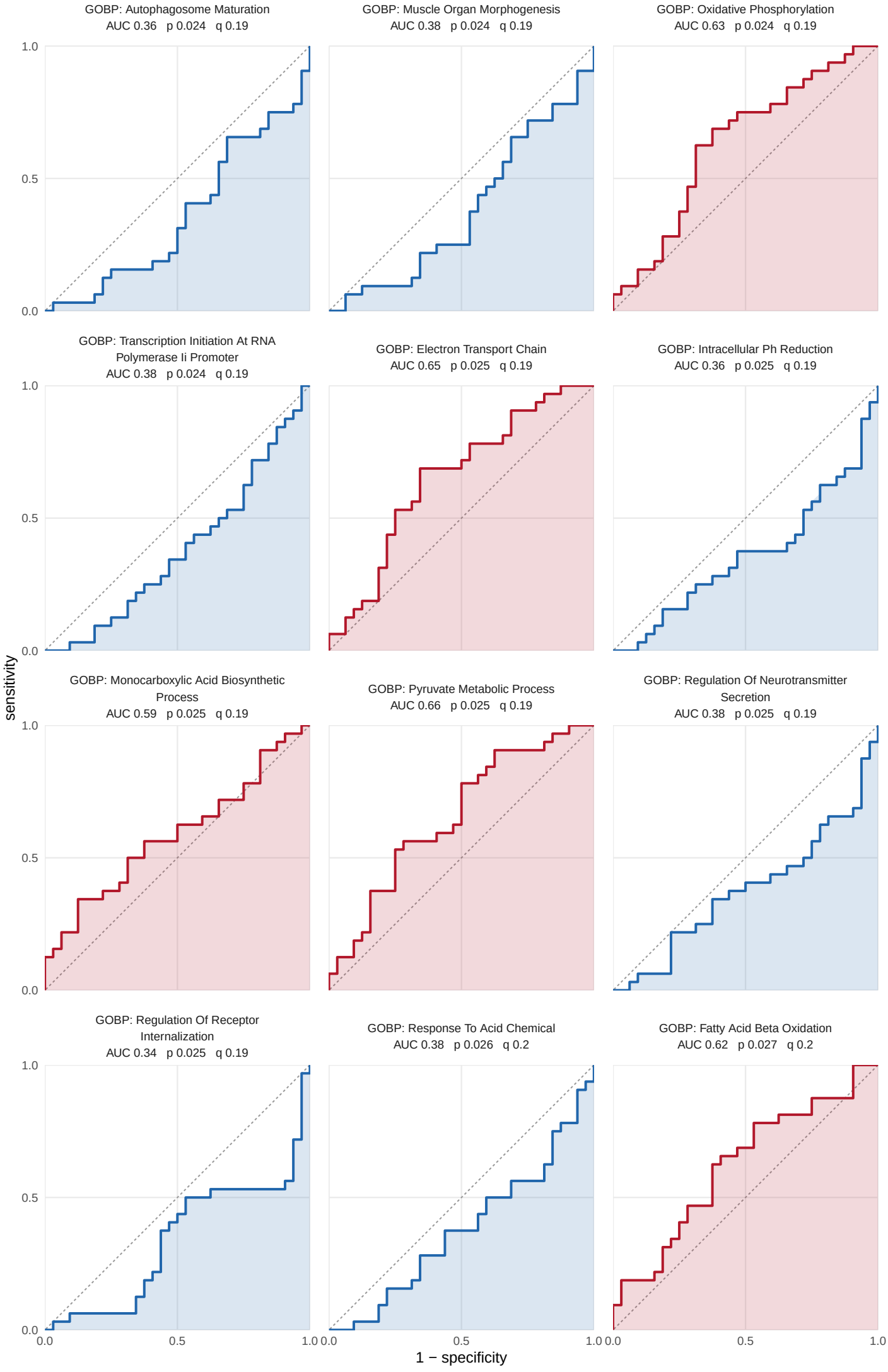

GOBP: Regulation Of Protein Transport
AUC 0.32 p 0.011 q 0.14
GOBP: Tetrapyrrole Metabolic Process
AUC 0.61 p 0.011 q 0.14
GOBP: Transforming Growth Factor Beta Receptor Signaling Pathway
AUC 0.31 p 0.011 q 0.14
GOBP: Mitochondrion Organization
AUC 0.70 p 0.012 q 0.14
GOBP: Muscle Adaptation
AUC 0.34 p 0.012 q 0.14
GOBP: Positive Regulation Of Actin Filament Bundle Assembly
AUC 0.37 p 0.012 q 0.15
GOBP: Positive Regulation Of Protein Transport
AUC 0.32 p 0.012 q 0.15
GOBP: Regulation Of Membrane Potential
AUC 0.39 p 0.012 q 0.15
GOBP: Calcineurin Mediated Signaling
AUC 0.36 p 0.013 q 0.15
GOBP: Protein Neddylation
AUC 0.41 p 0.013 q 0.15
GOBP: Regulation Of Bmp Signaling Pathway
AUC 0.34 p 0.013 q 0.15
GOBP: Regulation Of Calcineurin Mediated Signaling
AUC 0.36 p 0.013 q 0.15
sensitivity
1 − specificity
0.0
0.5
1.0

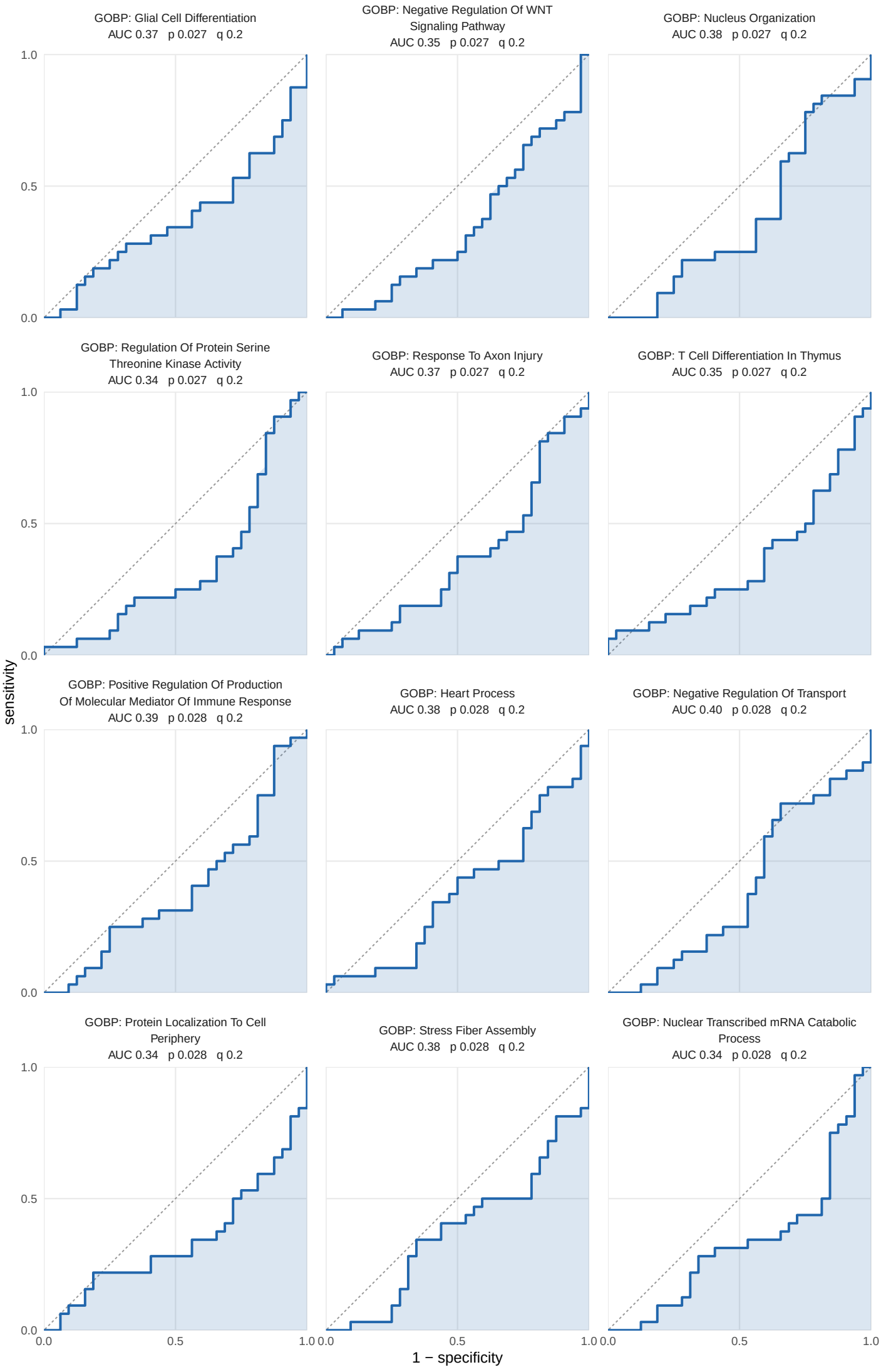

GOBP: Regulation Of Muscle Adaptation
AUC 0.35 p 0.013 q 0.15
GOBP: Regulation Of Reproductive Process
AUC 0.37 p 0.013 q 0.15
GOBP: Regulation Of Proteolysis
AUC 0.33 p 0.014 q 0.15
GOBP: Schwann Cell Differentiation
AUC 0.37 p 0.014 q 0.15
GOBP: Branched Chain Amino Acid Metabolic Process
AUC 0.61 p 0.015 q 0.16
GOBP: Neutral Lipid Metabolic Process
AUC 0.35 p 0.015 q 0.16
GOBP: Positive Regulation Of Protein Modification Process
AUC 0.32 p 0.015 q 0.16
GOBP: Positive Regulation Of WNT Signaling Pathway
AUC 0.38 p 0.015 q 0.16
GOBP: Protein Modification By Small Protein Removal
AUC 0.35 p 0.015 q 0.16
GOBP: Cellular Response To Ketone
AUC 0.64 p 0.015 q 0.16
GOBP: Amyloid Precursor Protein Metabolic Process
AUC 0.40 p 0.016 q 0.16
GOBP: Negative Regulation Of Canonical WNT Signaling Pathway
AUC 0.32 p 0.016 q 0.16
sensitivity
1 − specificity
0.0
0.5
1.0

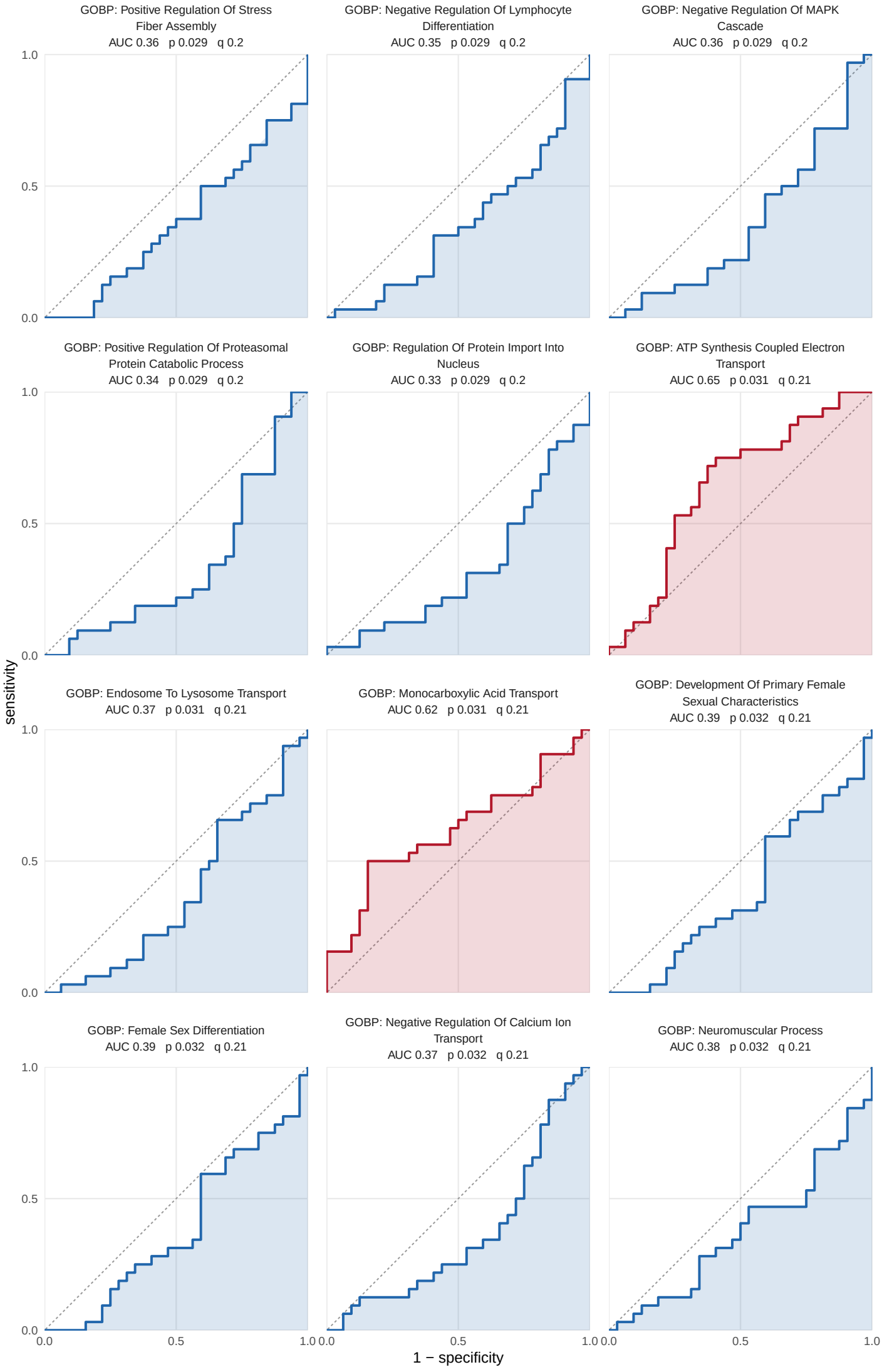

GOBP: Regulation Of Post Translational Protein Modification
AUC 0.32 p 0.016 q 0.16
GOBP: Detection Of Abiotic Stimulus
AUC 0.37 p 0.017 q 0.16
GOBP: Nucleotide Biosynthetic Process
AUC 0.61 p 0.017 q 0.16
GOBP: Positive Regulation Of Protein Containing Complex Disassembly
AUC 0.37 p 0.017 q 0.16
GOBP: Positive Regulation Of Ubiquitin Dependent Protein Catabolic Process
AUC 0.30 p 0.017 q 0.16
GOBP: Protein Localization To Mitochondrion
AUC 0.67 p 0.017 q 0.16
GOBP: Regulation Of Pattern Recognition Receptor Signaling Pathway
AUC 0.34 p 0.017 q 0.16
GOBP: Regulation Of Tumor Necrosis Factor Mediated Signaling Pathway
AUC 0.32 p 0.017 q 0.17
GOBP: Regulation Of Heart Rate
AUC 0.32 p 0.017 q 0.17
GOBP: Signal Transduction By p53 Class Mediator
AUC 0.34 p 0.018 q 0.17
GOBP: Negative Regulation Of Hemopoiesis
AUC 0.31 p 0.018 q 0.17
GOBP: Peripheral Nervous System Development
AUC 0.35 p 0.018 q 0.17
sensitivity
1 − specificity
0.0
0.5
1.0

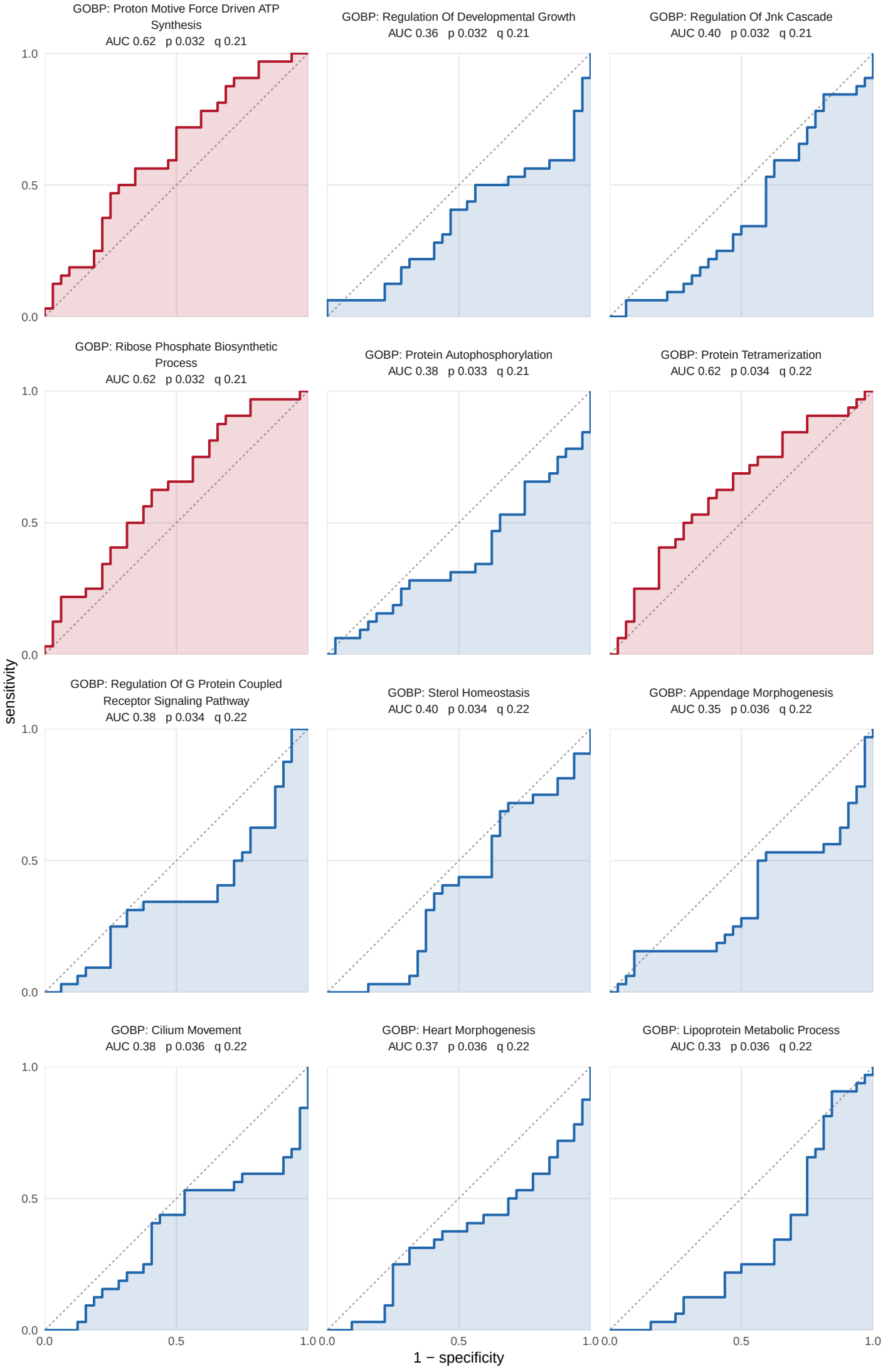

GOBP: Synapse Assembly
AUC 0.33 p 0.018 q 0.17
GOBP: Establishment Of Protein Localization To Mitochondrion
AUC 0.65 p 0.019 q 0.18
GOBP: Regulation Of Small Molecule Metabolic Process
AUC 0.65 p 0.019 q 0.18
GOBP: Sulfur Compound Biosynthetic Process
AUC 0.60 p 0.019 q 0.18
GOBP: Acetyl Coa Metabolic Process
AUC 0.61 p 0.021 q 0.18
GOBP: Endoplasmic Reticulum Organization
AUC 0.40 p 0.021 q 0.18
GOBP: Glycerolipid Metabolic Process
AUC 0.33 p 0.021 q 0.18
GOBP: Nuclear Export
AUC 0.32 p 0.021 q 0.18
GOBP: Positive Regulation Of Macroautophagy
AUC 0.38 p 0.021 q 0.18
GOBP: Positive Regulation Of Protein Localization
AUC 0.33 p 0.021 q 0.18
GOBP: Regulation Of Postsynaptic Membrane Neurotransmitter Receptor Levels
AUC 0.36 p 0.021 q 0.18
GOBP: Regulation Of Proteasomal Ubiquitin Dependent Protein Catabolic Process
AUC 0.34 p 0.021 q 0.18
sensitivity
1 − specificity
0.0
0.5
1.0

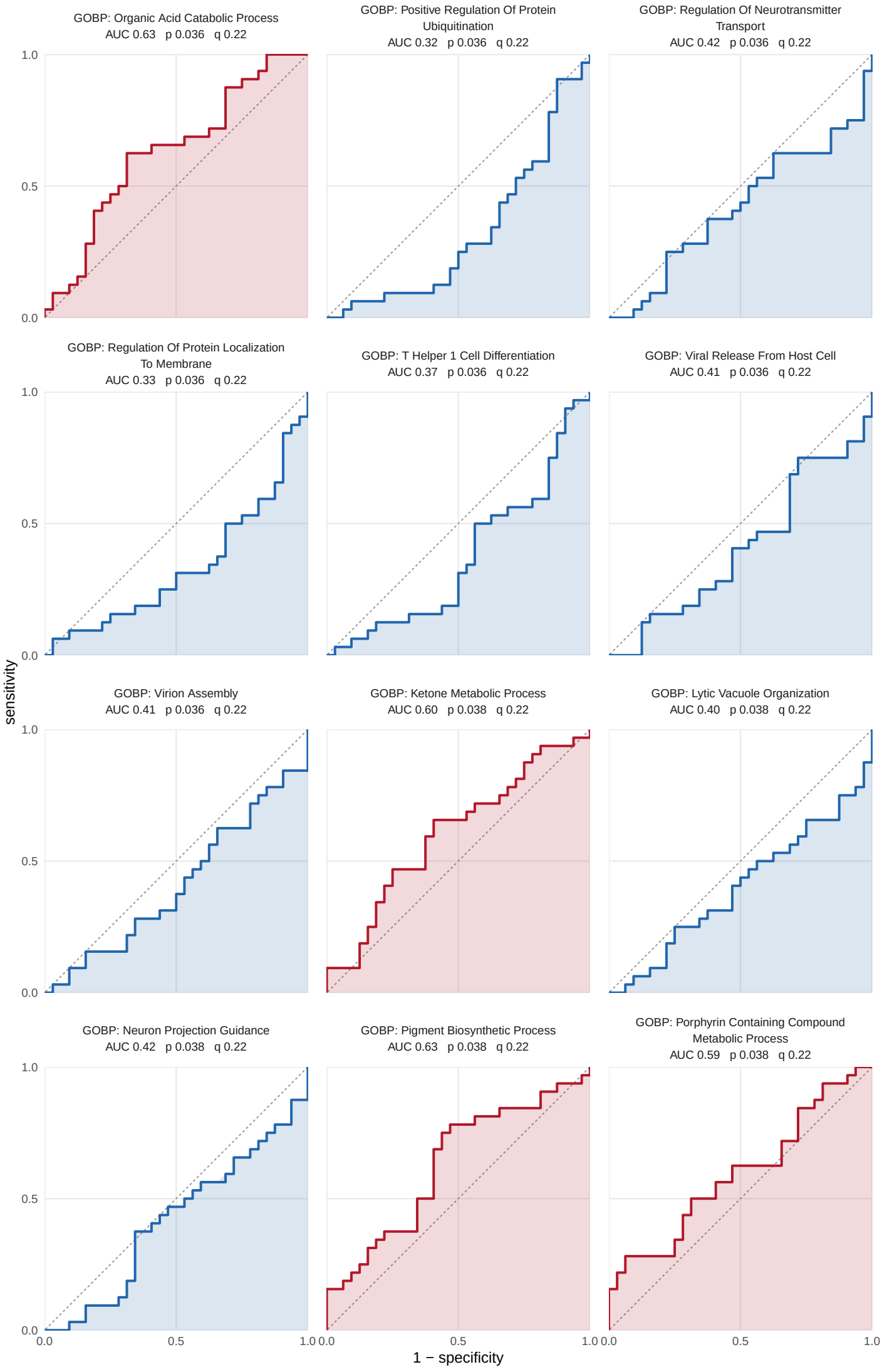

GOBP: Regulation Of Receptor Mediated Endocytosis
AUC 0.38 p 0.021 q 0.18
GOBP: Lysosomal Transport
AUC 0.38 p 0.022 q 0.18
GOBP: Positive Regulation Of Pattern Recognition Receptor Signaling Pathway
AUC 0.31 p 0.022 q 0.18
GOBP: Regulation Of Protein Catabolic Process
AUC 0.34 p 0.022 q 0.18
GOBP: Response To Transforming Growth Factor Beta
AUC 0.35 p 0.022 q 0.18
GOBP: T Cell Homeostasis
AUC 0.36 p 0.022 q 0.18
GOBP: NADH Dehydrogenase Complex Assembly
AUC 0.63 p 0.023 q 0.19
GOBP: Hepaticobiliary System Development
AUC 0.38 p 0.023 q 0.19
GOBP: Nucleoside Triphosphate Biosynthetic Process
AUC 0.63 p 0.023 q 0.19
GOBP: Positive Regulation Of Cytoskeleton Organization
AUC 0.39 p 0.023 q 0.19
GOBP: Regulation Of Protein Stability
AUC 0.35 p 0.023 q 0.19
GOBP: Hematopoietic Or Lymphoid Organ Development
AUC 0.38 p 0.024 q 0.19
sensitivity
1 − specificity
0.0
0.5
1.0

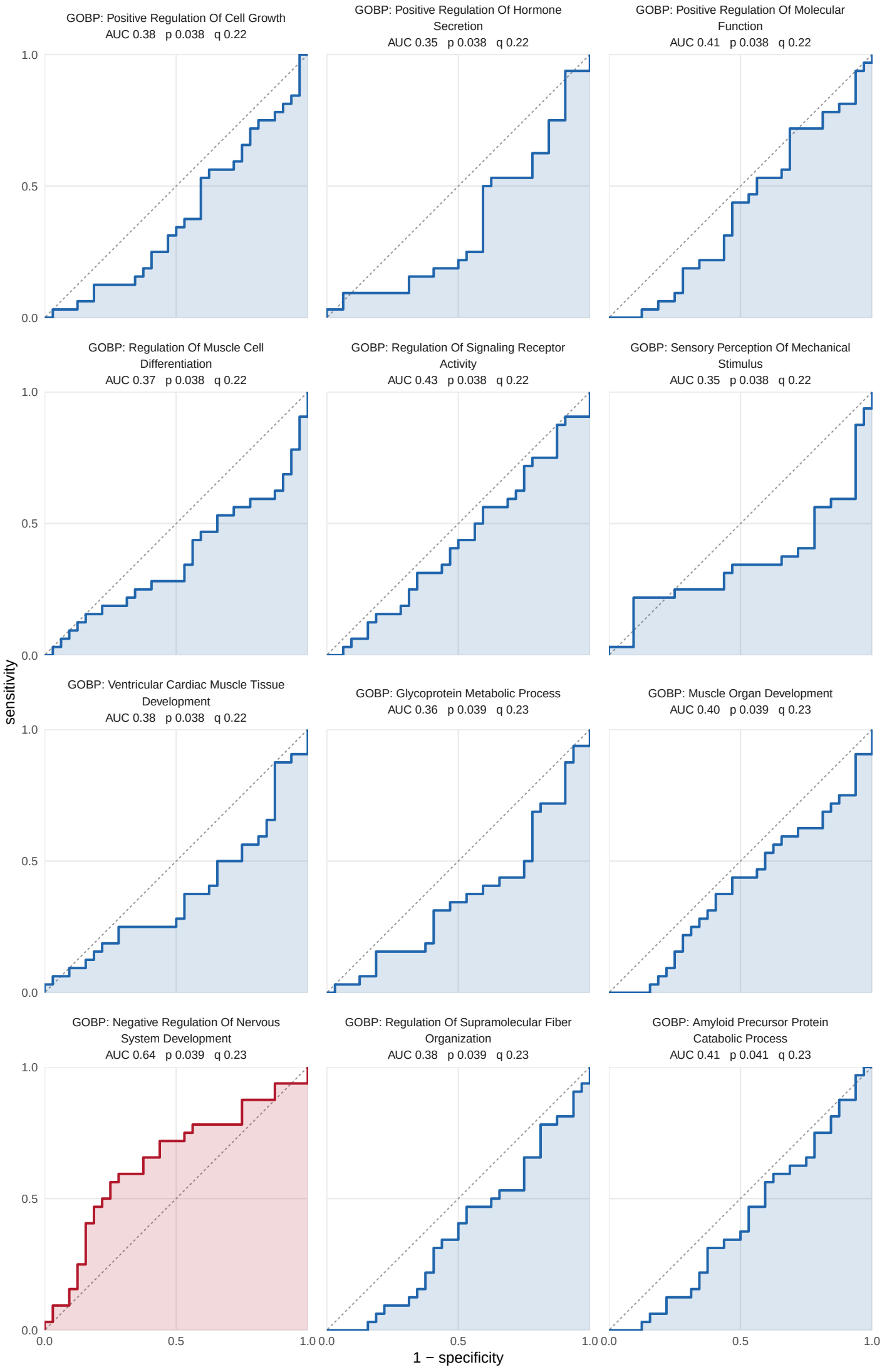

GOBP: Autophagosome Maturation
AUC 0.36 p 0.024 q 0.19
GOBP: Muscle Organ Morphogenesis
AUC 0.38 p 0.024 q 0.19
GOBP: Oxidative Phosphorylation
AUC 0.63 p 0.024 q 0.19
GOBP: Transcription Initiation At RNA Polymerase Ii Promoter
AUC 0.38 p 0.024 q 0.19
GOBP: Electron Transport Chain
AUC 0.65 p 0.025 q 0.19
GOBP: Intracellular Ph Reduction
AUC 0.36 p 0.025 q 0.19
GOBP: Monocarboxylic Acid Biosynthetic Process
AUC 0.59 p 0.025 q 0.19
GOBP: Pyruvate Metabolic Process
AUC 0.66 p 0.025 q 0.19
GOBP: Regulation Of Neurotransmitter Secretion
AUC 0.38 p 0.025 q 0.19
GOBP: Regulation Of Receptor Internalization
AUC 0.34 p 0.025 q 0.19
GOBP: Response To Acid Chemical
AUC 0.38 p 0.026 q 0.2
GOBP: Fatty Acid Beta Oxidation
AUC 0.62 p 0.027 q 0.2
sensitivity
1 − specificity
0.0
0.5
1.0

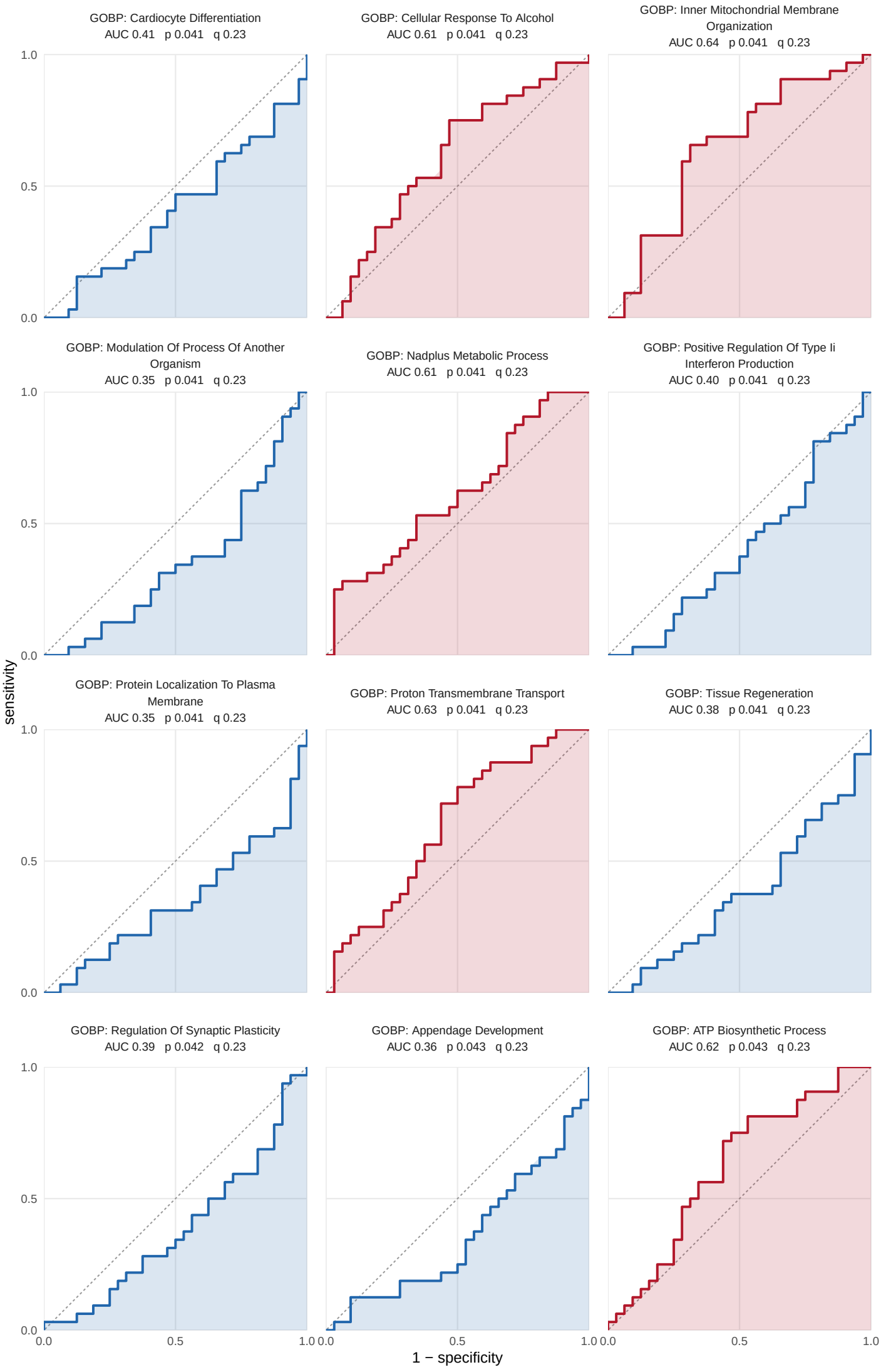

GOBP: Glial Cell Differentiation
AUC 0.37 p 0.027 q 0.2
GOBP: Negative Regulation Of WNT Signaling Pathway
AUC 0.35 p 0.027 q 0.2
GOBP: Nucleus Organization
AUC 0.38 p 0.027 q 0.2
GOBP: Regulation Of Protein Serine Threonine Kinase Activity
AUC 0.34 p 0.027 q 0.2
GOBP: Response To Axon Injury
AUC 0.37 p 0.027 q 0.2
GOBP: T Cell Differentiation In Thymus
AUC 0.35 p 0.027 q 0.2
GOBP: Positive Regulation Of Production Of Molecular Mediator Of Immune Response
AUC 0.39 p 0.028 q 0.2
GOBP: Heart Process
AUC 0.38 p 0.028 q 0.2
GOBP: Negative Regulation Of Transport
AUC 0.40 p 0.028 q 0.2
GOBP: Protein Localization To Cell Periphery
AUC 0.34 p 0.028 q 0.2
GOBP: Stress Fiber Assembly
AUC 0.38 p 0.028 q 0.2
GOBP: Nuclear Transcribed mRNA Catabolic Process
AUC 0.34 p 0.028 q 0.2
sensitivity
1 − specificity
0.0
0.5
1.0

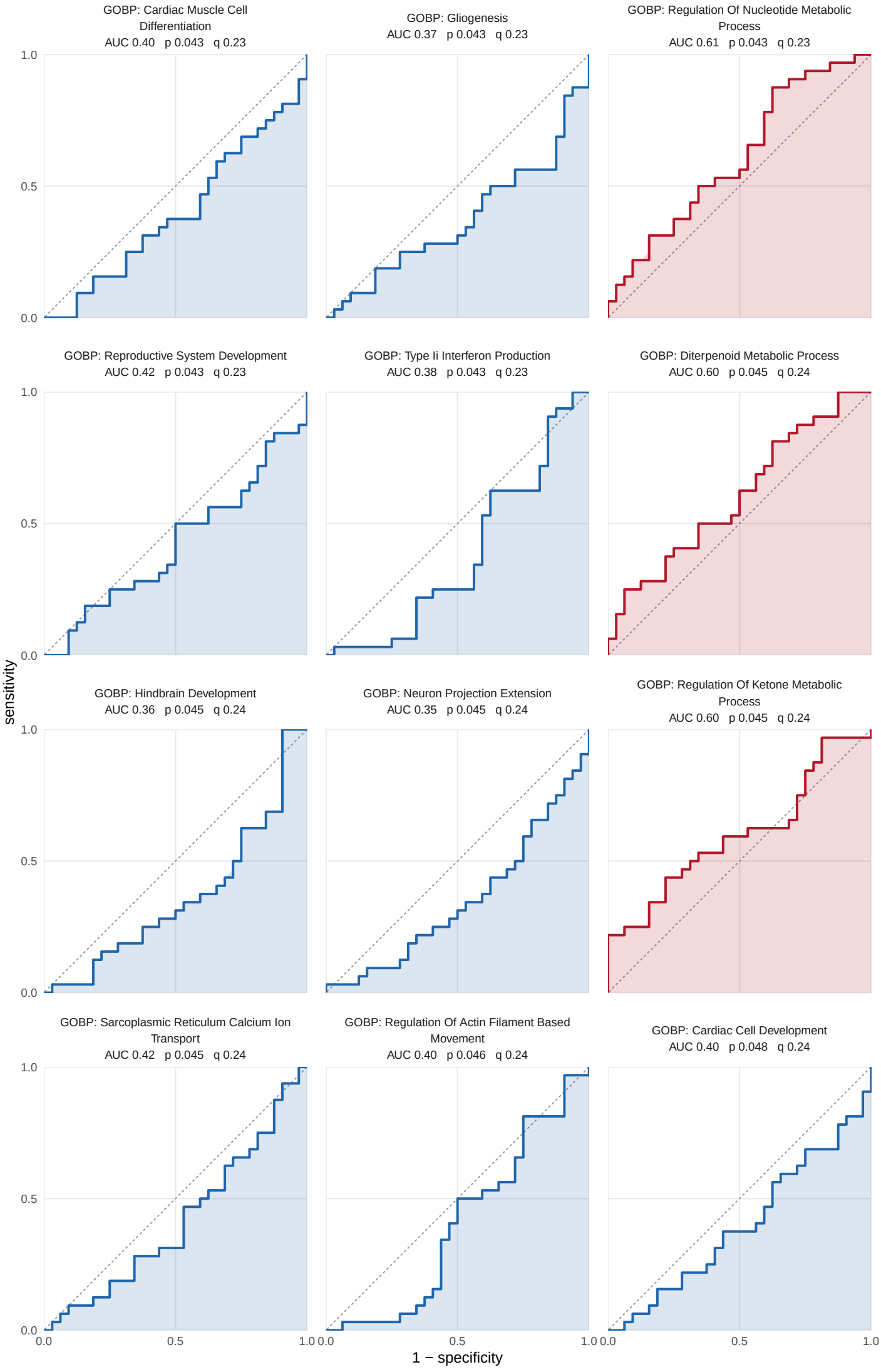

GOBP: Positive Regulation Of Stress Fiber Assembly
AUC 0.36 p 0.029 q 0.2
GOBP: Negative Regulation Of Lymphocyte Differentiation
AUC 0.35 p 0.029 q 0.2
GOBP: Negative Regulation Of MAPK Cascade
AUC 0.36 p 0.029 q 0.2
GOBP: Positive Regulation Of Proteasomal Protein Catabolic Process
AUC 0.34 p 0.029 q 0.2
GOBP: Regulation Of Protein Import Into Nucleus
AUC 0.33 p 0.029 q 0.2
GOBP: ATP Synthesis Coupled Electron Transport
AUC 0.65 p 0.031 q 0.21
GOBP: Endosome To Lysosome Transport
AUC 0.37 p 0.031 q 0.21
GOBP: Monocarboxylic Acid Transport
AUC 0.62 p 0.031 q 0.21
GOBP: Development Of Primary Female Sexual Characteristics
AUC 0.39 p 0.032 q 0.21
GOBP: Female Sex Differentiation
AUC 0.39 p 0.032 q 0.21
GOBP: Negative Regulation Of Calcium Ion Transport
AUC 0.37 p 0.032 q 0.21
GOBP: Neuromuscular Process
AUC 0.38 p 0.032 q 0.21
sensitivity
1 − specificity
0.0
0.5
1.0

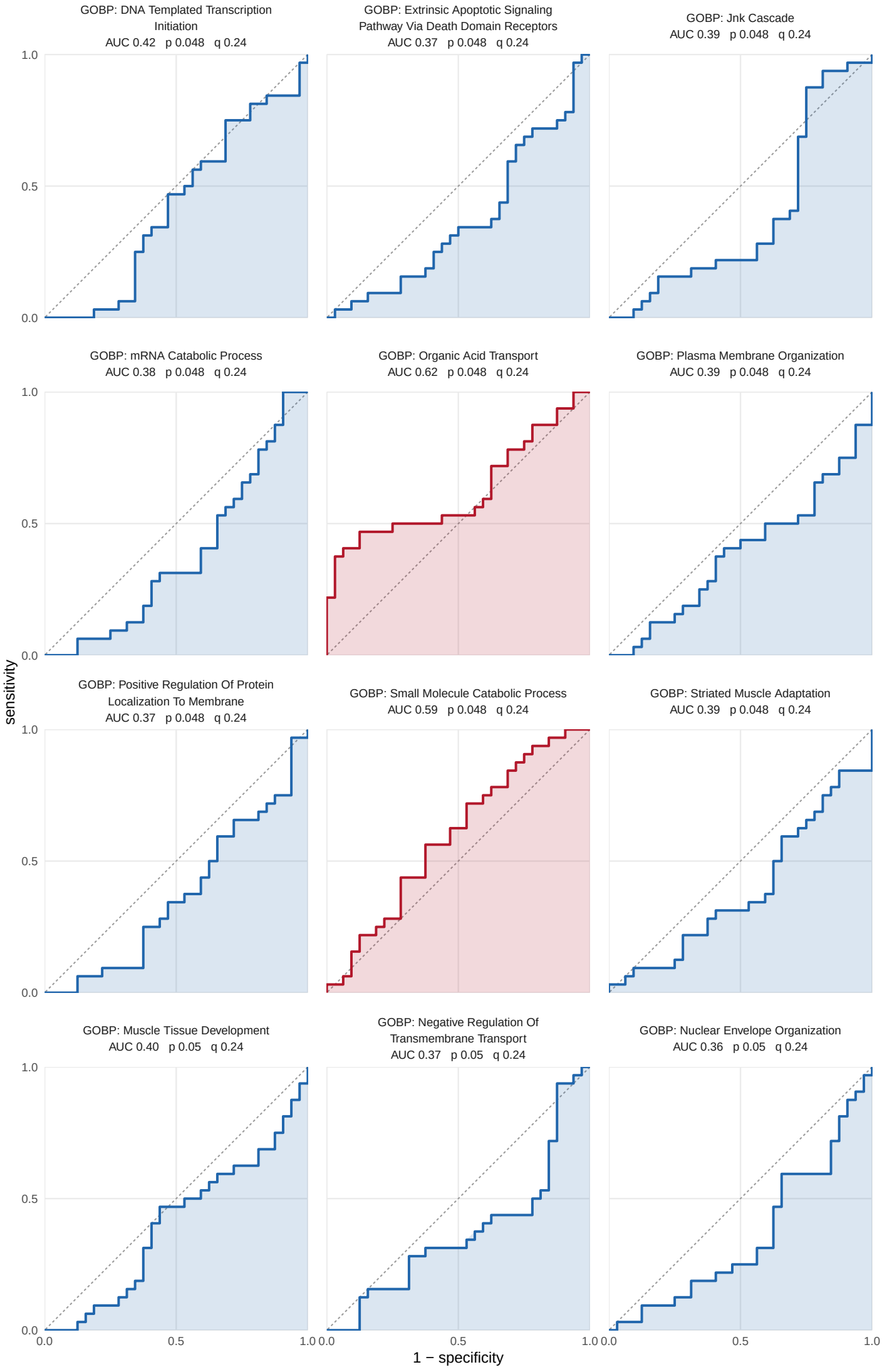

GOBP: Proton Motive Force Driven ATP Synthesis
AUC 0.62 p 0.032 q 0.21
GOBP: Regulation Of Developmental Growth
AUC 0.36 p 0.032 q 0.21
GOBP: Regulation Of Jnk Cascade
AUC 0.40 p 0.032 q 0.21
GOBP: Ribose Phosphate Biosynthetic Process
AUC 0.62 p 0.032 q 0.21
GOBP: Protein Autophosphorylation
AUC 0.38 p 0.033 q 0.21
GOBP: Protein Tetramerization
AUC 0.62 p 0.034 q 0.22
GOBP: Regulation Of G Protein Coupled Receptor Signaling Pathway
AUC 0.38 p 0.034 q 0.22
GOBP: Sterol Homeostasis
AUC 0.40 p 0.034 q 0.22
GOBP: Appendage Morphogenesis
AUC 0.35 p 0.036 q 0.22
GOBP: Cilium Movement
AUC 0.38 p 0.036 q 0.22
GOBP: Heart Morphogenesis
AUC 0.37 p 0.036 q 0.22
GOBP: Lipoprotein Metabolic Process
AUC 0.33 p 0.036 q 0.22
sensitivity
1 − specificity
0.0
0.5
1.0


GOBP: Organic Acid Catabolic Process
AUC 0.63 p 0.036 q 0.22
GOBP: Positive Regulation Of Protein Ubiquitination
AUC 0.32 p 0.036 q 0.22
GOBP: Regulation Of Neurotransmitter Transport
AUC 0.42 p 0.036 q 0.22
GOBP: Regulation Of Protein Localization To Membrane
AUC 0.33 p 0.036 q 0.22
GOBP: T Helper 1 Cell Differentiation
AUC 0.37 p 0.036 q 0.22
GOBP: Viral Release From Host Cell
AUC 0.41 p 0.036 q 0.22
GOBP: Virion Assembly
AUC 0.41 p 0.036 q 0.22
GOBP: Ketone Metabolic Process
AUC 0.60 p 0.038 q 0.22
GOBP: Lytic Vacuole Organization
AUC 0.40 p 0.038 q 0.22
GOBP: Neuron Projection Guidance
AUC 0.42 p 0.038 q 0.22
GOBP: Pigment Biosynthetic Process
AUC 0.63 p 0.038 q 0.22
GOBP: Porphyrin Containing Compound Metabolic Process
AUC 0.59 p 0.038 q 0.22
sensitivity
1 − specificity
0.0
0.5
1.0

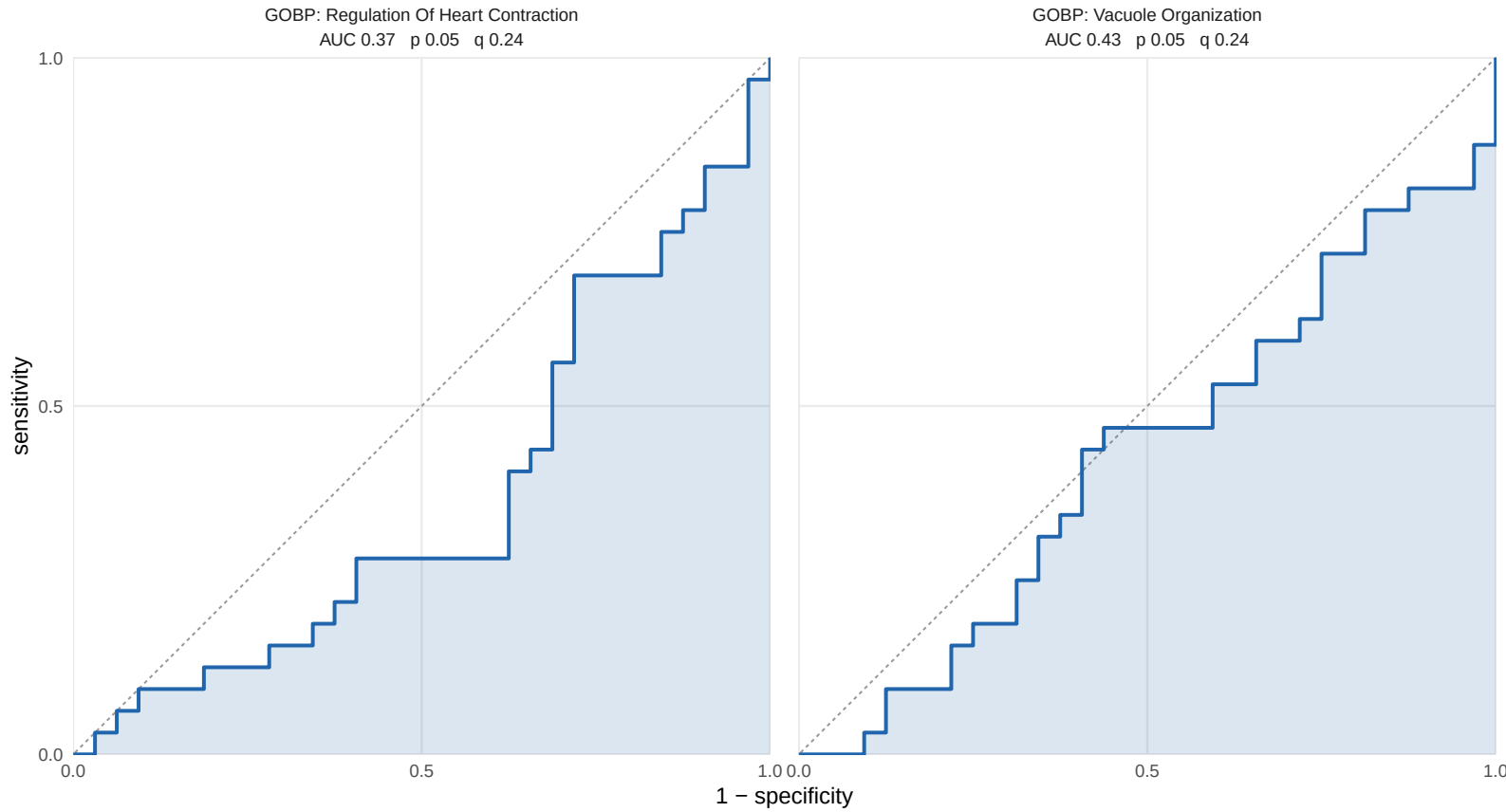

GOBP: Positive Regulation Of Cell Growth
AUC 0.38 p 0.038 q 0.22
GOBP: Positive Regulation Of Hormone Secretion
AUC 0.35 p 0.038 q 0.22
GOBP: Positive Regulation Of Molecular Function
AUC 0.41 p 0.038 q 0.22
GOBP: Regulation Of Muscle Cell Differentiation
AUC 0.37 p 0.038 q 0.22
GOBP: Regulation Of Signaling Receptor Activity
AUC 0.43 p 0.038 q 0.22
GOBP: Sensory Perception Of Mechanical Stimulus
AUC 0.35 p 0.038 q 0.22
GOBP: Ventricular Cardiac Muscle Tissue Development
AUC 0.38 p 0.038 q 0.22
GOBP: Glycoprotein Metabolic Process
AUC 0.36 p 0.039 q 0.23
GOBP: Muscle Organ Development
AUC 0.40 p 0.039 q 0.23
GOBP: Negative Regulation Of Nervous System Development
AUC 0.64 p 0.039 q 0.23
GOBP: Regulation Of Supramolecular Fiber Organization
AUC 0.38 p 0.039 q 0.23
GOBP: Amyloid Precursor Protein Catabolic Process
AUC 0.41 p 0.041 q 0.23
sensitivity
1 − specificity
0.0
0.5
1.0

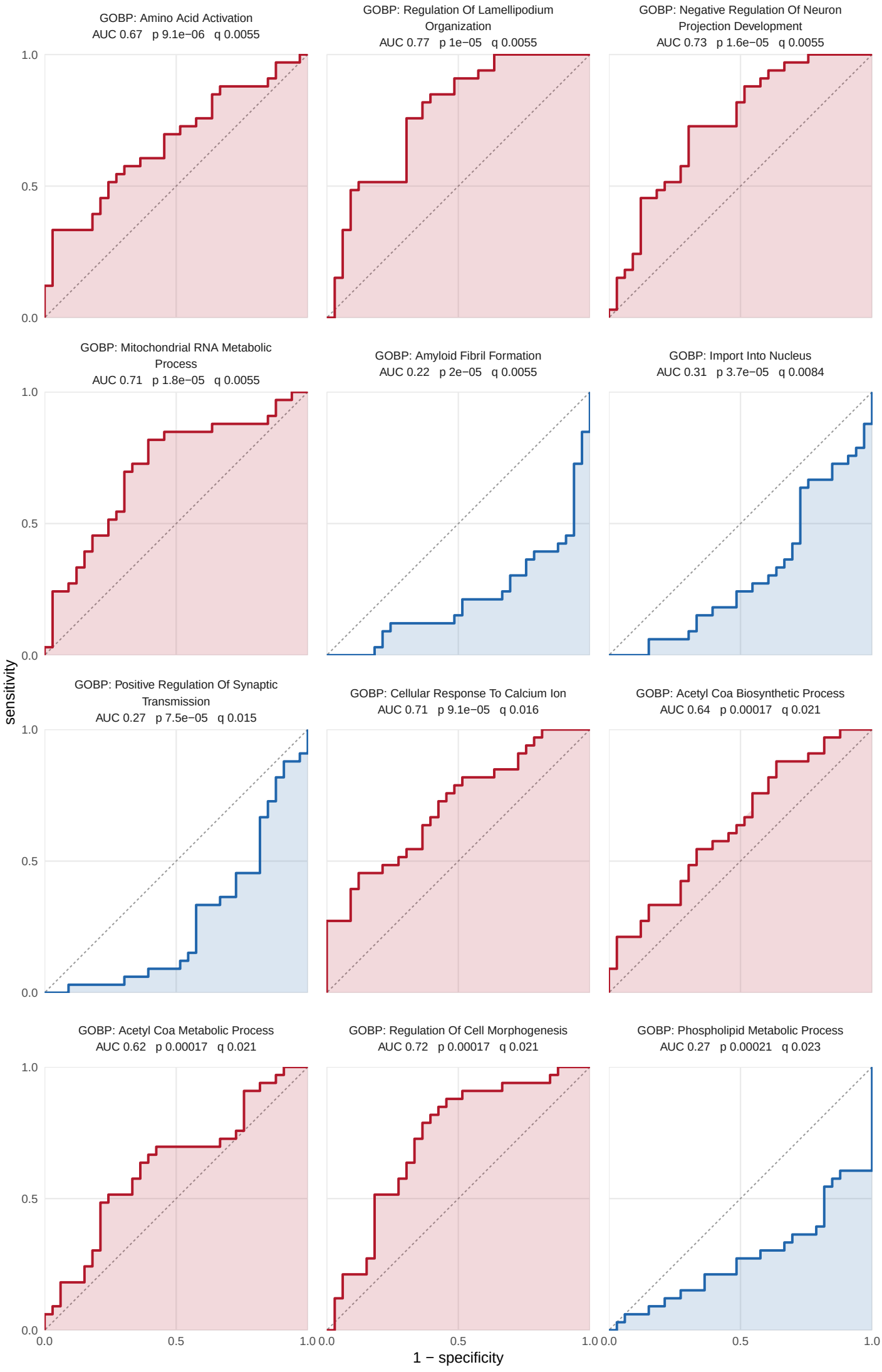

GOBP: Cardiocyte Differentiation
AUC 0.41 p 0.041 q 0.23
GOBP: Cellular Response To Alcohol
AUC 0.61 p 0.041 q 0.23
GOBP: Inner Mitochondrial Membrane Organization
AUC 0.64 p 0.041 q 0.23
GOBP: Modulation Of Process Of Another Organism
AUC 0.35 p 0.041 q 0.23
GOBP: Nadplus Metabolic Process
AUC 0.61 p 0.041 q 0.23
GOBP: Positive Regulation Of Type Ii Interferon Production
AUC 0.40 p 0.041 q 0.23
GOBP: Protein Localization To Plasma Membrane
AUC 0.35 p 0.041 q 0.23
GOBP: Proton Transmembrane Transport
AUC 0.63 p 0.041 q 0.23
GOBP: Tissue Regeneration
AUC 0.38 p 0.041 q 0.23
GOBP: Regulation Of Synaptic Plasticity
AUC 0.39 p 0.042 q 0.23
GOBP: Appendage Development
AUC 0.36 p 0.043 q 0.23
GOBP: ATP Biosynthetic Process
AUC 0.62 p 0.043 q 0.23
sensitivity
1 − specificity
0.0
0.5
1.0

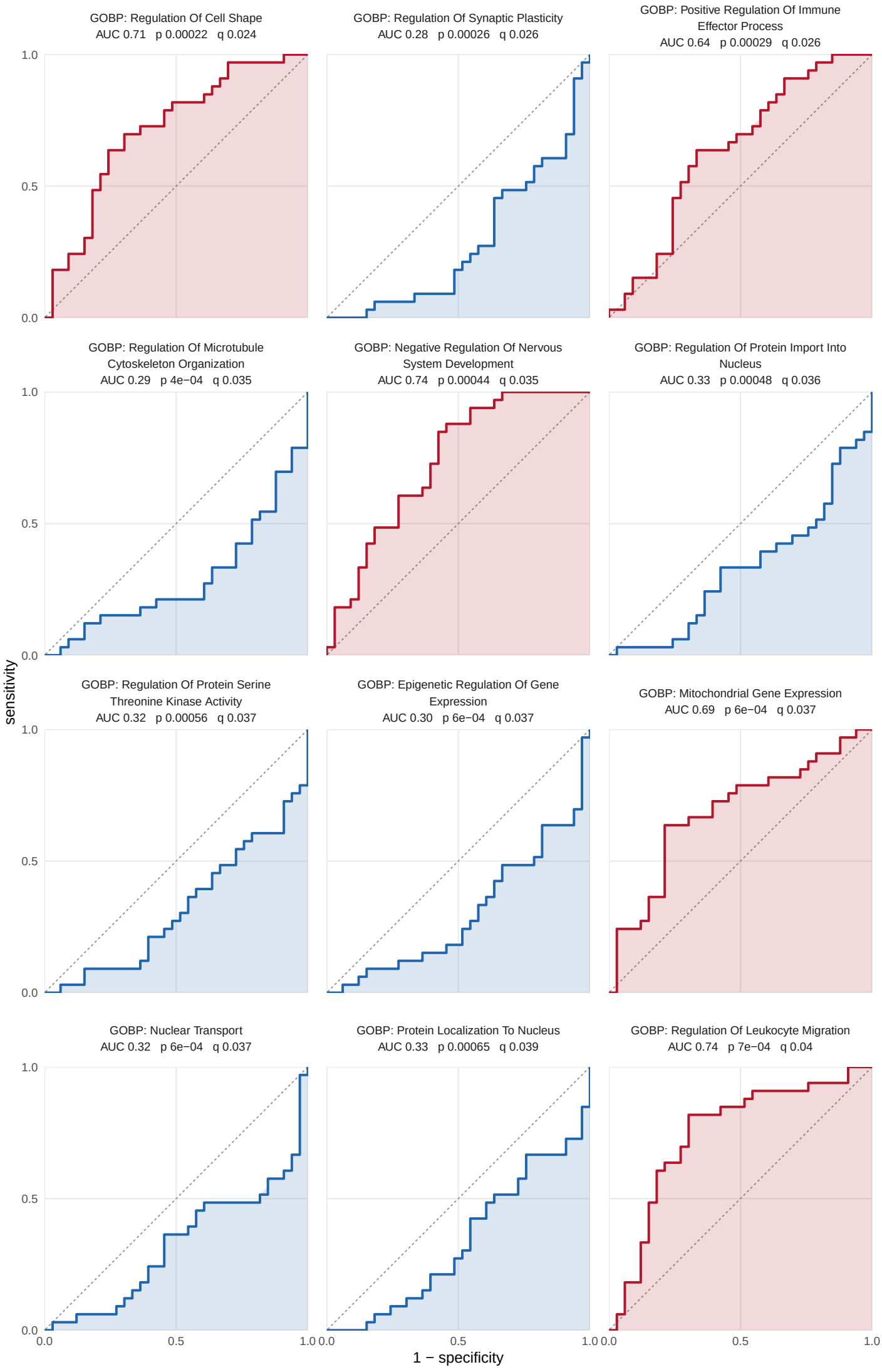

GOBP: Cardiac Muscle Cell Differentiation
AUC 0.40 p 0.043 q 0.23
GOBP: Gliogenesis
AUC 0.37 p 0.043 q 0.23
GOBP: Regulation Of Nucleotide Metabolic Process
AUC 0.61 p 0.043 q 0.23
GOBP: Reproductive System Development
AUC 0.42 p 0.043 q 0.23
GOBP: Type Ii Interferon Production
AUC 0.38 p 0.043 q 0.23
GOBP: Diterpenoid Metabolic Process
AUC 0.60 p 0.045 q 0.24
GOBP: Hindbrain Development
AUC 0.36 p 0.045 q 0.24
GOBP: Neuron Projection Extension
AUC 0.35 p 0.045 q 0.24
GOBP: Regulation Of Ketone Metabolic Process
AUC 0.60 p 0.045 q 0.24
GOBP: Sarcoplasmic Reticulum Calcium Ion Transport
AUC 0.42 p 0.045 q 0.24
GOBP: Regulation Of Actin Filament Based Movement
AUC 0.40 p 0.046 q 0.24
GOBP: Cardiac Cell Development
AUC 0.40 p 0.048 q 0.24
sensitivity
1 − specificity
0.0
0.5
1.0

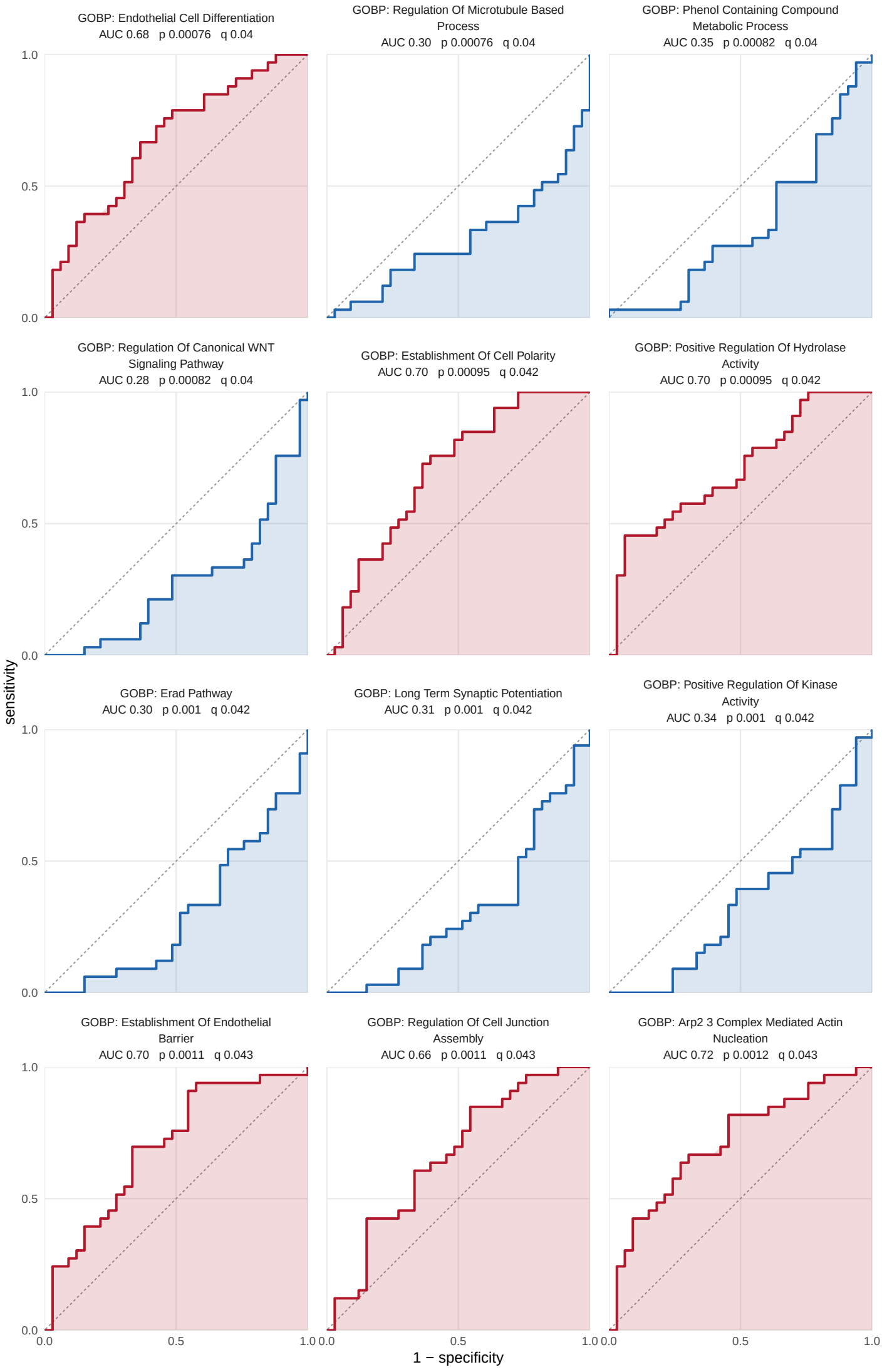

GOBP: DNA Templated Transcription Initiation
AUC 0.42 p 0.048 q 0.24
GOBP: Extrinsic Apoptotic Signaling Pathway Via Death Domain Receptors
AUC 0.37 p 0.048 q 0.24
GOBP: Jnk Cascade
AUC 0.39 p 0.048 q 0.24
GOBP: mRNA Catabolic Process
AUC 0.38 p 0.048 q 0.24
GOBP: Organic Acid Transport
AUC 0.62 p 0.048 q 0.24
GOBP: Plasma Membrane Organization
AUC 0.39 p 0.048 q 0.24
GOBP: Positive Regulation Of Protein Localization To Membrane
AUC 0.37 p 0.048 q 0.24
GOBP: Small Molecule Catabolic Process
AUC 0.59 p 0.048 q 0.24
GOBP: Striated Muscle Adaptation
AUC 0.39 p 0.048 q 0.24
GOBP: Muscle Tissue Development
AUC 0.40 p 0.05 q 0.24
GOBP: Negative Regulation Of Transmembrane Transport
AUC 0.37 p 0.05 q 0.24
GOBP: Nuclear Envelope Organization
AUC 0.36 p 0.05 q 0.24
sensitivity
1 − specificity
0.0
0.5
1.0

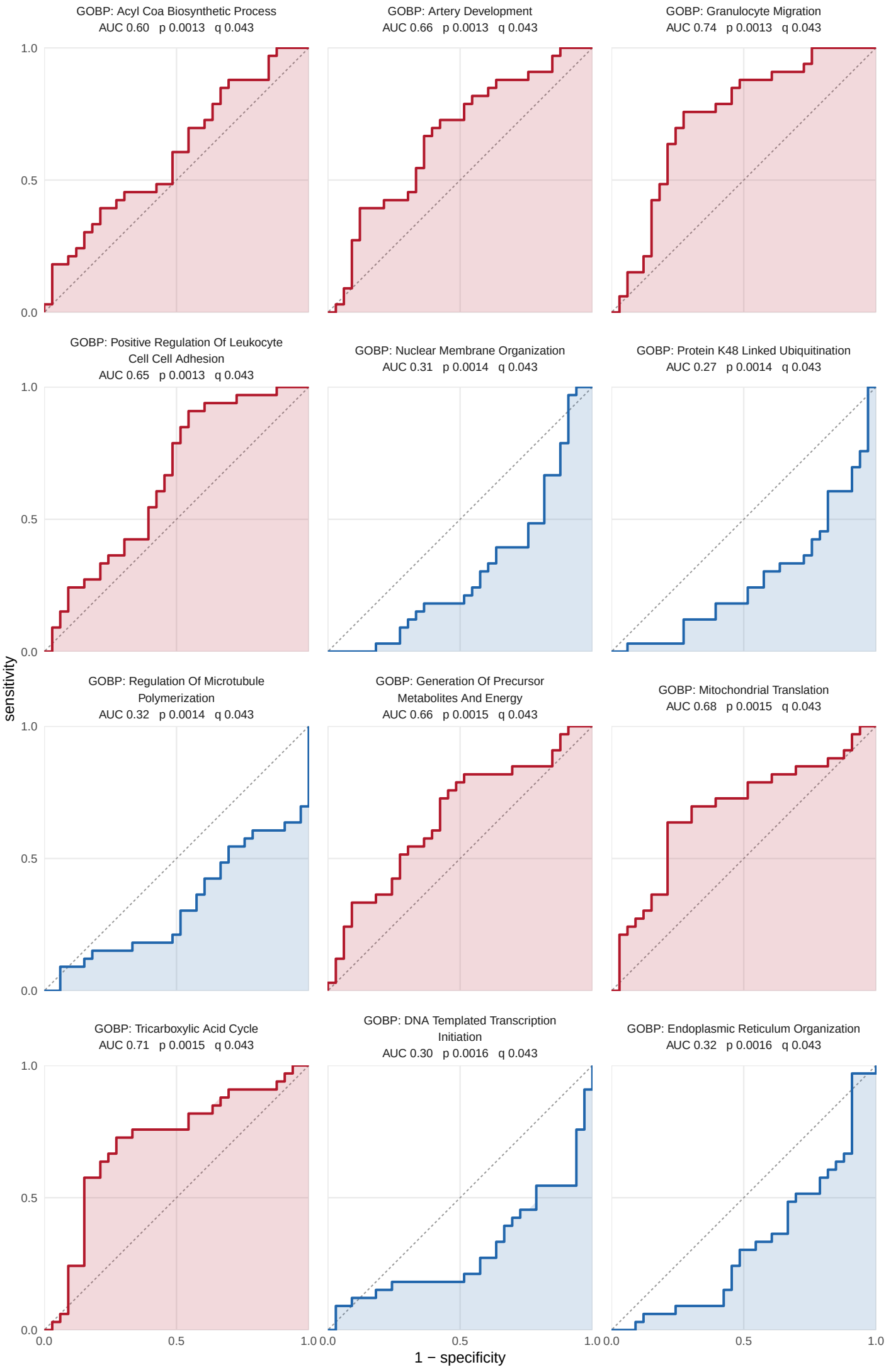
S6. ROC curves, Pre to post, BFR, GOBP (page 24 of 24)

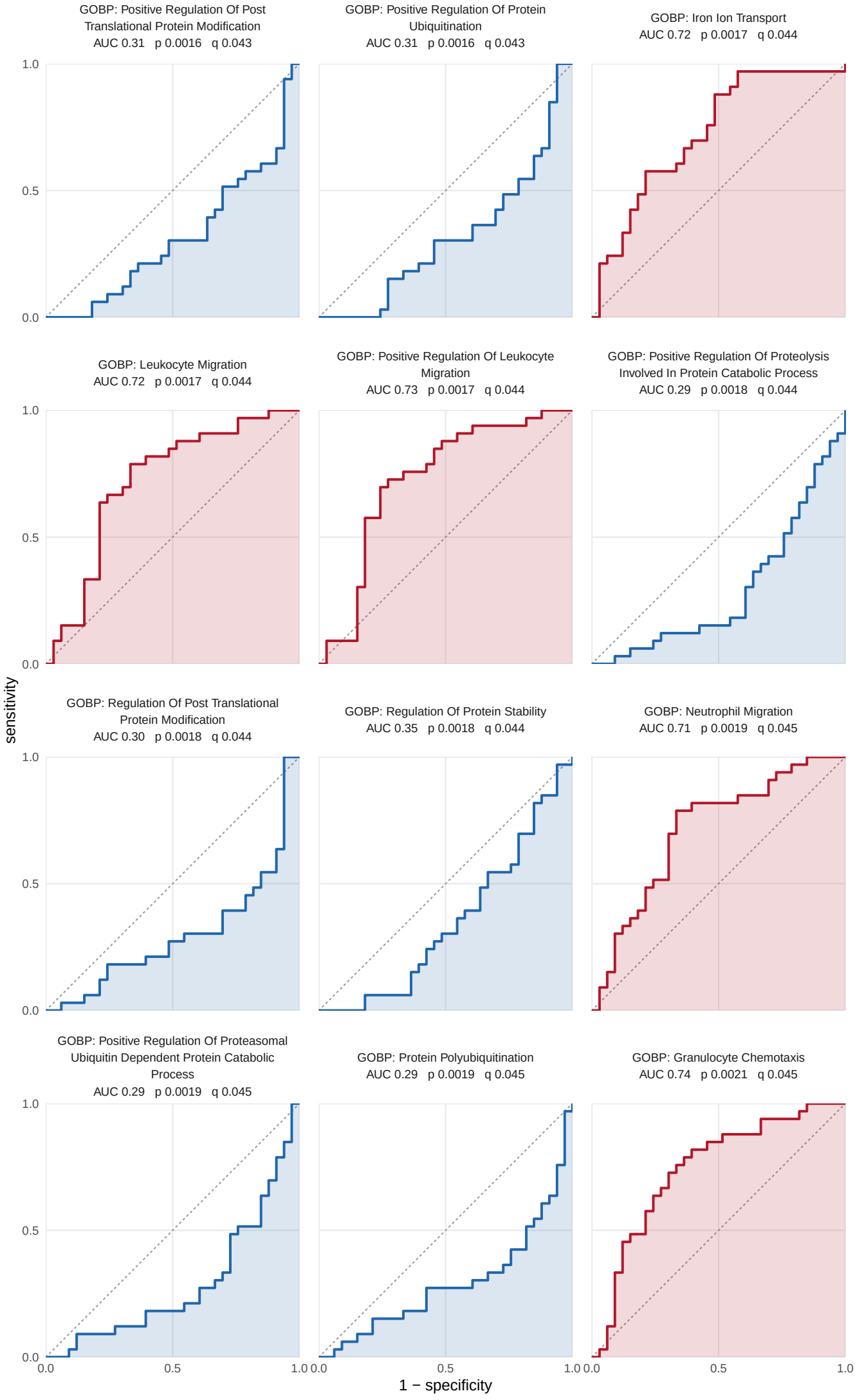


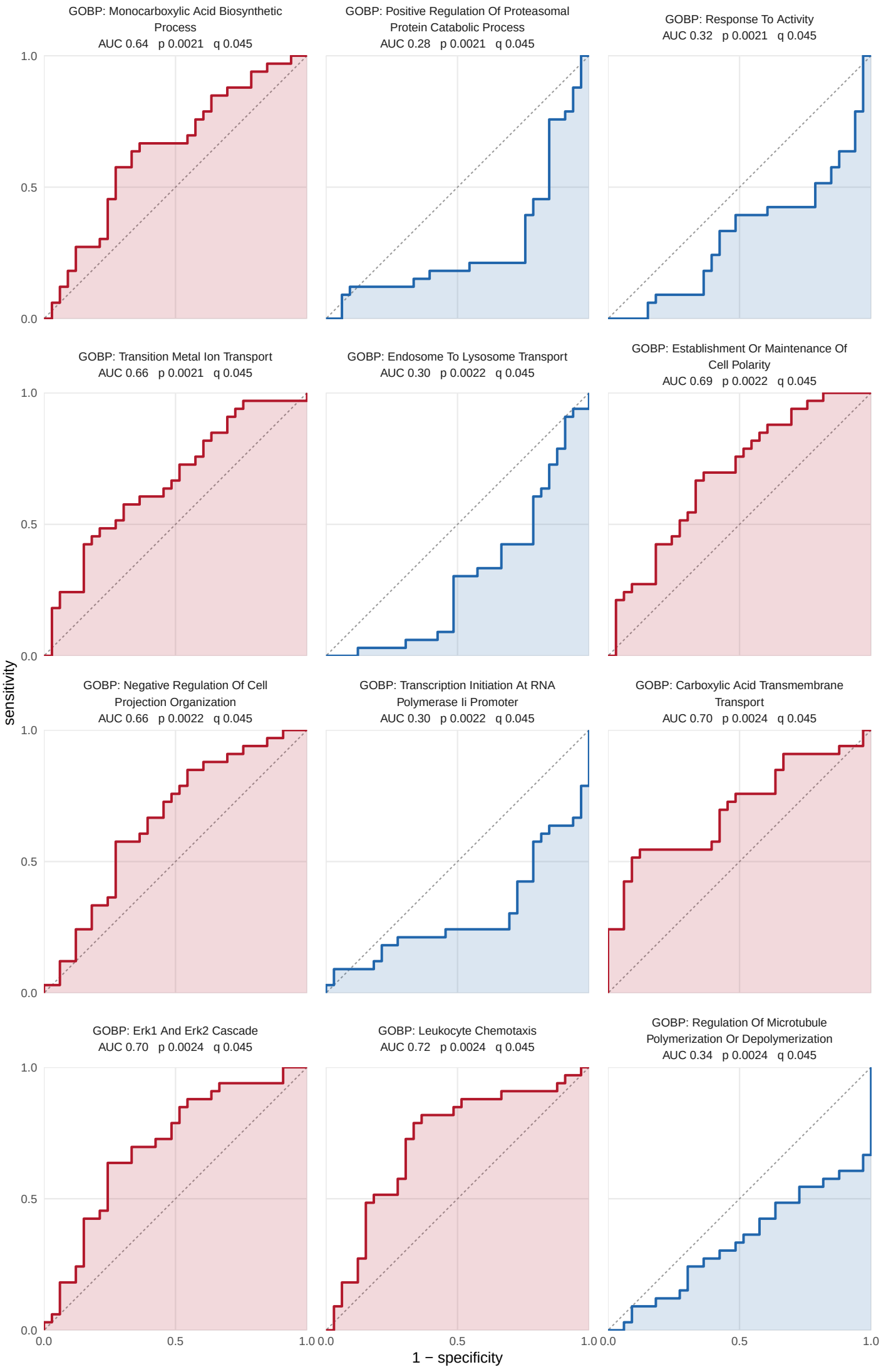

GOBP: Amino Acid Activation
AUC 0.67 p 9.1e−06 q 0.0055
GOBP: Regulation Of Lamellipodium Organization
AUC 0.77 p 1e−05 q 0.0055
GOBP: Negative Regulation Of Neuron Projection Development
AUC 0.73 p 1.6e−05 q 0.0055
GOBP: Mitochondrial RNA Metabolic Process
AUC 0.71 p 1.8e−05 q 0.0055
GOBP: Amyloid Fibril Formation
AUC 0.22 p 2e−05 q 0.0055
GOBP: Import Into Nucleus
AUC 0.31 p 3.7e−05 q 0.0084
GOBP: Positive Regulation Of Synaptic Transmission
AUC 0.27 p 7.5e−05 q 0.015
GOBP: Cellular Response To Calcium Ion
AUC 0.71 p 9.1e−05 q 0.016
GOBP: Acetyl Coa Biosynthetic Process
AUC 0.64 p 0.00017 q 0.021
GOBP: Acetyl Coa Metabolic Process
AUC 0.62 p 0.00017 q 0.021
GOBP: Regulation Of Cell Morphogenesis
AUC 0.72 p 0.00017 q 0.021
GOBP: Phospholipid Metabolic Process
AUC 0.27 p 0.00021 q 0.023
sensitivity
1 − specificity
0.0
0.5
1.0

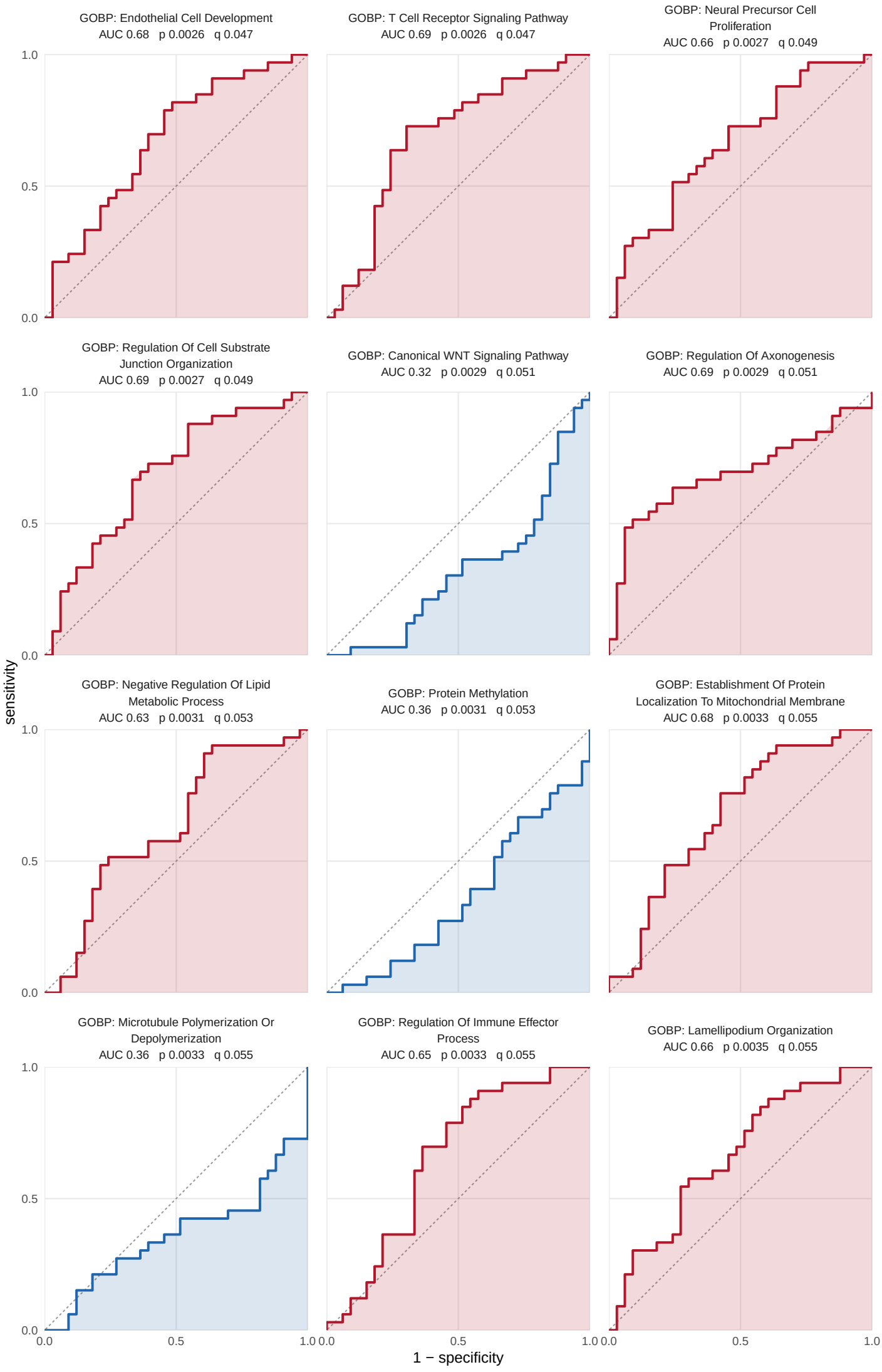

GOBP: Regulation Of Cell Shape
AUC 0.71 p 0.00022 q 0.024
GOBP: Regulation Of Synaptic Plasticity
AUC 0.28 p 0.00026 q 0.026
GOBP: Positive Regulation Of Immune Effector Process
AUC 0.64 p 0.00029 q 0.026
GOBP: Regulation Of Microtubule Cytoskeleton Organization
AUC 0.29 p 4e−04 q 0.035
GOBP: Negative Regulation Of Nervous System Development
AUC 0.74 p 0.00044 q 0.035
GOBP: Regulation Of Protein Import Into Nucleus
AUC 0.33 p 0.00048 q 0.036
GOBP: Regulation Of Protein Serine Threonine Kinase Activity
AUC 0.32 p 0.00056 q 0.037
GOBP: Epigenetic Regulation Of Gene Expression
AUC 0.30 p 6e−04 q 0.037
GOBP: Mitochondrial Gene Expression
AUC 0.69 p 6e−04 q 0.037
GOBP: Nuclear Transport
AUC 0.32 p 6e−04 q 0.037
GOBP: Protein Localization To Nucleus
AUC 0.33 p 0.00065 q 0.039
GOBP: Regulation Of Leukocyte Migration
AUC 0.74 p 7e−04 q 0.04
sensitivity
1 − specificity
0.0
0.5
1.0

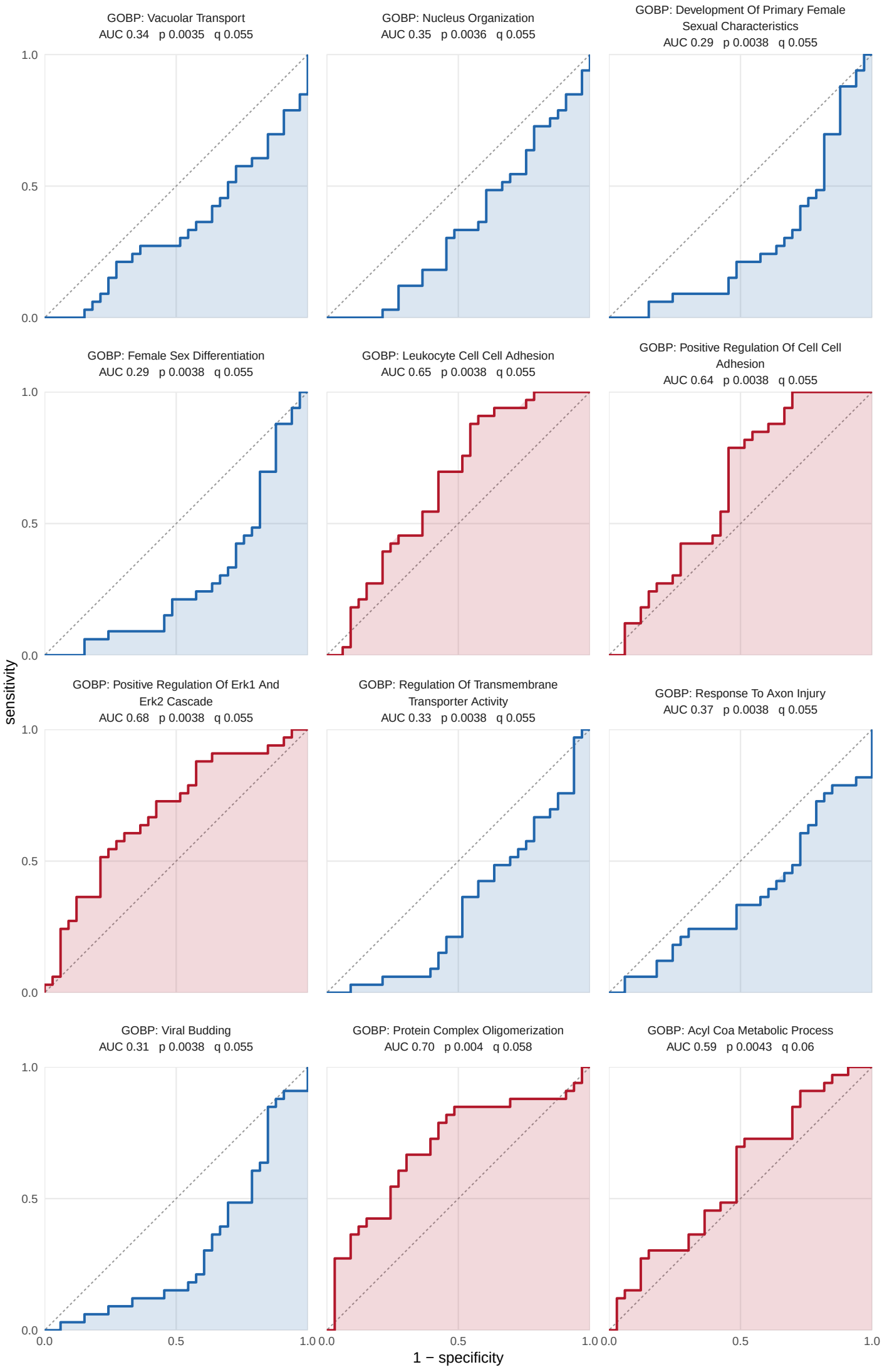

GOBP: Endothelial Cell Differentiation
AUC 0.68 p 0.00076 q 0.04
GOBP: Regulation Of Microtubule Based Process
AUC 0.30 p 0.00076 q 0.04
GOBP: Phenol Containing Compound Metabolic Process
AUC 0.35 p 0.00082 q 0.04
GOBP: Regulation Of Canonical WNT Signaling Pathway
AUC 0.28 p 0.00082 q 0.04
GOBP: Establishment Of Cell Polarity
AUC 0.70 p 0.00095 q 0.042
GOBP: Positive Regulation Of Hydrolase Activity
AUC 0.70 p 0.00095 q 0.042
GOBP: Erad Pathway
AUC 0.30 p 0.001 q 0.042
GOBP: Long Term Synaptic Potentiation
AUC 0.31 p 0.001 q 0.042
GOBP: Positive Regulation Of Kinase Activity
AUC 0.34 p 0.001 q 0.042
GOBP: Establishment Of Endothelial Barrier
AUC 0.70 p 0.0011 q 0.043
GOBP: Regulation Of Cell Junction Assembly
AUC 0.66 p 0.0011 q 0.043
GOBP: Arp2 3 Complex Mediated Actin Nucleation
AUC 0.72 p 0.0012 q 0.043
sensitivity
1 − specificity
0.0
0.5
1.0

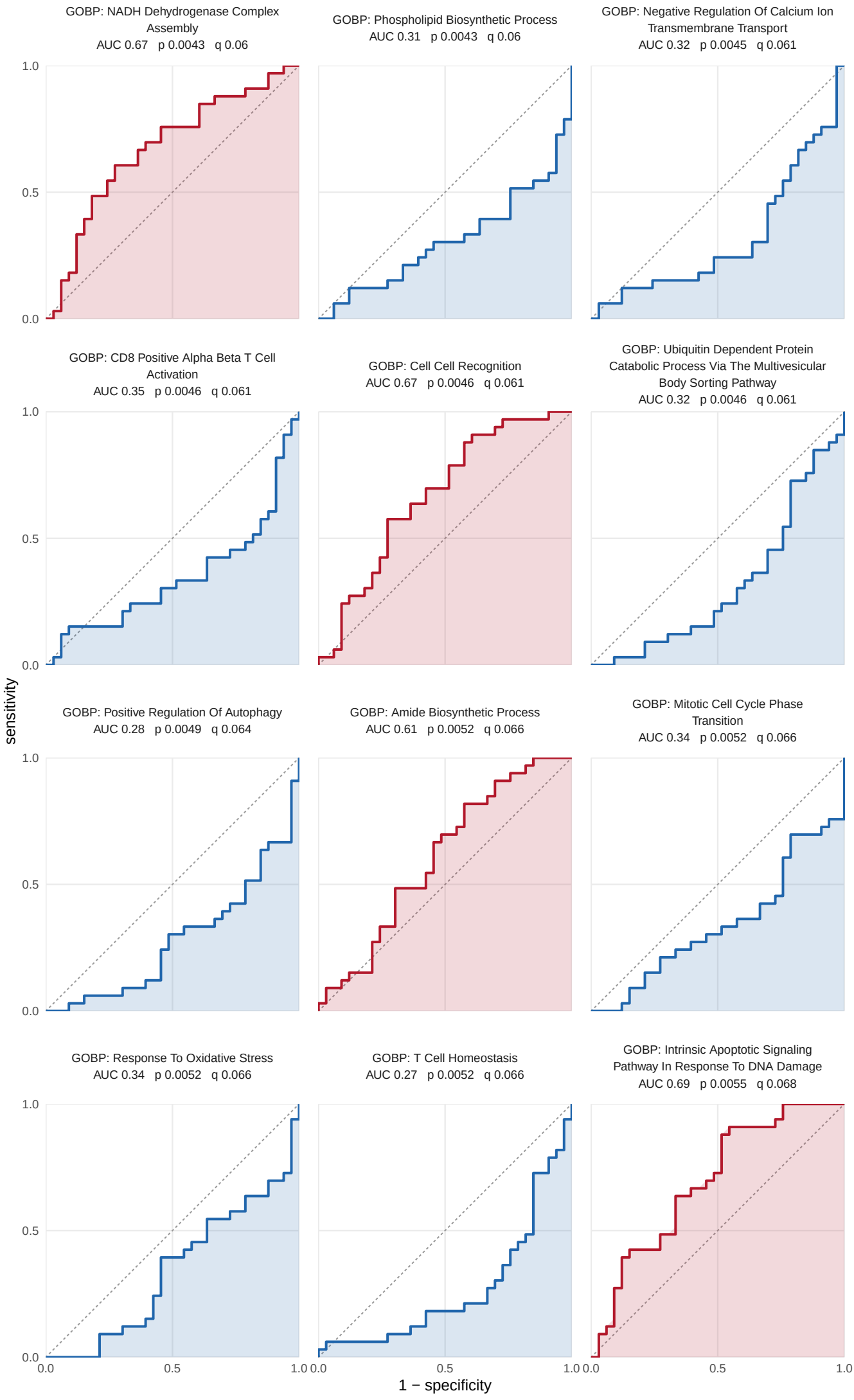

GOBP: Acyl Coa Biosynthetic Process
AUC 0.60 p 0.0013 q 0.043
GOBP: Artery Development
AUC 0.66 p 0.0013 q 0.043
GOBP: Granulocyte Migration
AUC 0.74 p 0.0013 q 0.043
GOBP: Positive Regulation Of Leukocyte Cell Cell Adhesion
AUC 0.65 p 0.0013 q 0.043
GOBP: Nuclear Membrane Organization
AUC 0.31 p 0.0014 q 0.043
GOBP: Protein K48 Linked Ubiquitination
AUC 0.27 p 0.0014 q 0.043
GOBP: Regulation Of Microtubule Polymerization
AUC 0.32 p 0.0014 q 0.043
GOBP: Generation Of Precursor Metabolites And Energy
AUC 0.66 p 0.0015 q 0.043
GOBP: Mitochondrial Translation
AUC 0.68 p 0.0015 q 0.043
GOBP: Tricarboxylic Acid Cycle
AUC 0.71 p 0.0015 q 0.043
GOBP: DNA Templated Transcription Initiation
AUC 0.30 p 0.0016 q 0.043
GOBP: Endoplasmic Reticulum Organization
AUC 0.32 p 0.0016 q 0.043
sensitivity
1 − specificity
0.0
0.5
1.0

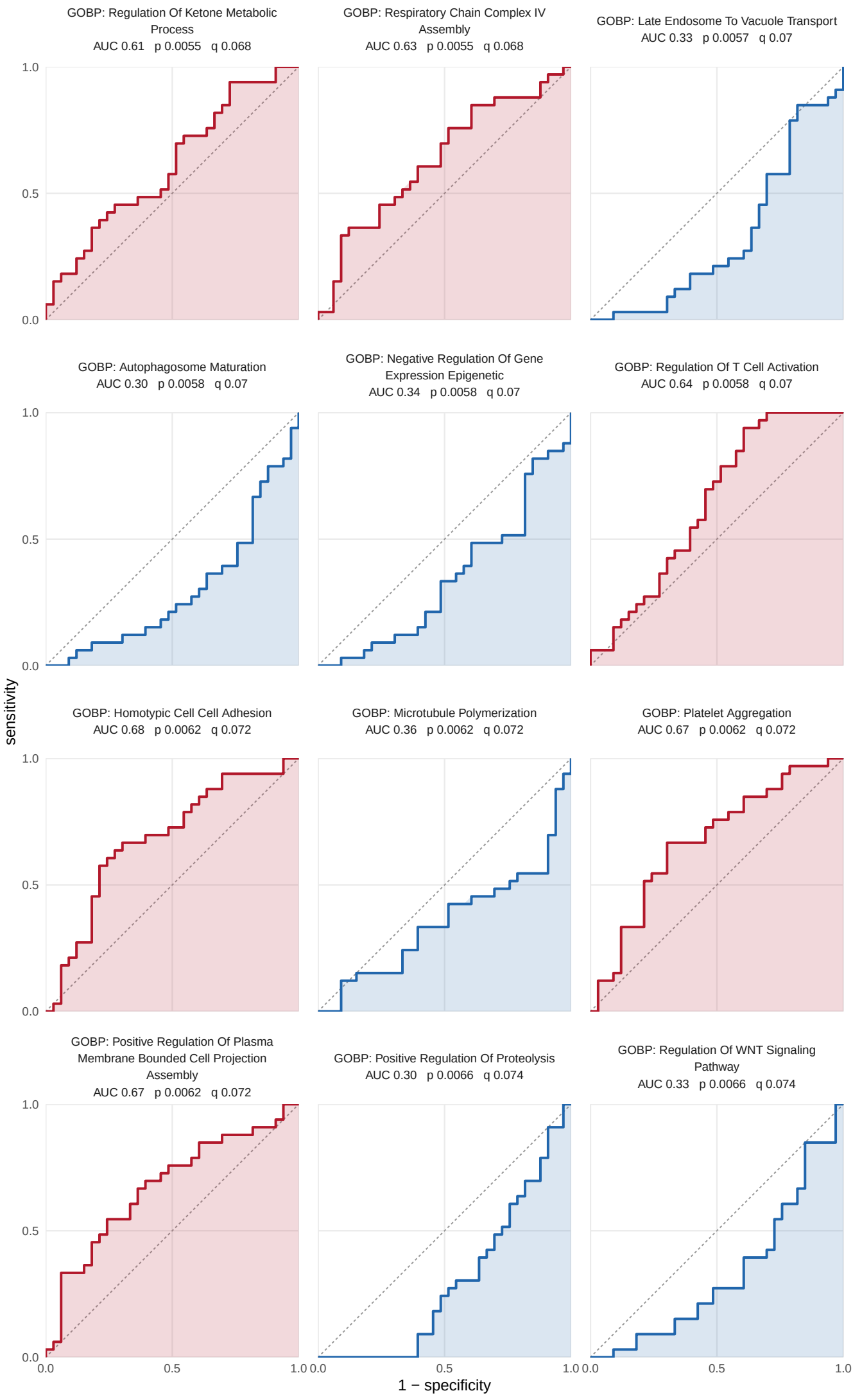

GOBP: Positive Regulation Of Post Translational Protein Modification
AUC 0.31 p 0.0016 q 0.043
GOBP: Positive Regulation Of Protein Ubiquitination
AUC 0.31 p 0.0016 q 0.043
GOBP: Iron Ion Transport
AUC 0.72 p 0.0017 q 0.044
GOBP: Leukocyte Migration
AUC 0.72 p 0.0017 q 0.044
GOBP: Positive Regulation Of Leukocyte Migration
AUC 0.73 p 0.0017 q 0.044
GOBP: Positive Regulation Of Proteolysis Involved In Protein Catabolic Process
AUC 0.29 p 0.0018 q 0.044
GOBP: Regulation Of Post Translational Protein Modification
AUC 0.30 p 0.0018 q 0.044
GOBP: Regulation Of Protein Stability
AUC 0.35 p 0.0018 q 0.044
GOBP: Neutrophil Migration
AUC 0.71 p 0.0019 q 0.045
GOBP: Positive Regulation Of Proteasomal Ubiquitin Dependent Protein Catabolic Process
AUC 0.29 p 0.0019 q 0.045
GOBP: Protein Polyubiquitination
AUC 0.29 p 0.0019 q 0.045
GOBP: Granulocyte Chemotaxis
AUC 0.74 p 0.0021 q 0.045
sensitivity
1 − specificity
0.0
0.5
1.0

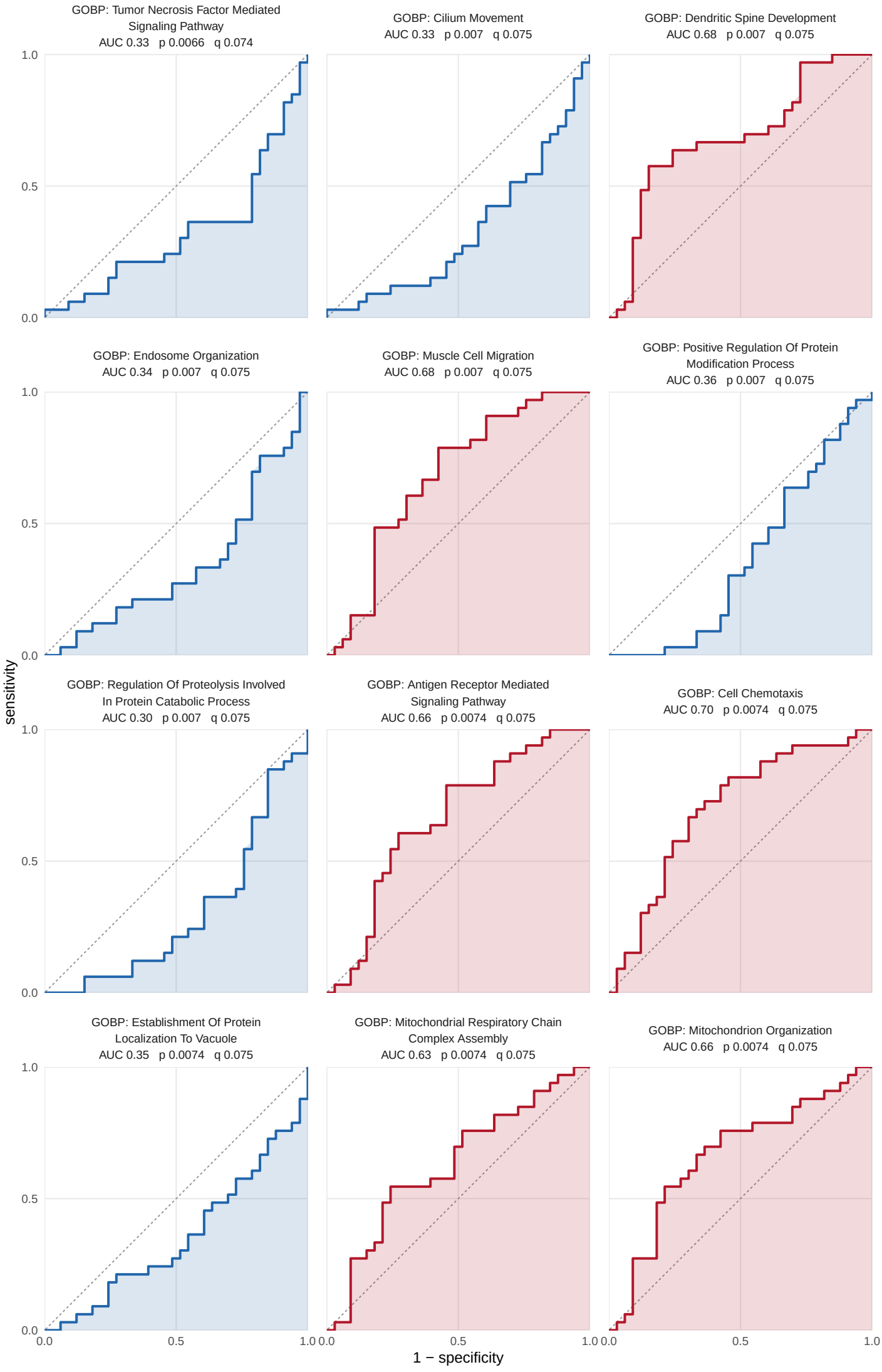

GOBP: Monocarboxylic Acid Biosynthetic Process
AUC 0.64 p 0.0021 q 0.045
GOBP: Positive Regulation Of Proteasomal Protein Catabolic Process
AUC 0.28 p 0.0021 q 0.045
GOBP: Response To Activity
AUC 0.32 p 0.0021 q 0.045
GOBP: Transition Metal Ion Transport
AUC 0.66 p 0.0021 q 0.045
GOBP: Endosome To Lysosome Transport
AUC 0.30 p 0.0022 q 0.045
GOBP: Establishment Or Maintenance Of Cell Polarity
AUC 0.69 p 0.0022 q 0.045
GOBP: Negative Regulation Of Cell Projection Organization
AUC 0.66 p 0.0022 q 0.045
GOBP: Transcription Initiation At RNA Polymerase Ii Promoter
AUC 0.30 p 0.0022 q 0.045
GOBP: Carboxylic Acid Transmembrane Transport
AUC 0.70 p 0.0024 q 0.045
GOBP: Erk1 And Erk2 Cascade
AUC 0.70 p 0.0024 q 0.045
GOBP: Leukocyte Chemotaxis
AUC 0.72 p 0.0024 q 0.045
GOBP: Regulation Of Microtubule Polymerization Or Depolymerization
AUC 0.34 p 0.0024 q 0.045
sensitivity
1 − specificity
0.0
0.5
1.0

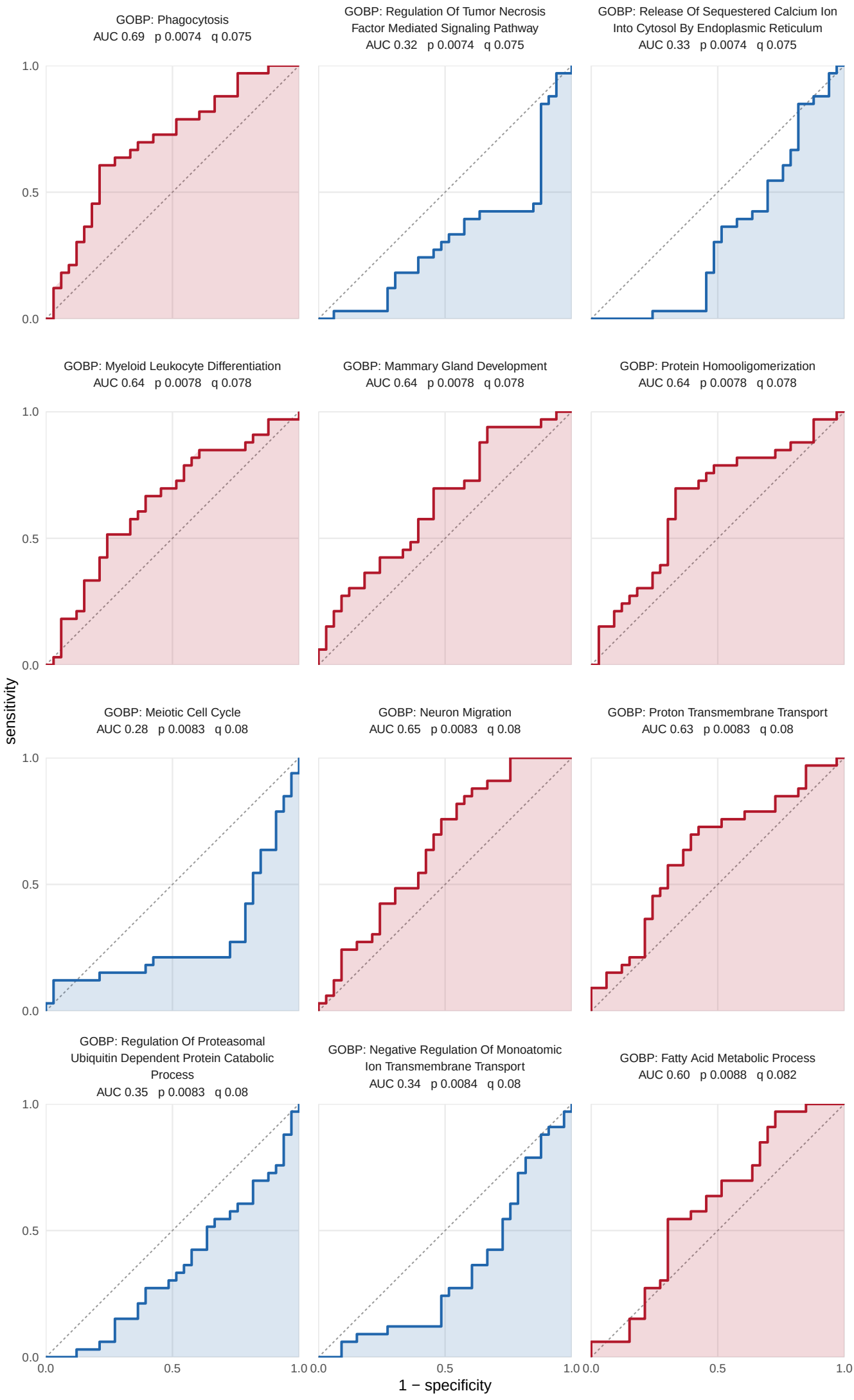

GOBP: Endothelial Cell Development
AUC 0.68 p 0.0026 q 0.047
GOBP: T Cell Receptor Signaling Pathway
AUC 0.69 p 0.0026 q 0.047
GOBP: Neural Precursor Cell Proliferation
AUC 0.66 p 0.0027 q 0.049
GOBP: Regulation Of Cell Substrate Junction Organization
AUC 0.69 p 0.0027 q 0.049
GOBP: Canonical WNT Signaling Pathway
AUC 0.32 p 0.0029 q 0.051
GOBP: Regulation Of Axonogenesis
AUC 0.69 p 0.0029 q 0.051
GOBP: Negative Regulation Of Lipid Metabolic Process
AUC 0.63 p 0.0031 q 0.053
GOBP: Protein Methylation
AUC 0.36 p 0.0031 q 0.053
GOBP: Establishment Of Protein Localization To Mitochondrial Membrane
AUC 0.68 p 0.0033 q 0.055
GOBP: Microtubule Polymerization Or Depolymerization
AUC 0.36 p 0.0033 q 0.055
GOBP: Regulation Of Immune Effector Process
AUC 0.65 p 0.0033 q 0.055
GOBP: Lamellipodium Organization
AUC 0.66 p 0.0035 q 0.055
sensitivity
1 − specificity
0.0
0.5
1.0

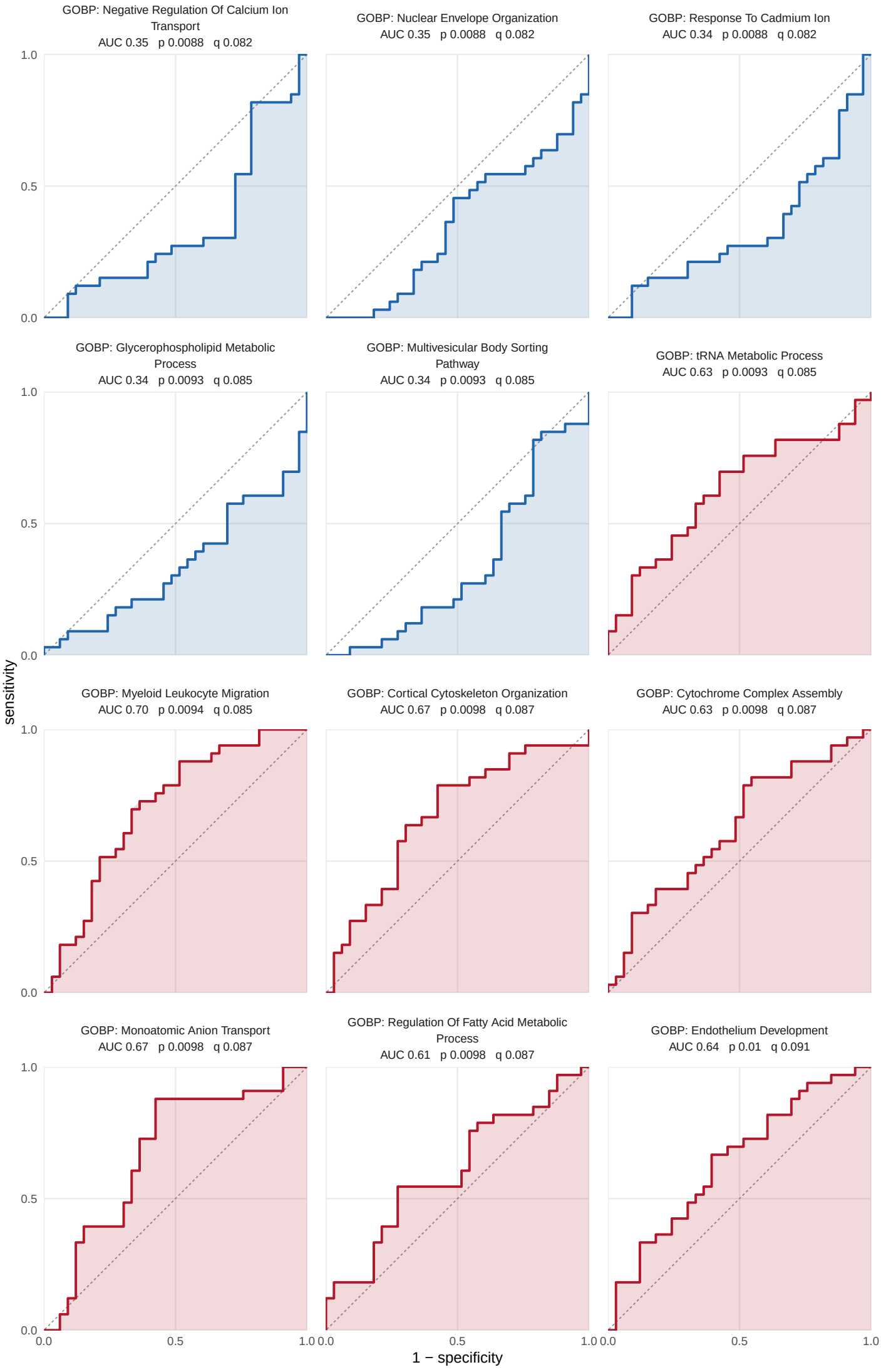

GOBP: Vacuolar Transport
AUC 0.34 p 0.0035 q 0.055
GOBP: Nucleus Organization
AUC 0.35 p 0.0036 q 0.055
GOBP: Development Of Primary Female Sexual Characteristics
AUC 0.29 p 0.0038 q 0.055
GOBP: Female Sex Differentiation
AUC 0.29 p 0.0038 q 0.055
GOBP: Leukocyte Cell Cell Adhesion
AUC 0.65 p 0.0038 q 0.055
GOBP: Positive Regulation Of Cell Cell Adhesion
AUC 0.64 p 0.0038 q 0.055
GOBP: Positive Regulation Of Erk1 And Erk2 Cascade
AUC 0.68 p 0.0038 q 0.055
GOBP: Regulation Of Transmembrane Transporter Activity
AUC 0.33 p 0.0038 q 0.055
GOBP: Response To Axon Injury
AUC 0.37 p 0.0038 q 0.055
GOBP: Viral Budding
AUC 0.31 p 0.0038 q 0.055
GOBP: Protein Complex Oligomerization
AUC 0.70 p 0.004 q 0.058
GOBP: Acyl Coa Metabolic Process
AUC 0.59 p 0.0043 q 0.06
sensitivity
1 − specificity
0.0
0.5
1.0

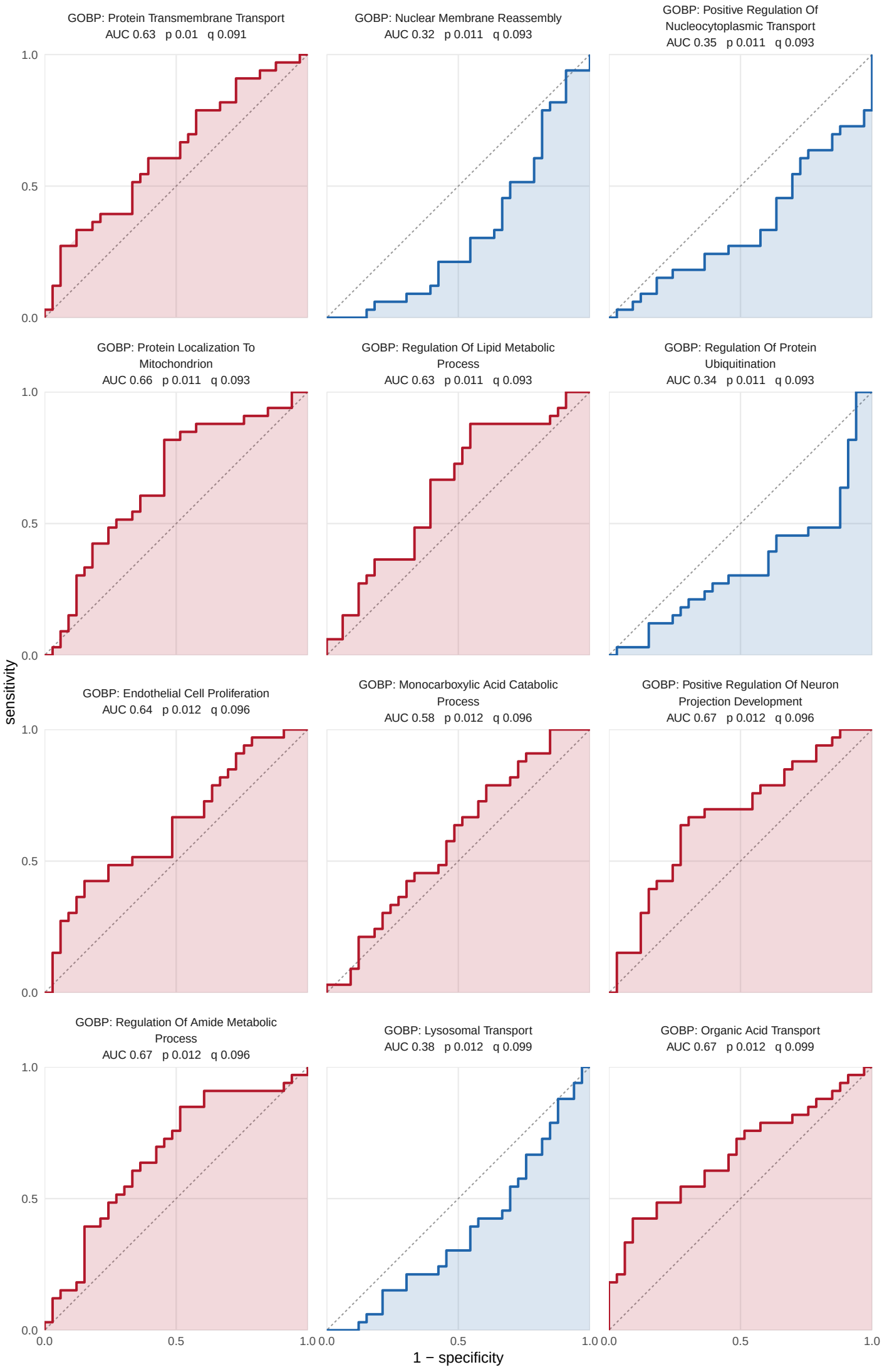

GOBP: NADH Dehydrogenase Complex Assembly
AUC 0.67 p 0.0043 q 0.06
GOBP: Phospholipid Biosynthetic Process
AUC 0.31 p 0.0043 q 0.06
GOBP: Negative Regulation Of Calcium Ion Transmembrane Transport
AUC 0.32 p 0.0045 q 0.061
GOBP: CD8 Positive Alpha Beta T Cell Activation
AUC 0.35 p 0.0046 q 0.061
GOBP: Cell Cell Recognition
AUC 0.67 p 0.0046 q 0.061
GOBP: Ubiquitin Dependent Protein Catabolic Process Via The Multivesicular Body Sorting Pathway
AUC 0.32 p 0.0046 q 0.061
GOBP: Positive Regulation Of Autophagy
AUC 0.28 p 0.0049 q 0.064
GOBP: Amide Biosynthetic Process
AUC 0.61 p 0.0052 q 0.066
GOBP: Mitotic Cell Cycle Phase Transition
AUC 0.34 p 0.0052 q 0.066
GOBP: Response To Oxidative Stress
AUC 0.34 p 0.0052 q 0.066
GOBP: T Cell Homeostasis
AUC 0.27 p 0.0052 q 0.066
GOBP: Intrinsic Apoptotic Signaling Pathway In Response To DNA Damage
AUC 0.69 p 0.0055 q 0.068
sensitivity
1 − specificity
0.0
0.5
1.0

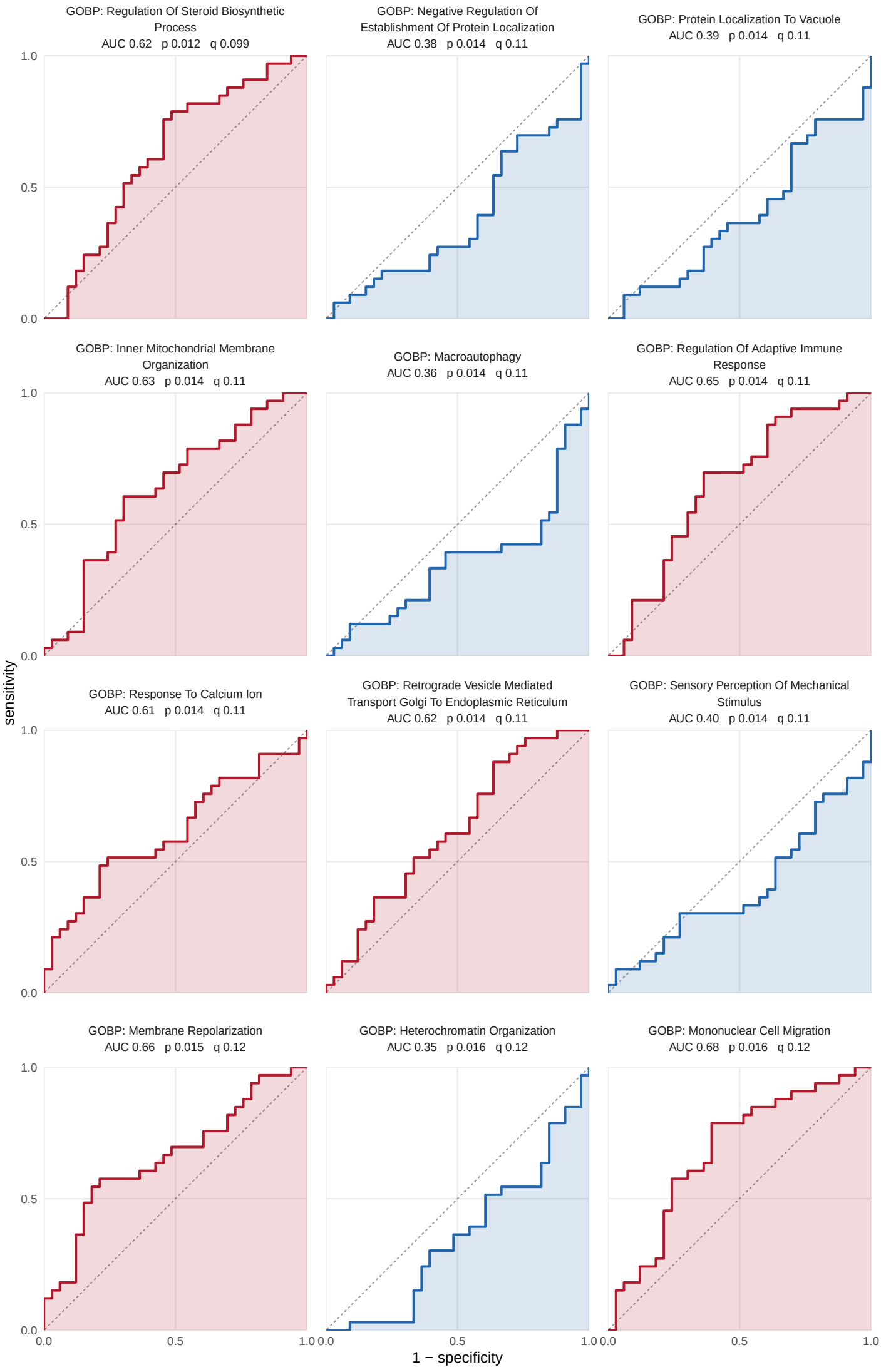

GOBP: Regulation Of Ketone Metabolic Process
AUC 0.61 p 0.0055 q 0.068
GOBP: Respiratory Chain Complex IV Assembly
AUC 0.63 p 0.0055 q 0.068
GOBP: Late Endosome To Vacuole Transport
AUC 0.33 p 0.0057 q 0.07
GOBP: Autophagosome Maturation
AUC 0.30 p 0.0058 q 0.07
GOBP: Negative Regulation Of Gene Expression Epigenetic
AUC 0.34 p 0.0058 q 0.07
GOBP: Regulation Of T Cell Activation
AUC 0.64 p 0.0058 q 0.07
GOBP: Homotypic Cell Cell Adhesion
AUC 0.68 p 0.0062 q 0.072
GOBP: Microtubule Polymerization
AUC 0.36 p 0.0062 q 0.072
GOBP: Platelet Aggregation
AUC 0.67 p 0.0062 q 0.072
GOBP: Positive Regulation Of Plasma Membrane Bounded Cell Projection Assembly
AUC 0.67 p 0.0062 q 0.072
GOBP: Positive Regulation Of Proteolysis
AUC 0.30 p 0.0066 q 0.074
GOBP: Regulation Of WNT Signaling Pathway
AUC 0.33 p 0.0066 q 0.074
sensitivity
1 − specificity
0.0
0.5
1.0

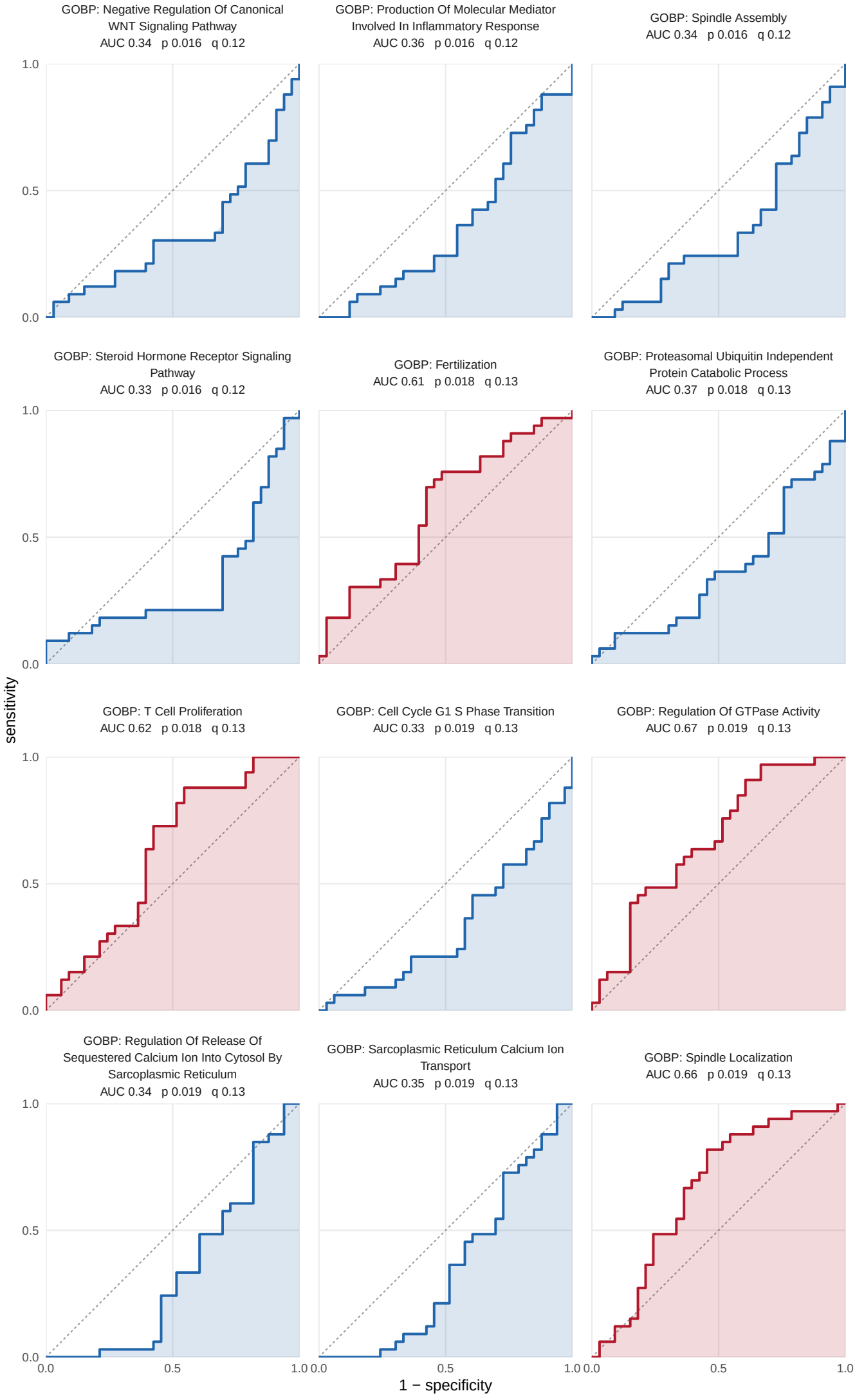

GOBP: Tumor Necrosis Factor Mediated Signaling Pathway
AUC 0.33 p 0.0066 q 0.074
GOBP: Cilium Movement
AUC 0.33 p 0.007 q 0.075
GOBP: Dendritic Spine Development
AUC 0.68 p 0.007 q 0.075
GOBP: Endosome Organization
AUC 0.34 p 0.007 q 0.075
GOBP: Muscle Cell Migration
AUC 0.68 p 0.007 q 0.075
GOBP: Positive Regulation Of Protein Modification Process
AUC 0.36 p 0.007 q 0.075
GOBP: Regulation Of Proteolysis Involved In Protein Catabolic Process
AUC 0.30 p 0.007 q 0.075
GOBP: Antigen Receptor Mediated Signaling Pathway
AUC 0.66 p 0.0074 q 0.075
GOBP: Cell Chemotaxis
AUC 0.70 p 0.0074 q 0.075
GOBP: Establishment Of Protein Localization To Vacuole
AUC 0.35 p 0.0074 q 0.075
GOBP: Mitochondrial Respiratory Chain Complex Assembly
AUC 0.63 p 0.0074 q 0.075
GOBP: Mitochondrion Organization
AUC 0.66 p 0.0074 q 0.075
sensitivity
1 − specificity
0.0
0.5
1.0

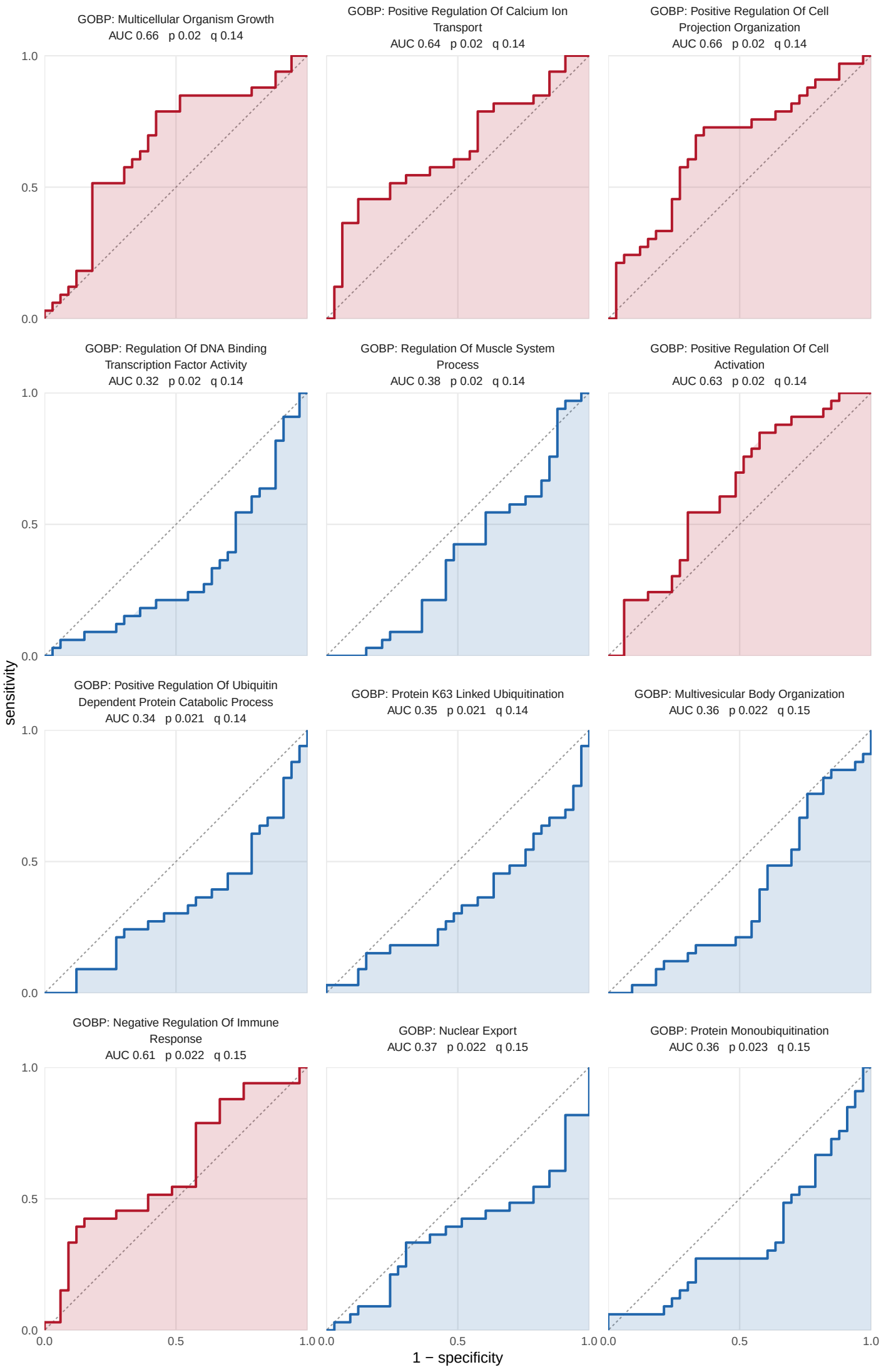

GOBP: Phagocytosis
AUC 0.69 p 0.0074 q 0.075
GOBP: Regulation Of Tumor Necrosis Factor Mediated Signaling Pathway
AUC 0.32 p 0.0074 q 0.075
GOBP: Release Of Sequestered Calcium Ion Into Cytosol By Endoplasmic Reticulum
AUC 0.33 p 0.0074 q 0.075
GOBP: Myeloid Leukocyte Differentiation
AUC 0.64 p 0.0078 q 0.078
GOBP: Mammary Gland Development
AUC 0.64 p 0.0078 q 0.078
GOBP: Protein Homooligomerization
AUC 0.64 p 0.0078 q 0.078
GOBP: Meiotic Cell Cycle
AUC 0.28 p 0.0083 q 0.08
GOBP: Neuron Migration
AUC 0.65 p 0.0083 q 0.08
GOBP: Proton Transmembrane Transport
AUC 0.63 p 0.0083 q 0.08
GOBP: Regulation Of Proteasomal Ubiquitin Dependent Protein Catabolic Process
AUC 0.35 p 0.0083 q 0.08
GOBP: Negative Regulation Of Monoatomic Ion Transmembrane Transport
AUC 0.34 p 0.0084 q 0.08
GOBP: Fatty Acid Metabolic Process
AUC 0.60 p 0.0088 q 0.082
sensitivity
1 − specificity
0.0
0.5
1.0

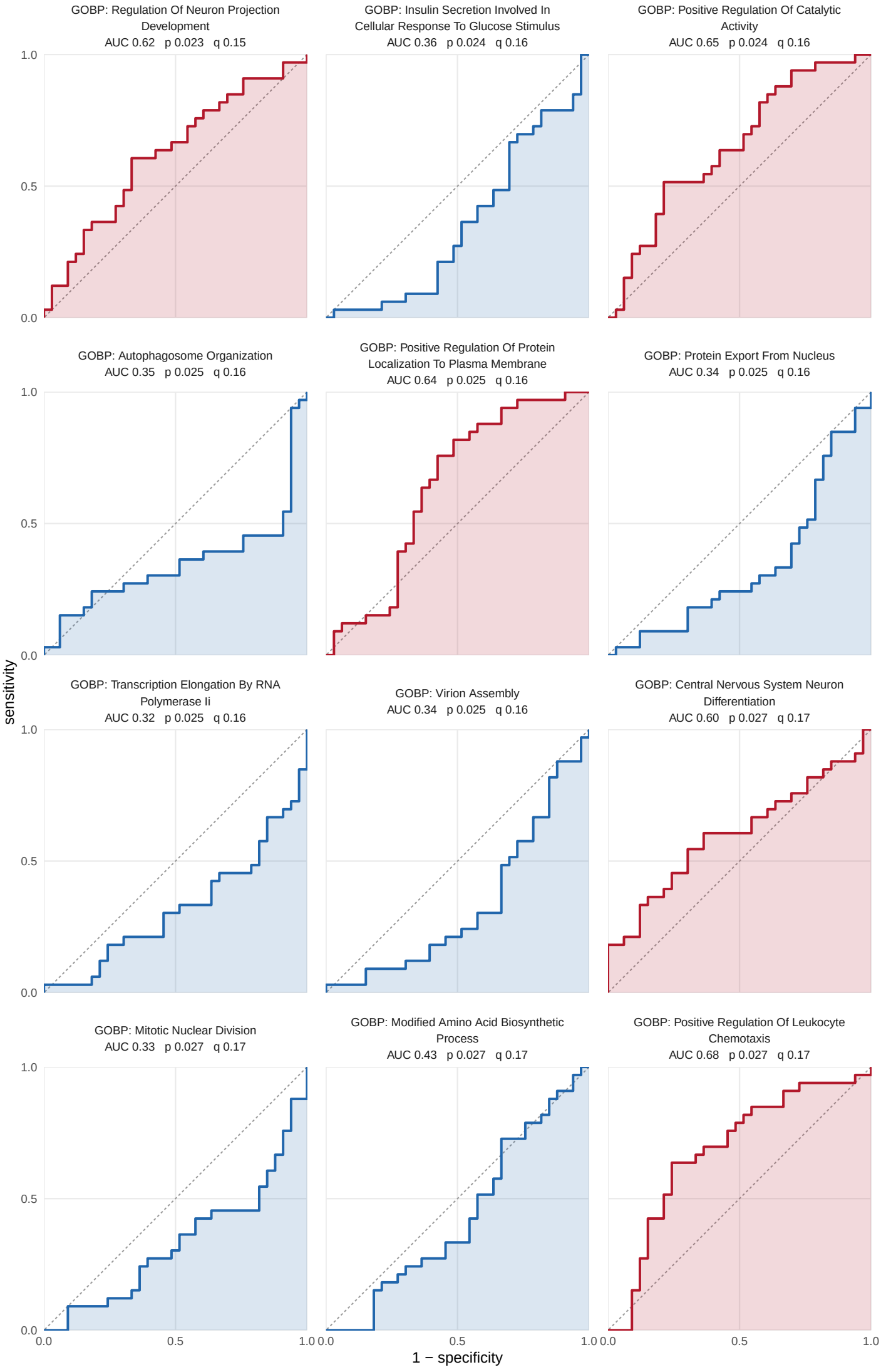

GOBP: Negative Regulation Of Calcium Ion Transport
AUC 0.35 p 0.0088 q 0.082
GOBP: Nuclear Envelope Organization
AUC 0.35 p 0.0088 q 0.082
GOBP: Response To Cadmium Ion
AUC 0.34 p 0.0088 q 0.082
GOBP: Glycerophospholipid Metabolic Process
AUC 0.34 p 0.0093 q 0.085
GOBP: Multivesicular Body Sorting Pathway
AUC 0.34 p 0.0093 q 0.085
GOBP: tRNA Metabolic Process
AUC 0.63 p 0.0093 q 0.085
GOBP: Myeloid Leukocyte Migration
AUC 0.70 p 0.0094 q 0.085
GOBP: Cortical Cytoskeleton Organization
AUC 0.67 p 0.0098 q 0.087
GOBP: Cytochrome Complex Assembly
AUC 0.63 p 0.0098 q 0.087
GOBP: Monoatomic Anion Transport
AUC 0.67 p 0.0098 q 0.087
GOBP: Regulation Of Fatty Acid Metabolic Process
AUC 0.61 p 0.0098 q 0.087
GOBP: Endothelium Development
AUC 0.64 p 0.01 q 0.091
sensitivity
1 − specificity
0.0
0.5
1.0

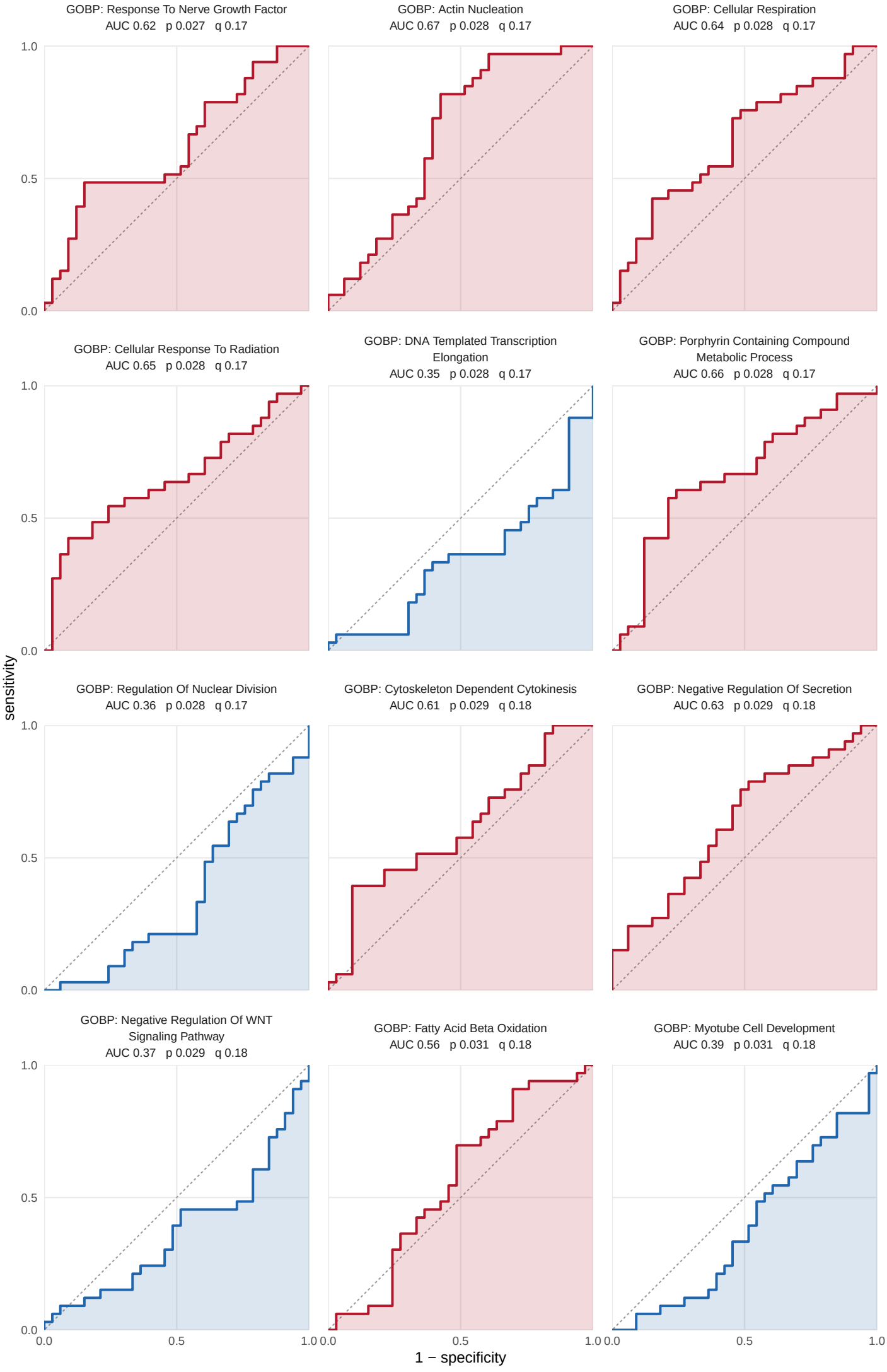

GOBP: Protein Transmembrane Transport
AUC 0.63 p 0.01 q 0.091
GOBP: Nuclear Membrane Reassembly
AUC 0.32 p 0.011 q 0.093
GOBP: Positive Regulation Of Nucleocytoplasmic Transport
AUC 0.35 p 0.011 q 0.093
GOBP: Protein Localization To Mitochondrion
AUC 0.66 p 0.011 q 0.093
GOBP: Regulation Of Lipid Metabolic Process
AUC 0.63 p 0.011 q 0.093
GOBP: Regulation Of Protein Ubiquitination
AUC 0.34 p 0.011 q 0.093
GOBP: Endothelial Cell Proliferation
AUC 0.64 p 0.012 q 0.096
GOBP: Monocarboxylic Acid Catabolic Process
AUC 0.58 p 0.012 q 0.096
GOBP: Positive Regulation Of Neuron Projection Development
AUC 0.67 p 0.012 q 0.096
GOBP: Regulation Of Amide Metabolic Process
AUC 0.67 p 0.012 q 0.096
GOBP: Lysosomal Transport
AUC 0.38 p 0.012 q 0.099
GOBP: Organic Acid Transport
AUC 0.67 p 0.012 q 0.099
sensitivity
1 − specificity
0.0
0.5
1.0

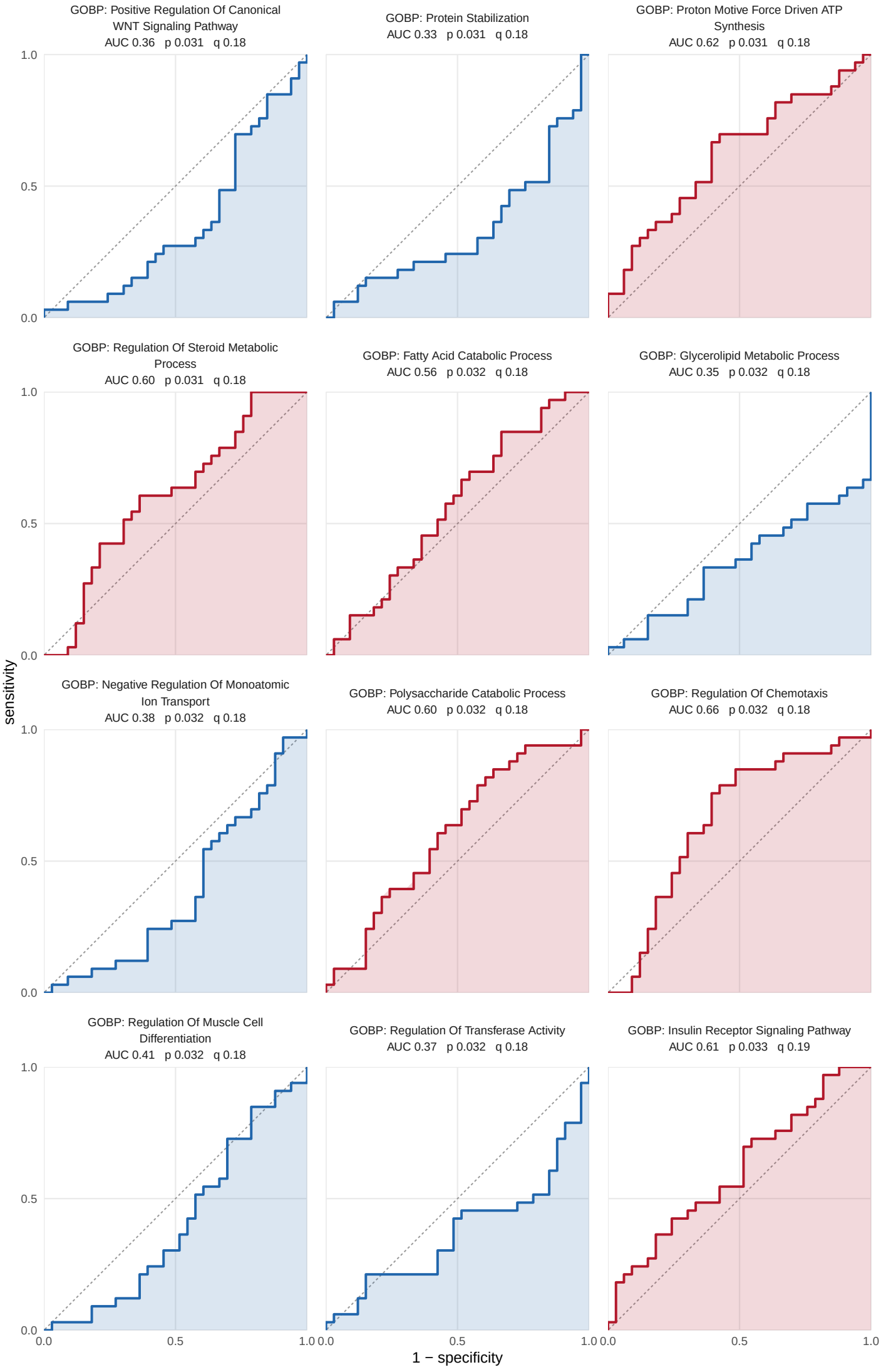

GOBP: Regulation Of Steroid Biosynthetic Process
AUC 0.62 p 0.012 q 0.099
GOBP: Negative Regulation Of Establishment Of Protein Localization
AUC 0.38 p 0.014 q 0.11
GOBP: Protein Localization To Vacuole
AUC 0.39 p 0.014 q 0.11
GOBP: Inner Mitochondrial Membrane Organization
AUC 0.63 p 0.014 q 0.11
GOBP: Macroautophagy
AUC 0.36 p 0.014 q 0.11
GOBP: Regulation Of Adaptive Immune Response
AUC 0.65 p 0.014 q 0.11
GOBP: Response To Calcium Ion
AUC 0.61 p 0.014 q 0.11
GOBP: Retrograde Vesicle Mediated Transport Golgi To Endoplasmic Reticulum
AUC 0.62 p 0.014 q 0.11
GOBP: Sensory Perception Of Mechanical Stimulus
AUC 0.40 p 0.014 q 0.11
GOBP: Membrane Repolarization
AUC 0.66 p 0.015 q 0.12
GOBP: Heterochromatin Organization
AUC 0.35 p 0.016 q 0.12
GOBP: Mononuclear Cell Migration
AUC 0.68 p 0.016 q 0.12
sensitivity
1 − specificity
0.0
0.5
1.0

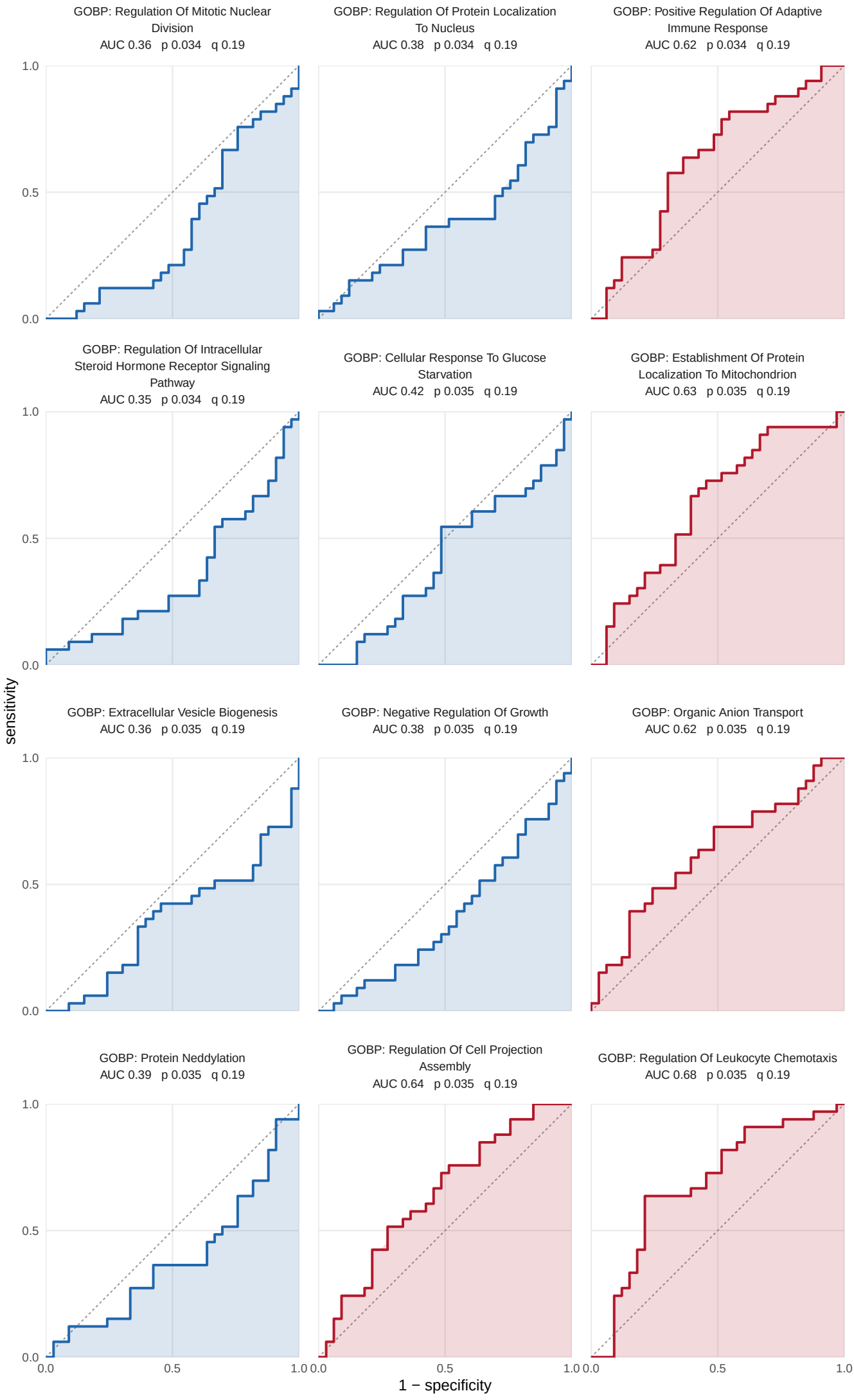

GOBP: Negative Regulation Of Canonical WNT Signaling Pathway
AUC 0.34 p 0.016 q 0.12
GOBP: Production Of Molecular Mediator Involved In Inflammatory Response
AUC 0.36 p 0.016 q 0.12
GOBP: Spindle Assembly
AUC 0.34 p 0.016 q 0.12
GOBP: Steroid Hormone Receptor Signaling Pathway
AUC 0.33 p 0.016 q 0.12
GOBP: Fertilization
AUC 0.61 p 0.018 q 0.13
GOBP: Proteasomal Ubiquitin Independent Protein Catabolic Process
AUC 0.37 p 0.018 q 0.13
GOBP: T Cell Proliferation
AUC 0.62 p 0.018 q 0.13
GOBP: Cell Cycle G1 S Phase Transition
AUC 0.33 p 0.019 q 0.13
GOBP: Regulation Of GTPase Activity
AUC 0.67 p 0.019 q 0.13
GOBP: Regulation Of Release Of Sequestered Calcium Ion Into Cytosol By Sarcoplasmic Reticulum
AUC 0.34 p 0.019 q 0.13
GOBP: Sarcoplasmic Reticulum Calcium Ion Transport
AUC 0.35 p 0.019 q 0.13
GOBP: Spindle Localization
AUC 0.66 p 0.019 q 0.13
sensitivity
1 − specificity
0.0
0.5
1.0

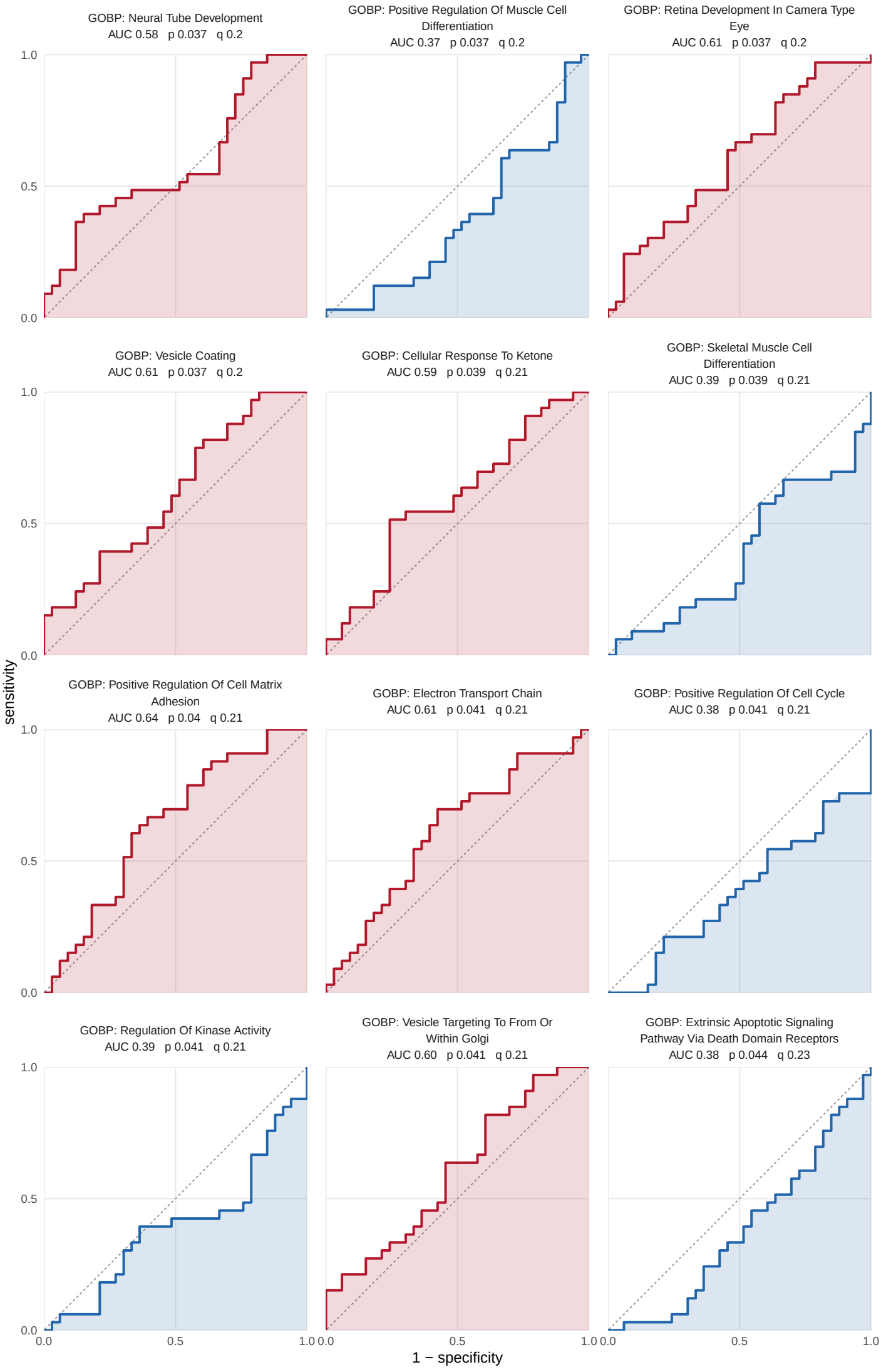

GOBP: Multicellular Organism Growth
AUC 0.66 p 0.02 q 0.14
GOBP: Positive Regulation Of Calcium Ion Transport
AUC 0.64 p 0.02 q 0.14
GOBP: Positive Regulation Of Cell Projection Organization
AUC 0.66 p 0.02 q 0.14
GOBP: Regulation Of DNA Binding Transcription Factor Activity
AUC 0.32 p 0.02 q 0.14
GOBP: Regulation Of Muscle System Process
AUC 0.38 p 0.02 q 0.14
GOBP: Positive Regulation Of Cell Activation
AUC 0.63 p 0.02 q 0.14
GOBP: Positive Regulation Of Ubiquitin Dependent Protein Catabolic Process
AUC 0.34 p 0.021 q 0.14
GOBP: Protein K63 Linked Ubiquitination
AUC 0.35 p 0.021 q 0.14
GOBP: Multivesicular Body Organization
AUC 0.36 p 0.022 q 0.15
GOBP: Negative Regulation Of Immune Response
AUC 0.61 p 0.022 q 0.15
GOBP: Nuclear Export
AUC 0.37 p 0.022 q 0.15
GOBP: Protein Monoubiquitination
AUC 0.36 p 0.023 q 0.15
sensitivity
1 − specificity
0.0
0.5
1.0

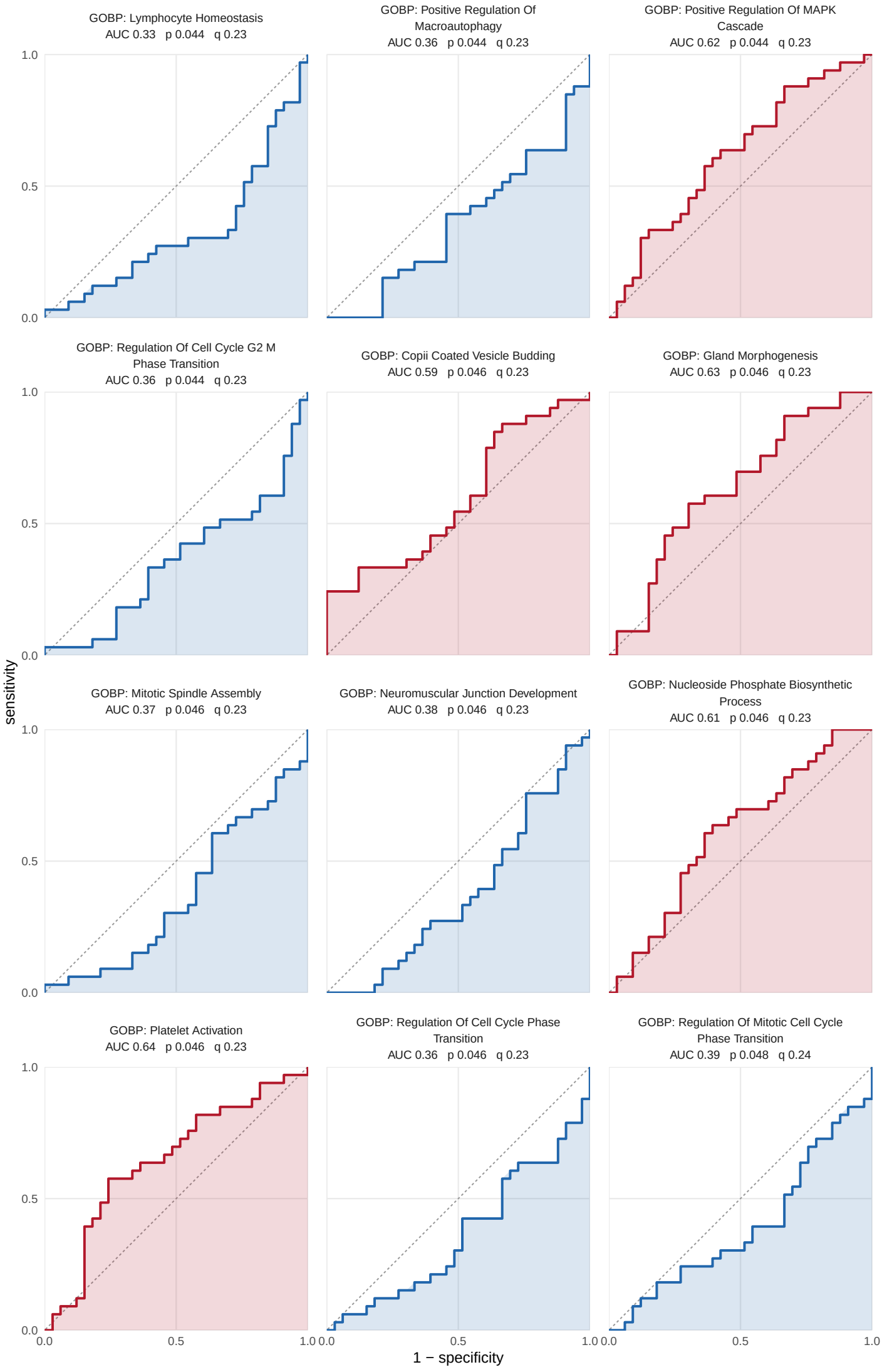

GOBP: Regulation Of Neuron Projection Development
AUC 0.62 p 0.023 q 0.15
GOBP: Insulin Secretion Involved In Cellular Response To Glucose Stimulus
AUC 0.36 p 0.024 q 0.16
GOBP: Positive Regulation Of Catalytic Activity
AUC 0.65 p 0.024 q 0.16
GOBP: Autophagosome Organization
AUC 0.35 p 0.025 q 0.16
GOBP: Positive Regulation Of Protein Localization To Plasma Membrane
AUC 0.64 p 0.025 q 0.16
GOBP: Protein Export From Nucleus
AUC 0.34 p 0.025 q 0.16
GOBP: Transcription Elongation By RNA Polymerase Ii
AUC 0.32 p 0.025 q 0.16
GOBP: Virion Assembly
AUC 0.34 p 0.025 q 0.16
GOBP: Central Nervous System Neuron Differentiation
AUC 0.60 p 0.027 q 0.17
GOBP: Mitotic Nuclear Division
AUC 0.33 p 0.027 q 0.17
GOBP: Modified Amino Acid Biosynthetic Process
AUC 0.43 p 0.027 q 0.17
GOBP: Positive Regulation Of Leukocyte Chemotaxis
AUC 0.68 p 0.027 q 0.17
sensitivity
1 − specificity
0.0
0.5
1.0


GOBP: Response To Nerve Growth Factor
AUC 0.62 p 0.027 q 0.17
GOBP: Actin Nucleation
AUC 0.67 p 0.028 q 0.17
GOBP: Cellular Respiration
AUC 0.64 p 0.028 q 0.17
GOBP: Cellular Response To Radiation
AUC 0.65 p 0.028 q 0.17
GOBP: DNA Templated Transcription Elongation
AUC 0.35 p 0.028 q 0.17
GOBP: Porphyrin Containing Compound Metabolic Process
AUC 0.66 p 0.028 q 0.17
GOBP: Regulation Of Nuclear Division
AUC 0.36 p 0.028 q 0.17
GOBP: Cytoskeleton Dependent Cytokinesis
AUC 0.61 p 0.029 q 0.18
GOBP: Negative Regulation Of Secretion
AUC 0.63 p 0.029 q 0.18
GOBP: Negative Regulation Of WNT Signaling Pathway
AUC 0.37 p 0.029 q 0.18
GOBP: Fatty Acid Beta Oxidation
AUC 0.56 p 0.031 q 0.18
GOBP: Myotube Cell Development
AUC 0.39 p 0.031 q 0.18
sensitivity
1 − specificity
0.0
0.5
1.0

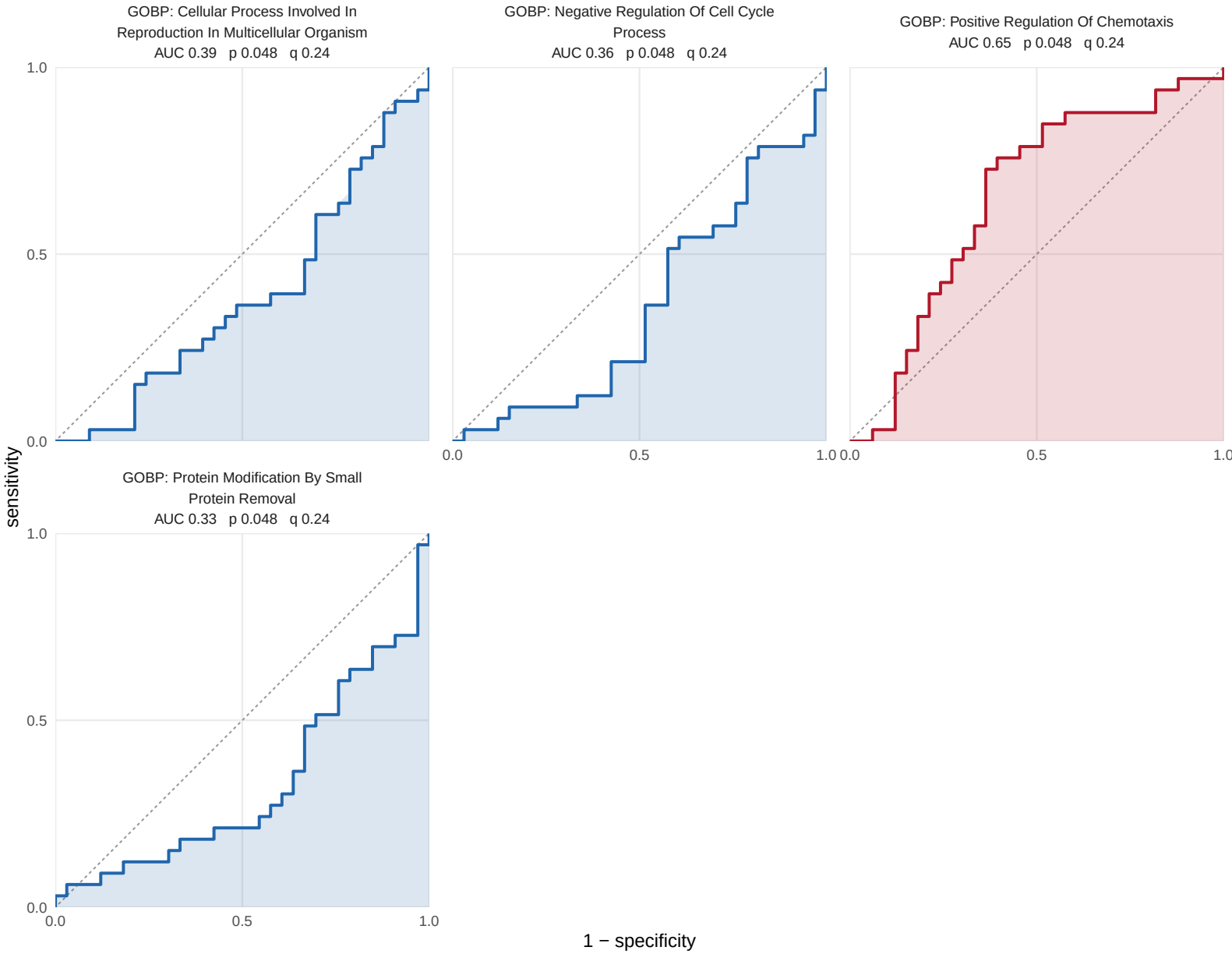

GOBP: Positive Regulation Of Canonical WNT Signaling Pathway
AUC 0.36 p 0.031 q 0.18
GOBP: Protein Stabilization
AUC 0.33 p 0.031 q 0.18
GOBP: Proton Motive Force Driven ATP Synthesis
AUC 0.62 p 0.031 q 0.18
GOBP: Regulation Of Steroid Metabolic Process
AUC 0.60 p 0.031 q 0.18
GOBP: Fatty Acid Catabolic Process
AUC 0.56 p 0.032 q 0.18
GOBP: Glycerolipid Metabolic Process
AUC 0.35 p 0.032 q 0.18
GOBP: Negative Regulation Of Monoatomic Ion Transport
AUC 0.38 p 0.032 q 0.18
GOBP: Polysaccharide Catabolic Process
AUC 0.60 p 0.032 q 0.18
GOBP: Regulation Of Chemotaxis
AUC 0.66 p 0.032 q 0.18
GOBP: Regulation Of Muscle Cell Differentiation
AUC 0.41 p 0.032 q 0.18
GOBP: Regulation Of Transferase Activity
AUC 0.37 p 0.032 q 0.18
GOBP: Insulin Receptor Signaling Pathway
AUC 0.61 p 0.033 q 0.19
sensitivity
1 − specificity
0.0
0.5
1.0


GOBP: Regulation Of Mitotic Nuclear Division
AUC 0.36 p 0.034 q 0.19
GOBP: Regulation Of Protein Localization To Nucleus
AUC 0.38 p 0.034 q 0.19
GOBP: Positive Regulation Of Adaptive Immune Response
AUC 0.62 p 0.034 q 0.19
GOBP: Regulation Of Intracellular Steroid Hormone Receptor Signaling Pathway
AUC 0.35 p 0.034 q 0.19
GOBP: Cellular Response To Glucose Starvation
AUC 0.42 p 0.035 q 0.19
GOBP: Establishment Of Protein Localization To Mitochondrion
AUC 0.63 p 0.035 q 0.19
GOBP: Extracellular Vesicle Biogenesis
AUC 0.36 p 0.035 q 0.19
GOBP: Negative Regulation Of Growth
AUC 0.38 p 0.035 q 0.19
GOBP: Organic Anion Transport
AUC 0.62 p 0.035 q 0.19
GOBP: Protein Neddylation
AUC 0.39 p 0.035 q 0.19
GOBP: Regulation Of Cell Projection Assembly
AUC 0.64 p 0.035 q 0.19
GOBP: Regulation Of Leukocyte Chemotaxis
AUC 0.68 p 0.035 q 0.19
sensitivity
1 − specificity
0.0
0.5
1.0

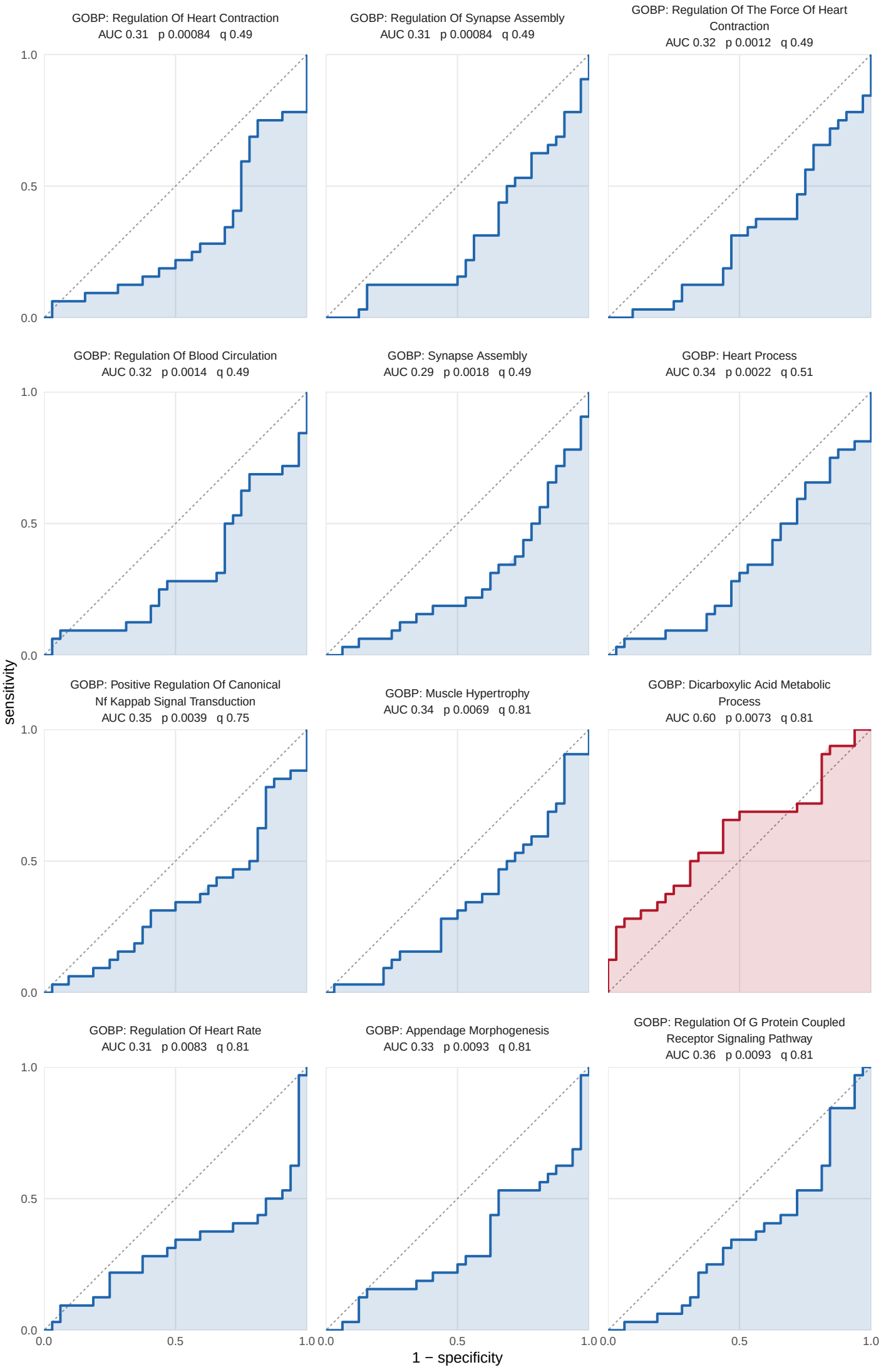

GOBP: Neural Tube Development
AUC 0.58 p 0.037 q 0.2
GOBP: Positive Regulation Of Muscle Cell Differentiation
AUC 0.37 p 0.037 q 0.2
GOBP: Retina Development In Camera Type Eye
AUC 0.61 p 0.037 q 0.2
GOBP: Vesicle Coating
AUC 0.61 p 0.037 q 0.2
GOBP: Cellular Response To Ketone
AUC 0.59 p 0.039 q 0.21
GOBP: Skeletal Muscle Cell Differentiation
AUC 0.39 p 0.039 q 0.21
GOBP: Positive Regulation Of Cell Matrix Adhesion
AUC 0.64 p 0.04 q 0.21
GOBP: Electron Transport Chain
AUC 0.61 p 0.041 q 0.21
GOBP: Positive Regulation Of Cell Cycle
AUC 0.38 p 0.041 q 0.21
GOBP: Regulation Of Kinase Activity
AUC 0.39 p 0.041 q 0.21
GOBP: Vesicle Targeting To From Or Within Golgi
AUC 0.60 p 0.041 q 0.21
GOBP: Extrinsic Apoptotic Signaling Pathway Via Death Domain Receptors
AUC 0.38 p 0.044 q 0.23
sensitivity
1 − specificity
0.0
0.5
1.0


GOBP: Lymphocyte Homeostasis
AUC 0.33 p 0.044 q 0.23
GOBP: Positive Regulation Of Macroautophagy
AUC 0.36 p 0.044 q 0.23
GOBP: Positive Regulation Of MAPK Cascade
AUC 0.62 p 0.044 q 0.23
GOBP: Regulation Of Cell Cycle G2 M Phase Transition
AUC 0.36 p 0.044 q 0.23
GOBP: Copii Coated Vesicle Budding
AUC 0.59 p 0.046 q 0.23
GOBP: Gland Morphogenesis
AUC 0.63 p 0.046 q 0.23
GOBP: Mitotic Spindle Assembly
AUC 0.37 p 0.046 q 0.23
GOBP: Neuromuscular Junction Development
AUC 0.38 p 0.046 q 0.23
GOBP: Nucleoside Phosphate Biosynthetic Process
AUC 0.61 p 0.046 q 0.23
GOBP: Platelet Activation
AUC 0.64 p 0.046 q 0.23
GOBP: Regulation Of Cell Cycle Phase Transition
AUC 0.36 p 0.046 q 0.23
GOBP: Regulation Of Mitotic Cell Cycle Phase Transition
AUC 0.39 p 0.048 q 0.24
sensitivity
1 − specificity
0.0
0.5
1.0

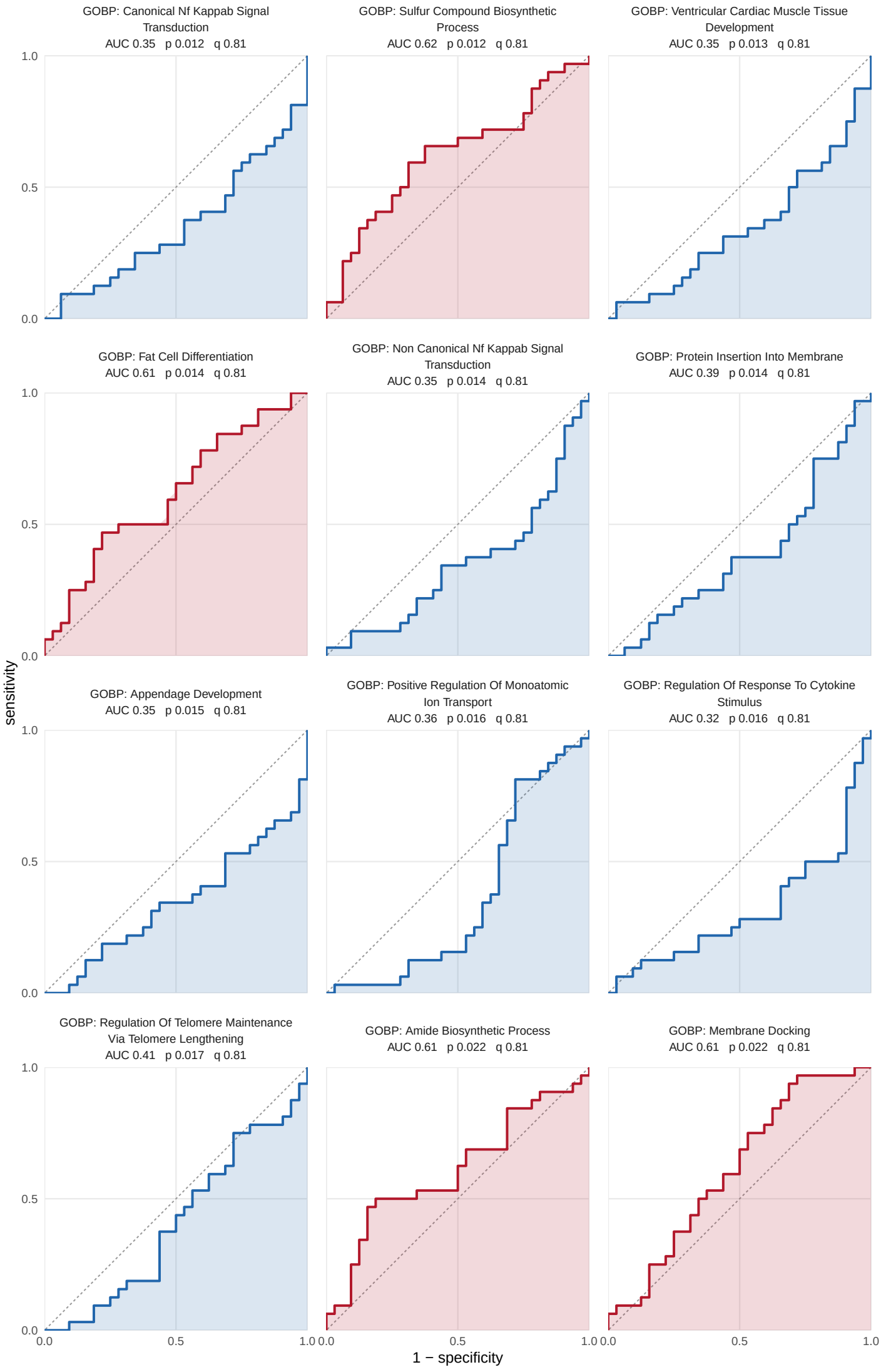
S7. ROC curves, Pre to post, HLRT, GOBP (page 24 of 24)



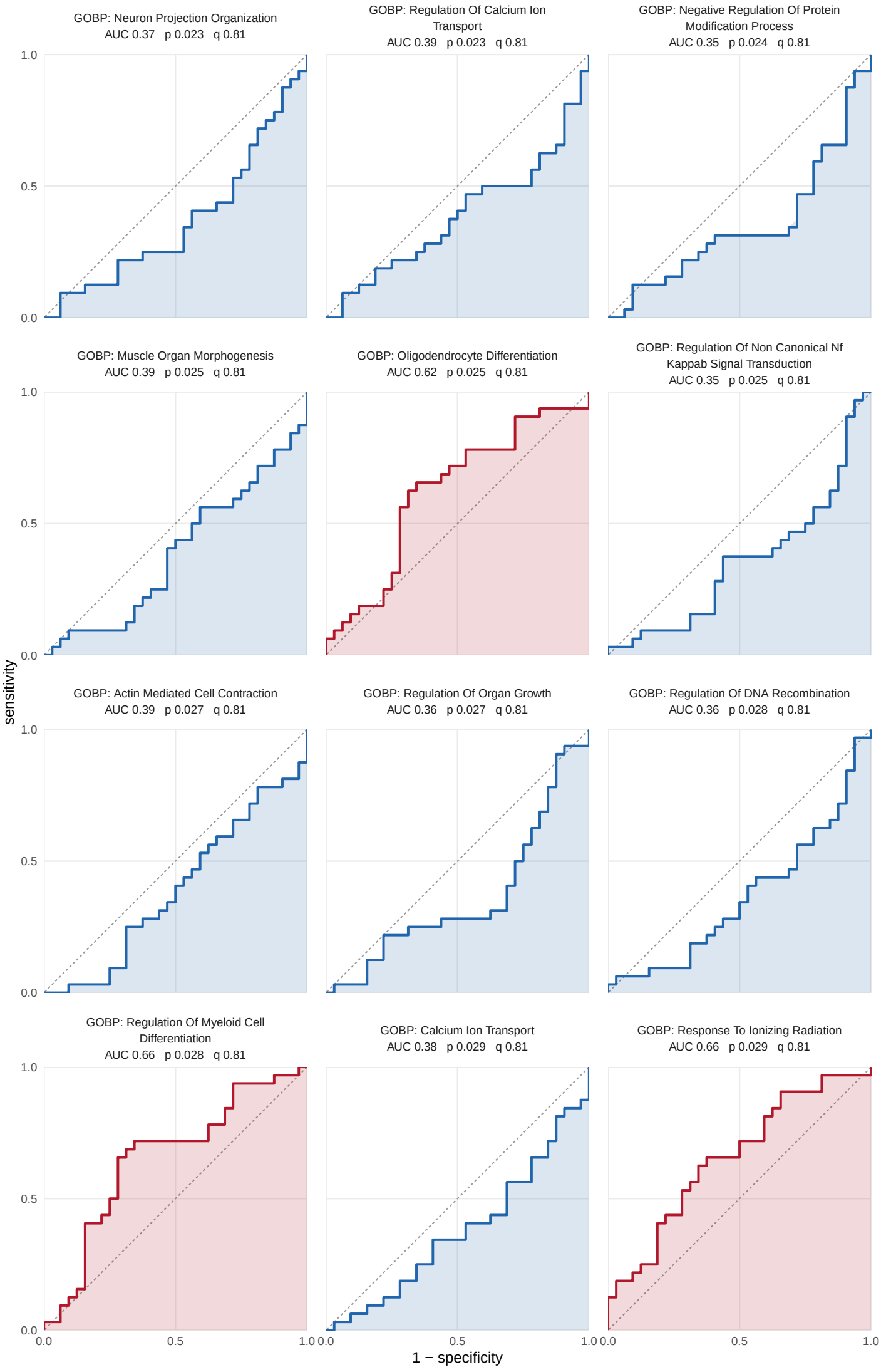
S8. ROC curves, Post BFR vs HLRT, GOBP (page 1 of 6)

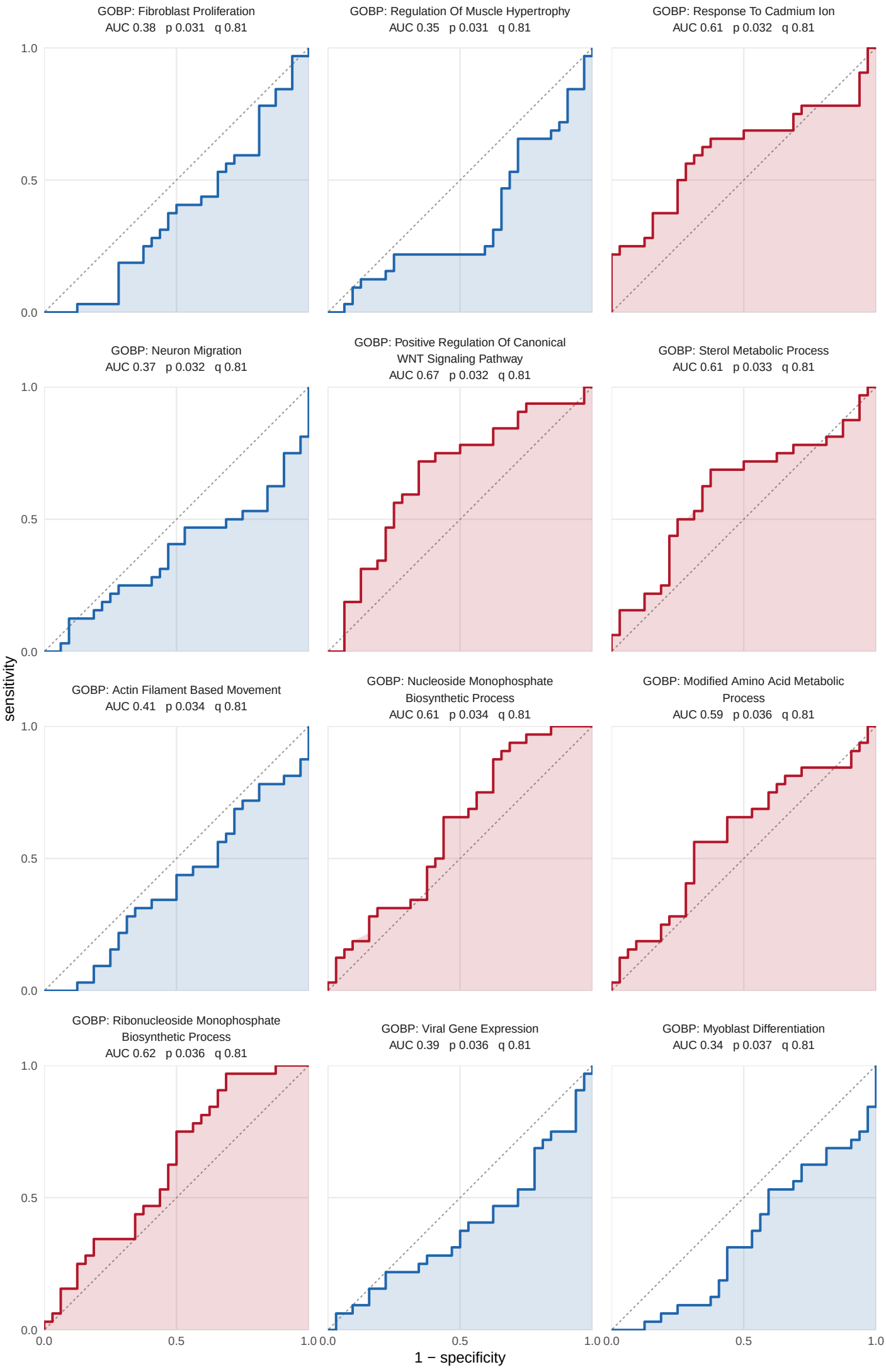


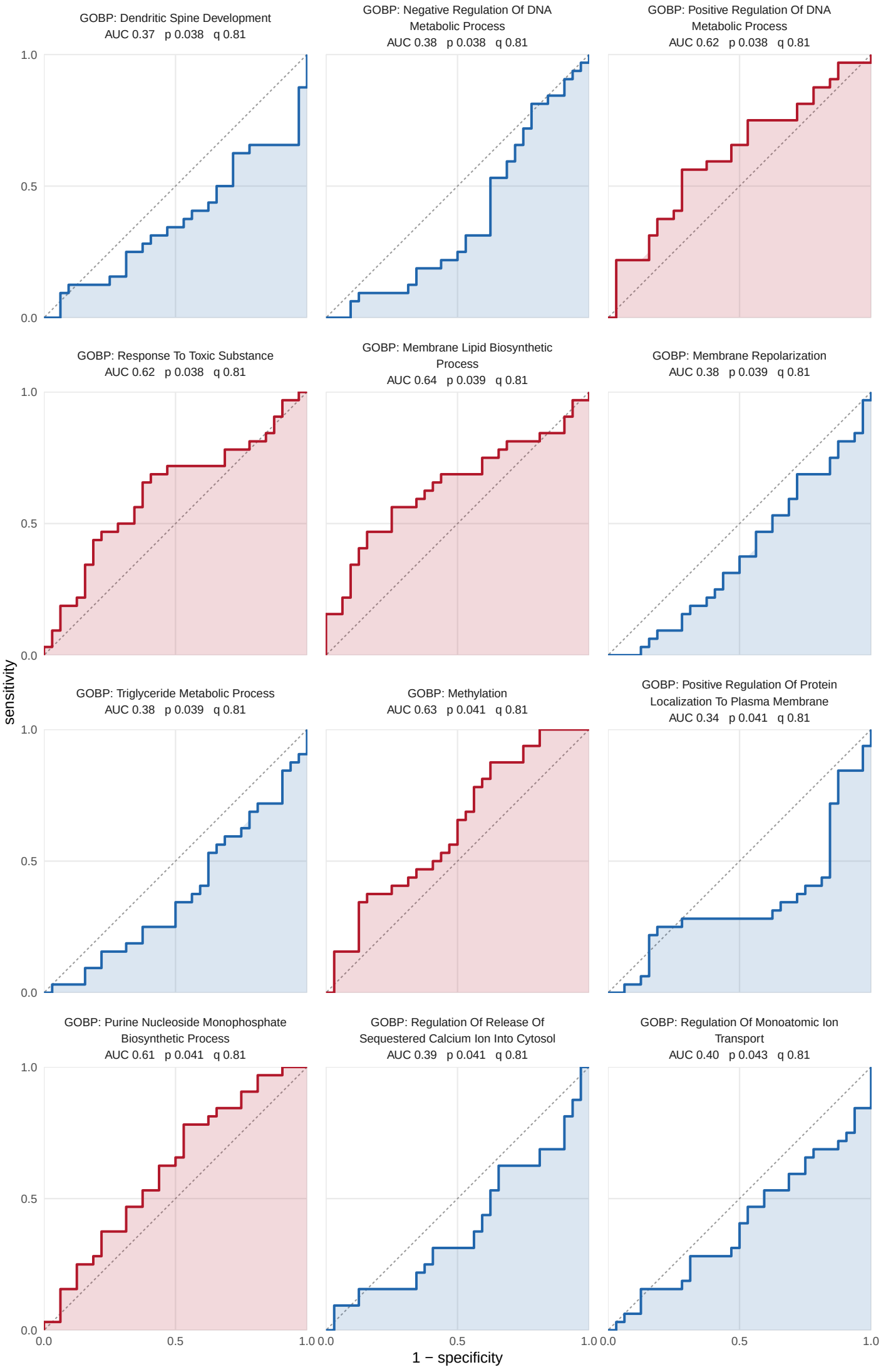
S8. ROC curves, Post BFR vs HLRT, GOBP (page 2 of 6)

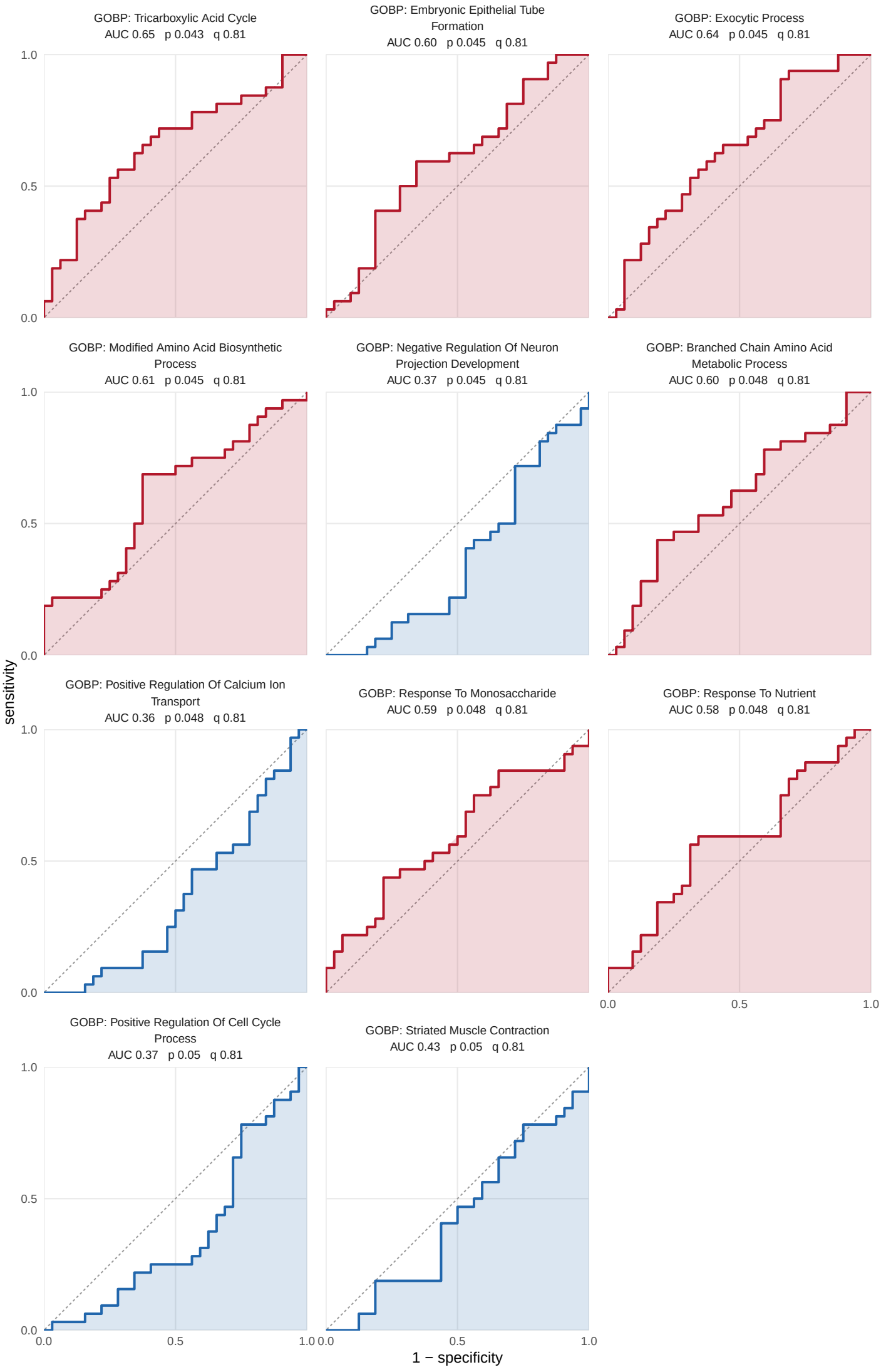


S8. ROC curves, Post BFR vs HLRT, GOBP (page 3 of 6)

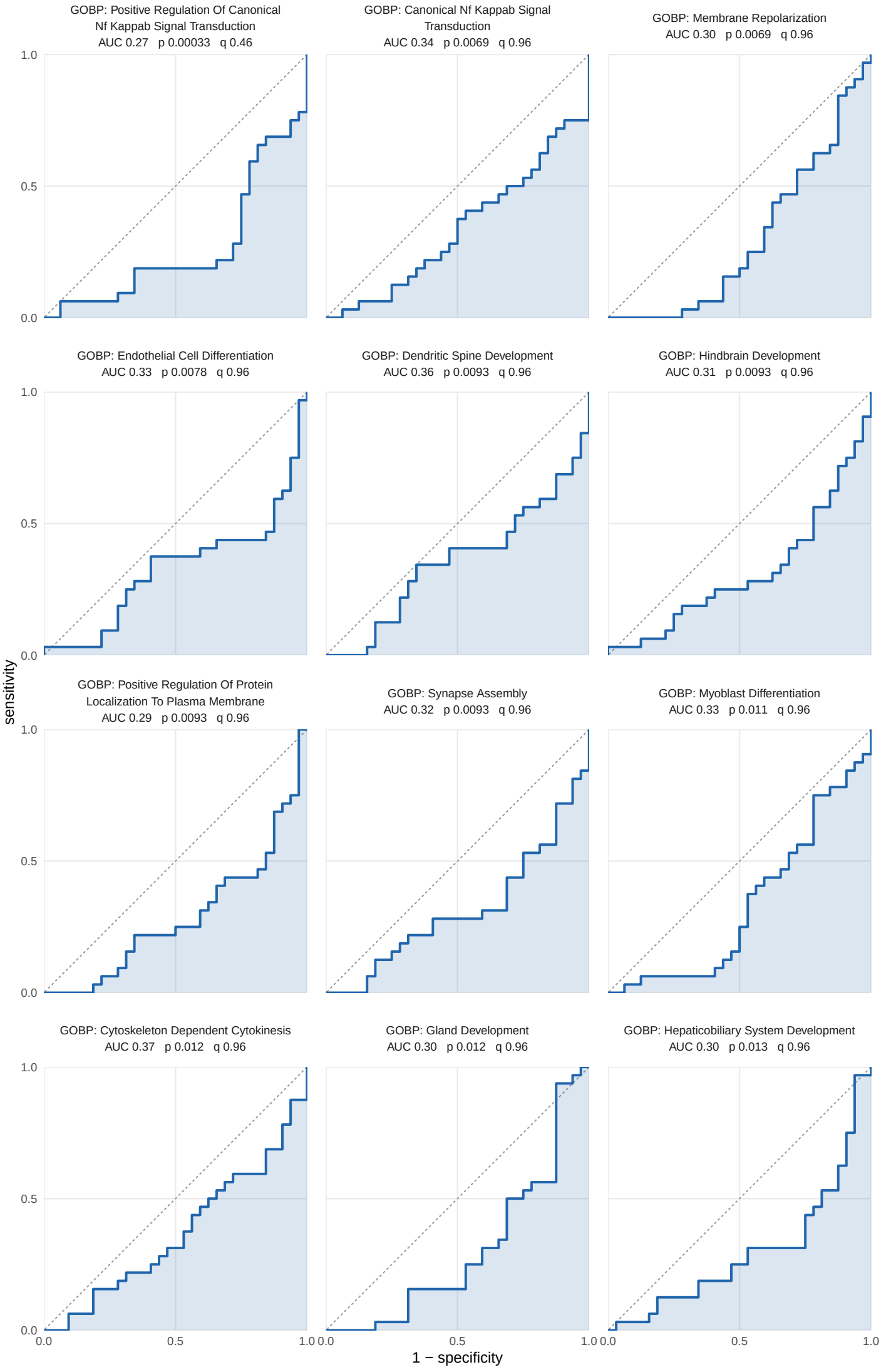


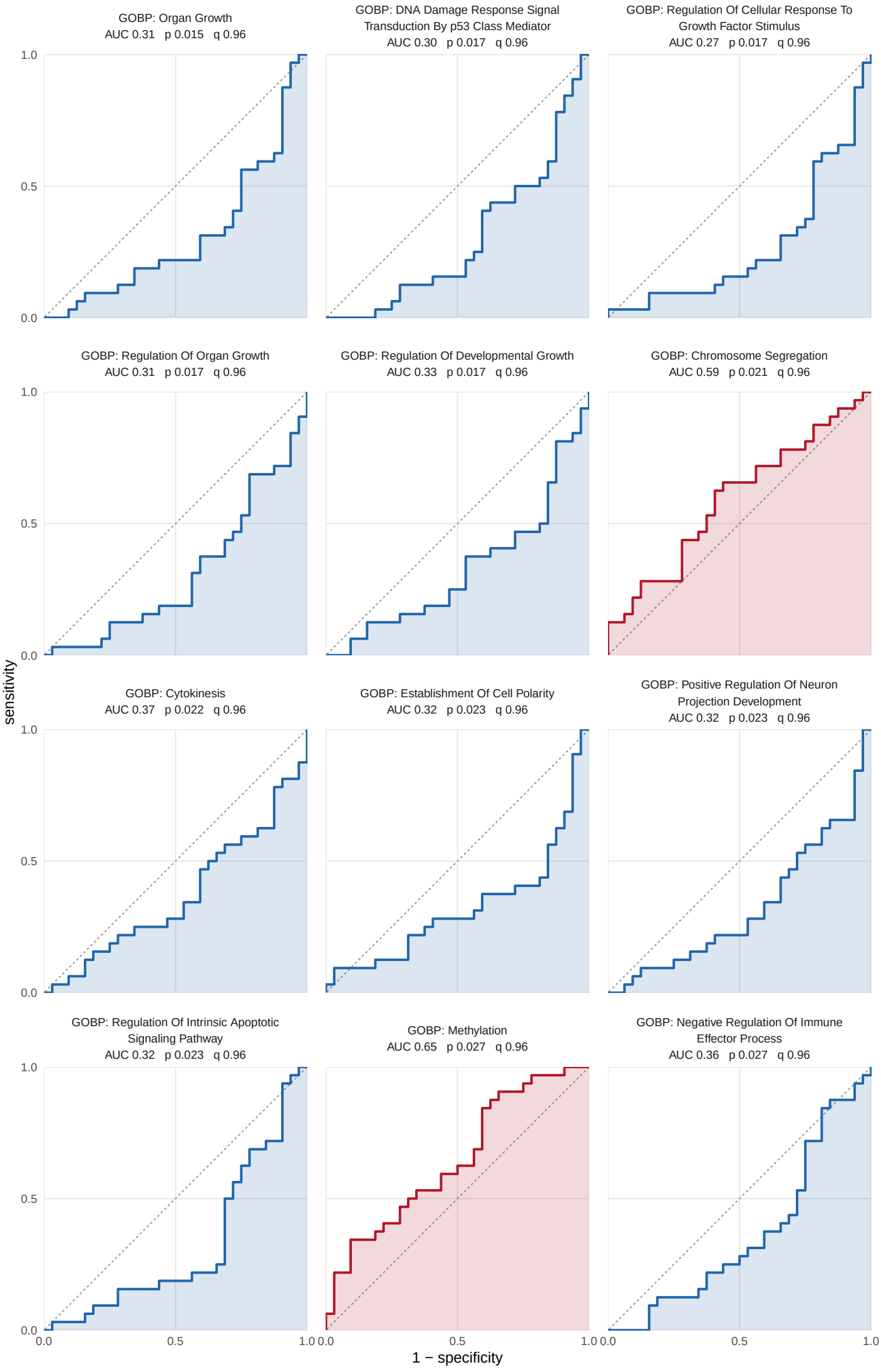

GOBP: Fibroblast Proliferation
AUC 0.38 p 0.031 q 0.81
GOBP: Regulation Of Muscle Hypertrophy
AUC 0.35 p 0.031 q 0.81
GOBP: Response To Cadmium Ion
AUC 0.61 p 0.032 q 0.81
GOBP: Neuron Migration
AUC 0.37 p 0.032 q 0.81
GOBP: Positive Regulation Of Canonical WNT Signaling Pathway
AUC 0.67 p 0.032 q 0.81
GOBP: Sterol Metabolic Process
AUC 0.61 p 0.033 q 0.81
GOBP: Actin Filament Based Movement
AUC 0.41 p 0.034 q 0.81
GOBP: Nucleoside Monophosphate Biosynthetic Process
AUC 0.61 p 0.034 q 0.81
GOBP: Modified Amino Acid Metabolic Process
AUC 0.59 p 0.036 q 0.81
GOBP: Ribonucleoside Monophosphate Biosynthetic Process
AUC 0.62 p 0.036 q 0.81
GOBP: Viral Gene Expression
AUC 0.39 p 0.036 q 0.81
GOBP: Myoblast Differentiation
AUC 0.34 p 0.037 q 0.81
sensitivity
1 − specificity
0.0
0.5
1.0


GOBP: Dendritic Spine Development
AUC 0.37 p 0.038 q 0.81
GOBP: Negative Regulation Of DNA Metabolic Process
AUC 0.38 p 0.038 q 0.81
GOBP: Positive Regulation Of DNA Metabolic Process
AUC 0.62 p 0.038 q 0.81
GOBP: Response To Toxic Substance
AUC 0.62 p 0.038 q 0.81
GOBP: Membrane Lipid Biosynthetic Process
AUC 0.64 p 0.039 q 0.81
GOBP: Membrane Repolarization
AUC 0.38 p 0.039 q 0.81
GOBP: Triglyceride Metabolic Process
AUC 0.38 p 0.039 q 0.81
GOBP: Methylation
AUC 0.63 p 0.041 q 0.81
GOBP: Positive Regulation Of Protein Localization To Plasma Membrane
AUC 0.34 p 0.041 q 0.81
GOBP: Purine Nucleoside Monophosphate Biosynthetic Process
AUC 0.61 p 0.041 q 0.81
GOBP: Regulation Of Release Of Sequestered Calcium Ion Into Cytosol
AUC 0.39 p 0.041 q 0.81
GOBP: Regulation Of Monoatomic Ion Transport
AUC 0.40 p 0.043 q 0.81
sensitivity
1 − specificity
0.0
0.5
1.0

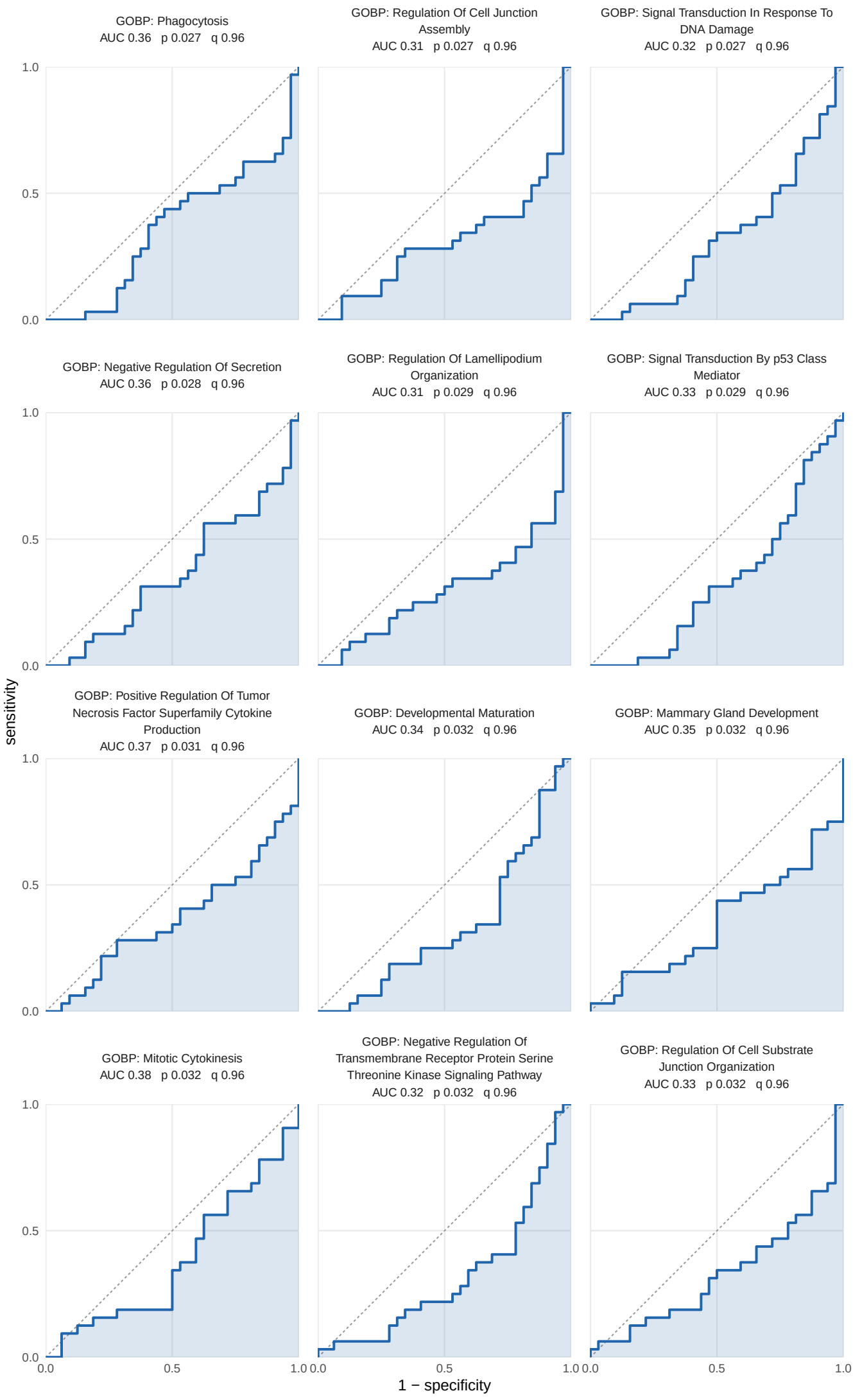

GOBP: Tricarboxylic Acid Cycle
AUC 0.65 p 0.043 q 0.81
GOBP: Embryonic Epithelial Tube Formation
AUC 0.60 p 0.045 q 0.81
GOBP: Exocytic Process
AUC 0.64 p 0.045 q 0.81
GOBP: Modified Amino Acid Biosynthetic Process
AUC 0.61 p 0.045 q 0.81
GOBP: Negative Regulation Of Neuron Projection Development
AUC 0.37 p 0.045 q 0.81
GOBP: Branched Chain Amino Acid Metabolic Process
AUC 0.60 p 0.048 q 0.81
GOBP: Positive Regulation Of Calcium Ion Transport
AUC 0.36 p 0.048 q 0.81
GOBP: Response To Monosaccharide
AUC 0.59 p 0.048 q 0.81
GOBP: Response To Nutrient
AUC 0.58 p 0.048 q 0.81
GOBP: Positive Regulation Of Cell Cycle Process
AUC 0.37 p 0.05 q 0.81
GOBP: Striated Muscle Contraction
AUC 0.43 p 0.05 q 0.81
sensitivity
1 − specificity
0.0
0.5
1.0

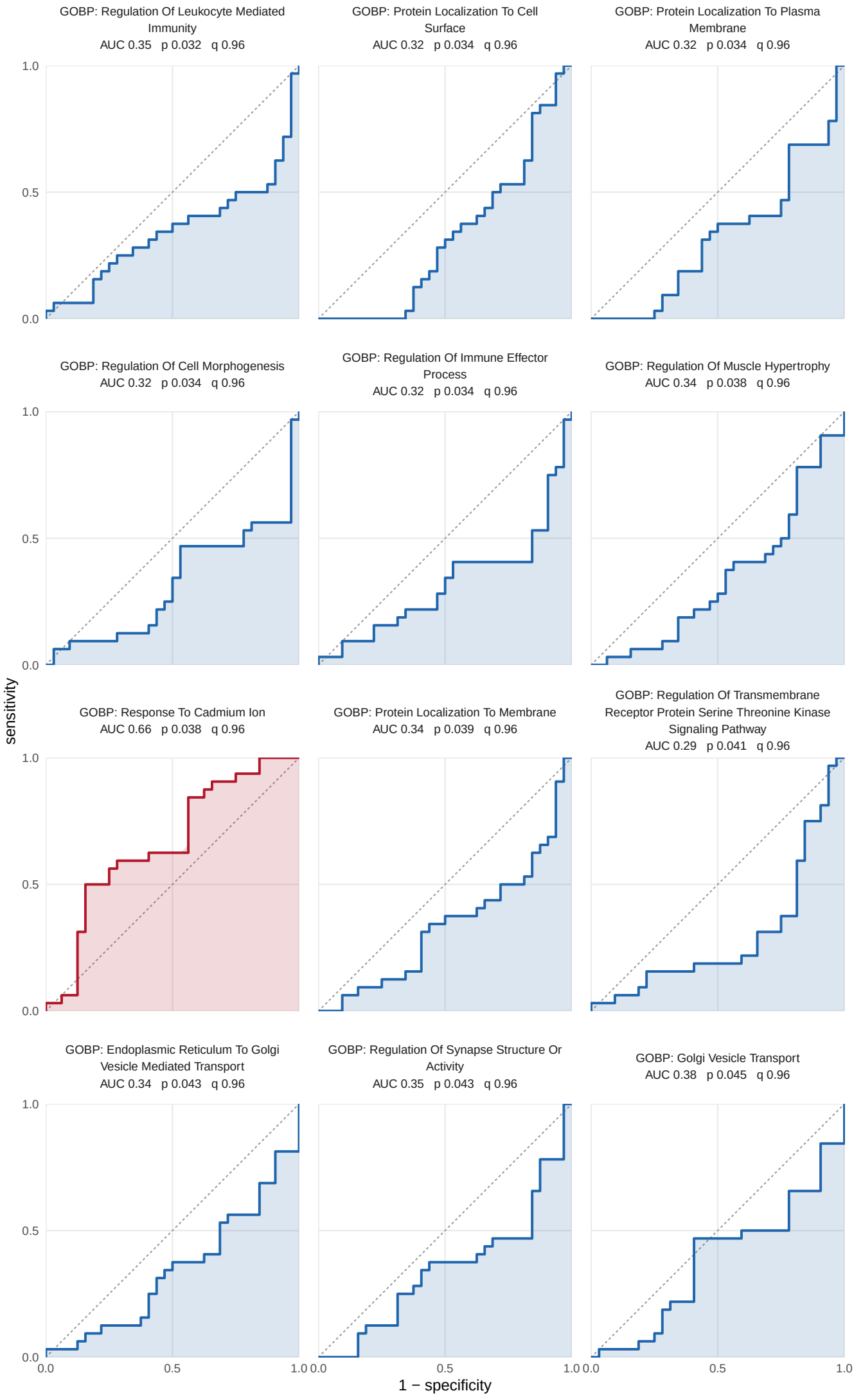
S9. ROC curves, Change, BFR vs HLRT, GOBP (page 1 of 5)




GOBP: Organ Growth
AUC 0.31 p 0.015 q 0.96
GOBP: DNA Damage Response Signal Transduction By p53 Class Mediator
AUC 0.30 p 0.017 q 0.96
GOBP: Regulation Of Cellular Response To Growth Factor Stimulus
AUC 0.27 p 0.017 q 0.96
GOBP: Regulation Of Organ Growth
AUC 0.31 p 0.017 q 0.96
GOBP: Regulation Of Developmental Growth
AUC 0.33 p 0.017 q 0.96
GOBP: Chromosome Segregation
AUC 0.59 p 0.021 q 0.96
GOBP: Cytokinesis
AUC 0.37 p 0.022 q 0.96
GOBP: Establishment Of Cell Polarity
AUC 0.32 p 0.023 q 0.96
GOBP: Positive Regulation Of Neuron Projection Development
AUC 0.32 p 0.023 q 0.96
GOBP: Regulation Of Intrinsic Apoptotic Signaling Pathway
AUC 0.32 p 0.023 q 0.96
GOBP: Methylation
AUC 0.65 p 0.027 q 0.96
GOBP: Negative Regulation Of Immune Effector Process
AUC 0.36 p 0.027 q 0.96
sensitivity
1 − specificity
0.0
0.5
1.0

S9. ROC curves, Change, BFR vs HLRT, GOBP (page 3 of 5)




GOBP: Regulation Of Leukocyte Mediated Immunity
AUC 0.35 p 0.032 q 0.96
GOBP: Protein Localization To Cell Surface
AUC 0.32 p 0.034 q 0.96
GOBP: Protein Localization To Plasma Membrane
AUC 0.32 p 0.034 q 0.96
GOBP: Regulation Of Cell Morphogenesis
AUC 0.32 p 0.034 q 0.96
GOBP: Regulation Of Immune Effector Process
AUC 0.32 p 0.034 q 0.96
GOBP: Regulation Of Muscle Hypertrophy
AUC 0.34 p 0.038 q 0.96
GOBP: Response To Cadmium Ion
AUC 0.66 p 0.038 q 0.96
GOBP: Protein Localization To Membrane
AUC 0.34 p 0.039 q 0.96
GOBP: Regulation Of Transmembrane Receptor Protein Serine Threonine Kinase Signaling Pathway
AUC 0.29 p 0.041 q 0.96
GOBP: Endoplasmic Reticulum To Golgi Vesicle Mediated Transport
AUC 0.34 p 0.043 q 0.96
GOBP: Regulation Of Synapse Structure Or Activity
AUC 0.35 p 0.043 q 0.96
GOBP: Golgi Vesicle Transport
AUC 0.38 p 0.045 q 0.96
sensitivity
1 − specificity
0.0
0.5
1.0

S9. ROC curves, Change, BFR vs HLRT, GOBP (page 5 of 5)

